B787 IOE Workbook

Answered, in flashcard form

Source: 787/A321/A330 Fleets OE Workbook, Version 3, June 2026. 787 content only; Airbus-only questions removed.

Workbook questions	**342**
Answered from the manuals	**249**
Routed to source to verify	**93**
Critical Observable Items	**120**
Drill cards / walkthrough items	**323 / 19**
Manuals used	**FOM Rev 125.1 (8/12/26), FCOM R10, QRH R7**
Compiled	**August 16, 2026**

How to read this document

Each question carries the answer in green, and nothing else. That green line is what you should be able to say cold. Everything under it is the backup: the full answer, the manual section, a verbatim quote where one exists, a note, and the workbook's own wording of the question where it differed.

An answer in amber has no source in the FOM, FCOM or QRH. It names the manual, card or system that holds it, or says plainly that no requirement is published. Those were routed, never guessed. Most need FCTM R9, MEL R5, the PRC, FD Pro, Comply365 or the OpSpecs.

Two questions ask for the flight deck door entry code and the crew rest door lock combination. Those are deliberately not recorded. They are security sensitive under FOM Chapter 15 and do not belong in a study document.

1.2.3 Perform Interior/Exterior Inspection

EMERGENCY EQUIPMENT

What flight deck emergency equipment must you check?

Extinguisher, crash axe, overhead door, EDDs, PBE, gloves, AVSAX, vest, flashlights

Under the Preliminary Preflight Procedure (FCOM NP.21.1) the emergency equipment check is: fire extinguisher checked and stowed, crash axe stowed, flight deck overhead door closed and latched, Emergency Descent Devices stowed, PBE stowed, gloves stowed, AVSAX stowed, life vest stowed, flashlights stowed. The fire extinguisher check means seal on the trigger, pull pin intact, gauge in the green (the same standard used for the cabin check in FCOM SP.1.4). The preceding lines also require oxygen pressure, hydraulic quantity and engine oil quantity to be sufficient for flight, and CREW OXYGEN LOW to be blank. Locations are shown in the FCOM 1.45.7 Emergency Equipment Locations - Flight Deck diagram.

> **FCOM NP.21.1 Preliminary Preflight Procedure - Captain or First Officer**
>
> *“Emergency equipment .... Check Fire extinguisher ....Checked and stowed Crash axe ....Stowed Flight deck overhead door .... Closed and latched Emergency Descent Devices ....Stowed PBE ....Stowed Gloves ....Stowed”*
>
> The list continues on NP.21.2 with AVSAX, Life Vest and Flashlights, all 'Stowed'. This block plus the Maintenance Logbook review is only required after a crew change or maintenance action; the flight deck equipment location diagram is FCOM 1.45.7.
>
> *Workbook wording: What flight deck emergency items are required to be checked?*

INTERNATIONAL THEATERS

Where is the aircraft Sanitation Certificate?

In the aircraft document folio and the OpSpecs

Japanese JCAB ramp-check items (Sanitation Certificate location, NADP 1 OpSpec evidence) are carried in the aircraft document folio and the OpSpec pages, not the FOM. Confirm the physical location on the aircraft and which OpSpec page satisfies the inspector. FOM 5.4.7 gives Ops Spec C068 as the NADP authority.

> **Aircraft document folio / OpSpecs**
>
> *Workbook wording: Where is the aircraft Sanitation Certificate located? (inspected during Japanese JCAB Ramp Checks)*

INTERNATIONAL THEATERS

Is HA approved for the NADP 1 departure?

Yes. Normal takeoff procedures satisfy it

FOM 5.4.7 carries the noise abatement departure procedure under Ops Spec C068. Compliance with published noise abatement procedures is required at noise-sensitive airports. Because normal takeoff procedures already satisfy both the close-in (NADP 1) and distant (NADP 2) profiles, there is no separate NADP profile to fly. The JCAB ramp inspector is looking for the OpSpec page that shows the authority.

> **FOM 5.4.7 Noise Abatement Departure Procedure (NADP), Ops Spec C068**
>
> *"(AS) Normal takeoff procedures satisfy either close-in or distant NADP profile requirements."*
>
> The FOM tags this (AS), and post-merger it is the governing procedure. Specific gradients and configuration-change altitudes come from the fleet manuals, the 10-7, or the Jeppesen Airway Manual noise abatement pages. Deviation is allowed for severe weather, traffic conflict or safety, and then the Captain requests an ATC clearance.
>
> *Workbook wording: Is HA allowed to perform the NADP 1 Departure Procedure? (Japanese JCAB Ramp Check question; they*

SECURITY AND DOORS

Where is the Jumpseat and Rest Facility Briefing Card?

On the aircraft. Policy is in the FOM

The Cockpit Jumpseat and Rest Facility Briefing Card is carried on the aircraft. FOM 5.2.6.2 covers the Crew Rest Facility Briefing and 5.1.23 the crew rest facility policy, including 5.1.23.3 Captain's Briefings Regarding Crew Rest Facility. Confirm the physical stowage location on the flight deck during your walkthrough.

> **FOM 5.1.23 / 5.2.6.2 / aircraft**
>
> *Workbook wording: Where is the Cockpit Jumpseat and/or Rest Facility Briefing Card located?*

SECURITY AND DOORS

How do you escape through the 787 overhead door?

Cover off, rotate the handle, the door falls inward, then descent devices

The flight deck overhead door is used for emergency egress using the installed descent devices. It can only be opened on the ground with the airplane depressurized. To open: remove the protective cover from the overhead door, rotate the door lock handle to the open position, and the door opens inside the flight deck and hangs from the ceiling on hinges. The EICAS advisory DOOR FD OVHD displays if the door is not closed and secure.

> **FCOM 1.40.19 Flight Deck Overhead Door**
>
> *""The flight deck overhead door is hinged to fall inward when opened.""*
>
> That WARNING is the practical gotcha: the door falls INWARD, so stand clear when you rotate the handle. Ground and depressurized only. Additional description at FCOM 1.30 Flight Deck Overhead Door and Vent. FCOM Bulletin HWI-26 R1 notes that after the flight deck door is set to normal closed, the overhead escape hatch remains available for egress in an evacuation.
>
> *Workbook wording: How would you operate/escape through the 787 Flight Deck Overhead Door?*

Crew Rest Procedures/Door Operation

CREW REST

Where are the crew rest requirements?

In the FOM crew rest facility section

Chapter 5 carries the crew rest policy. 5.1.23 covers the facility generally, 5.1.23.1 limitations and restrictions, 5.1.23.3 the Captain's briefing requirement, and 5.1.23.7 the 787 facility. Use is restricted to Flight Crew on the HA System Seniority list, and in some circumstances non-revenue pilots per 5.1.23.2.

> **FOM 5.1.23 Crew Rest Facility**
>
> *""The purpose of the crew rest facility is to provide a quiet place for the crew to rest. It is not a lounge, a meal area, or a place to stow personal carry-on items.""*
>
> RENUMBERED: this was 5.1.21 in FOM 123.1 and is 5.1.23 in Rev 125.1. Controlling access is called out as critical to the federally-mandated Fatigue Risk Management Plan, so this is a Captain enforcement item, not a courtesy. One occupant per bunk, one emergency oxygen mask per bunk, no food or beverage except bottled water, no alcohol in the cabin before use.
>
> *Workbook wording: Where can the Crew Rest procedures/requirements be found?*

CREW REST

What is the OFCR/OFAR door lock combination?

Not recorded. Security sensitive

FCOM 1.46 (OFCR) and FCOM 1.47 (OFAR) describe the entrance enclosure keyless lock: a door handle, a button keypad and a reset button. To unlock you enter the airline-programmed code and rotate the handle either direction; releasing the handle re-latches the bolt so the door can be closed without retracting it. The code itself is distributed through company security channels, not the manuals, and FOM 15.1.1.1 specifically forbids passing security access codes over ACARS. Do not write it down or transmit it.

> **FCOM 1.46 Overhead Flight Crew Rest - Door Lock Operation (also FCOM 1.47 OFAR)**
>
> *"To unlock the door, enter the airline-programmed code and rotate the door handle in either direction to retract the door bolt. When the door handle is released, the"*
>
> The FCOM only describes the keyless lock as taking an 'airline-programmed code' and never prints the digits. Actual combinations are Sensitive Security Information under FOM 15.1 Security Policy and are not in the FCOM, FOM or QRH.
>
> *Workbook wording: 787 - What is the OFCR/OFAR Door lock combination?*

CREW REST

How many may use the OFCR?

2 occupants

The primary 787 crew rest facility is the Overhead Flight Crew Rest (OFCR); the backup is the Overhead Flight Attendant Rest (OFAR). If flight time or FDP requires it the Company may provide up to two bunks in the OFAR, labeled alternate pilot bunks, with a laminated briefing card supplied to the Cabin Crew. That is limited to a maximum of one flight segment, extendable by one more segment if the OFCR deferral was discovered within 90 minutes of scheduled departure.

> **FOM 5.1.23.7 787 Crew Rest Facility**
>
> *""No more than 2 occupants are authorized to use the OFCR at any time.""*
>
> The OFCR seat, bunks and common area may not be occupied during taxi, takeoff or landing. 787s departing a domicile or outstation with an Augmented, Double Augmented or Double Crew must carry at least 4 prepackaged bedding packets (sheet, blanket, pillow and case). DESKTOP: the specific crew rest scheduling app is a company tool, not FOM content.
>
> *Workbook wording: Do you have a CREW REST APP? (for Long-Range flying w/more than 2 pilots)*

CREW REST

What is the flight deck door entry code?

Not recorded. Security sensitive

FOM 15.2.6.3 states the keypad control device shall not be used for normal entry and that entry attempts using the keypad shall be denied by the Flight Crew. FOM 15.2.6.3 also warns that an unexpected keypad activation signal must be treated as a Level 4 hijack or attempted hijack in progress until the Flight Attendants say otherwise. Normal access is by interphone plus visual identification through the view port (FOM 15.2.6). The flight deck door access system test, which uses the emergency access code, is FCOM SP.1.3.

> **FOM 15.2.6.3 Keypad Control**
>
> *"The keypad control device shall not be used for normal entry. Entry attempts using the keypad shall be denied by the Flight Crew. For A330F exception, see 15.2.6.10 – Flight Deck Access without Functioning Service Interphone."*
>
> The code is not published in FOM, FCOM or QRH; it is SSI (FOM 15.1.1.1 lists security access codes as SSI). Note the policy point: the keypad is not for normal entry and keypad attempts must be denied.
>
> *Workbook wording: What is the Flight Deck Door entry code?*

Cabin Emergency Equipment Locations (Megaphone, ELT, etc.)

EMERGENCY EQUIPMENT

Where are the passenger cabin emergency equipment locations?

FCOM Emergency Equipment section, passenger cabin diagram

FCOM Chapter 1 Section 45 covers emergency equipment, and the passenger cabin location diagram sits at FCOM 1.45.8, immediately after the flight deck diagram at 1.45.7 and the symbols page at 1.45.6. The ELT text at FCOM 1.45.1 explicitly refers you to that diagram for ELT locations. Crew rest facilities carry their own equipment lists and diagrams (FCOM 1.46.16-1.46.17 for the OFCR, 1.47.12-1.47.13 for the OFAR), so do not use the cabin diagram for crew rest items.

> **FCOM 1.45.8 Emergency Equipment Locations - Passenger Cabin**
>
> *"ELTs are installed in the passenger cabin, as shown in the Emergency Equipment Locations – Passenger Cabin diagram."*
>
> It is a graphic, so text search will not find individual items. The flight deck companion diagram is FCOM 1.45.7 and the symbol legend is FCOM 1.45.6. Crew rest equipment is separate: OFCR at FCOM 1.46.17, OFAR at FCOM 1.47.13.
>
> *Workbook wording: Where can the "Emergency Equipment Locations – Passenger Cabin" be found?*

1.2.1 Perform Preflight Preparation

Flight Deck Door/Overhead Door Operation (787)

SECURITY AND DOORS

Where is the 787 overhead door described?

FCOM Doors systems description, Flight Deck Overhead Door

FCOM 1.40.19 gives the description and operation: the door is for emergency egress using the installed descent devices, can only be opened on the ground with the airplane depressurized, and is opened by removing the protective cover and rotating the door lock handle, after which it swings down inside the flight deck on hinges. EICAS advisory DOOR FD OVHD shows if it is not closed and secure. FCOM 1.40.20-1.40.21 covers the descent devices and the egress climb sequence: first observer's seat (right foot), observer's console (left foot), fold-down step in the bulkhead (right foot), then the overhead door at the hinged area, plus the exterior step used to escape to the left side if hazards exist on the right. Controls and components are pictured at FCOM 1.30.30.

> **FCOM 1.40.19 Flight Deck Overhead Door (controls at FCOM 1.30.30)**
>
> *"Flight Deck Overhead Door The flight deck overhead door is used for emergency egress using the installed descent devices. The door can only be opened on the ground with the airplane"*
>
> Two places: FCOM 1.30.30 Flight Deck Overhead Door and Vent for the cover, handle and components, and FCOM 1.40.19-1.40.21 for description, operation and the egress route. FCOM NP.21.1 requires it closed and latched at preflight.
>
> *Workbook wording: Where can you find the 787 forward overhead door description, operation, components and egress procedures?*

1.1.1 Perform Flight Planning [CRITICAL OBSERVABLE ITEM]

EFB Review

EFB AND FD PRO &MIDDOT; WALKTHROUGH ITEM, PERFORM RATHER THAN RECALL

EFB IOS current and updated?

In the FD Pro guide in Comply365

This is a Jeppesen FD Pro X or EFB workflow step, not FOM/FCOM content. Verify against the FD Pro Pubs Quick Reference guide in Comply365 and demonstrate on the actual device during OE. FOM 5.7.3.15 does require the ENTIRE cleared route be loaded into the approved charting app for ORCN segments.

> **FD Pro Pubs Quick Reference (Comply365)**
>
> *Workbook wording: EFB IOS - Current/updated.*

EFB AND FD PRO &MIDDOT; WALKTHROUGH ITEM, PERFORM RATHER THAN RECALL

FD Pro updates completed?

In the FD Pro guide in Comply365

This is a Jeppesen FD Pro X or EFB workflow step, not FOM/FCOM content. Verify against the FD Pro Pubs Quick Reference guide in Comply365 and demonstrate on the actual device during OE. FOM 5.7.3.15 does require the ENTIRE cleared route be loaded into the approved charting app for ORCN segments.

> **FD Pro Pubs Quick Reference (Comply365)**
>
> *Workbook wording: FD Pro: updates completed.*

EFB AND FD PRO &MIDDOT; WALKTHROUGH ITEM, PERFORM RATHER THAN RECALL

FD Pro: open Weather and the Weather Layers pane

In the FD Pro guide in Comply365

This is a Jeppesen FD Pro X or EFB workflow step, not FOM/FCOM content. Verify against the FD Pro Pubs Quick Reference guide in Comply365 and demonstrate on the actual device during OE. FOM 5.7.3.15 does require the ENTIRE cleared route be loaded into the approved charting app for ORCN segments.

> **FD Pro Pubs Quick Reference (Comply365)**
> *Workbook wording: FD Pro: “Weather” (bottom/left of screen), “Weather Layers” pane/”I” (greyed-out “info” button/s, RT*

EFB AND FD PRO &MIDDOT; WALKTHROUGH ITEM, PERFORM RATHER THAN RECALL

Select the layers you want on the Weather Layers pane

In the FD Pro guide in Comply365

This is a Jeppesen FD Pro X or EFB workflow step, not FOM/FCOM content. Verify against the FD Pro Pubs Quick Reference guide in Comply365 and demonstrate on the actual device during OE. FOM 5.7.3.15 does require the ENTIRE cleared route be loaded into the approved charting app for ORCN segments.

> **FD Pro Pubs Quick Reference (Comply365)**
> *Workbook wording: Select desired layer/s on “Weather Layers” pane*

EFB AND FD PRO &MIDDOT; WALKTHROUGH ITEM, PERFORM RATHER THAN RECALL

REFRESH on the Weather Layers pane

In the FD Pro guide in Comply365

This is a Jeppesen FD Pro X or EFB workflow step, not FOM/FCOM content. Verify against the FD Pro Pubs Quick Reference guide in Comply365 and demonstrate on the actual device during OE. FOM 5.7.3.15 does require the ENTIRE cleared route be loaded into the approved charting app for ORCN segments.

> **FD Pro Pubs Quick Reference (Comply365)**
> *Workbook wording: “REFRESH” (bottom/right) on “Weather Layers” pane*

EFB AND FD PRO &MIDDOT; WALKTHROUGH ITEM, PERFORM RATHER THAN RECALL

FD Pro: verify the correct aircraft

In the FD Pro guide in Comply365

This is a Jeppesen FD Pro X or EFB workflow step, not FOM/FCOM content. Verify against the FD Pro Pubs Quick Reference guide in Comply365 and demonstrate on the actual device during OE. FOM 5.7.3.15 does require the ENTIRE cleared route be loaded into the approved charting app for ORCN segments.

> **FD Pro Pubs Quick Reference (Comply365)**
> *Workbook wording: FD Pro: Verify correct “aircraft” on top/center of screen*

EFB AND FD PRO &MIDDOT; WALKTHROUGH ITEM, PERFORM RATHER THAN RECALL

FD Pro: flight plan upload demo

In the FD Pro guide in Comply365

This is a Jeppesen FD Pro X or EFB workflow step, not FOM/FCOM content. Verify against the FD Pro Pubs Quick Reference guide in Comply365 and demonstrate on the actual device during OE. FOM 5.7.3.15 does require the ENTIRE cleared route be loaded into the approved charting app for ORCN segments.

> **FD Pro Pubs Quick Reference (Comply365)**
> *Workbook wording: FD Pro Flight plan upload demo*

EFB AND FD PRO &MIDDOT; WALKTHROUGH ITEM, PERFORM RATHER THAN RECALL

FD Pro: transfer a flight plan to another EFB

In the FD Pro guide in Comply365

This is a Jeppesen FD Pro X or EFB workflow step, not FOM/FCOM content. Verify against the FD Pro Pubs Quick Reference guide in Comply365 and demonstrate on the actual device during OE. FOM 5.7.3.15 does require the ENTIRE cleared route be loaded into the approved charting app for ORCN segments.

> **FD Pro Pubs Quick Reference (Comply365)**
> *Workbook wording: FD Pro Flight plan transfer to another EFB demo*

Dispatch Release: Review all sections

FUEL PLANNING

Where is preflight planning and the international release?

FOM Dispatch chapter, Flight Information and Redispatch section

FOM Chapter 8 (Dispatch) holds the planning material: 8.1 authorized operations and airports, 8.2 alternates, 8.3 navigation, 8.4 weather, and 8.5 flight information documents and redispatch. FOM 8.5.1 defines the Flight Information Packet/Dispatch Release Package and makes the PIC responsible for having it prior to departure from an originating station. The HA Dispatch Release sample is FOM 8.5.7 with the International Dispatch Release example and legend at 8.5.7.1; AS samples and the ETOPS release are 8.5.4 through 8.5.6. Signing the release is covered by FOM 5.2.11.

> **FOM 8.5 Flight Information - Documents and Redispatch (8.5.1 Flight Information Packet/Dispatch Release Package)**
> *“Dispatch produces the Flight Information Packet/Dispatch Release Package containing required and supplemental data for flight release. The PIC shall review and ensure the following information (as applicable) is in their possession prior to departure from an originating station:”*
> The HA international release example and its legend are FOM 8.5.7 and 8.5.7.1. Preflight duties themselves are FOM 5.2; theater-specific international items are FOM chapters 19-24.
> *Workbook wording: FOM Preflight Planning and International Dispatch Release*

OCEANIC AND NAT

Named vs unnamed waypoint?

Named is in the database. Unnamed is lat/long or hand entered

Non-defined or user-named waypoints are waypoints that are either not in the aircraft database, unnamed such as a latitude/longitude like 26N170W, or manually entered by the crew. These may require specific verification procedures, and the FOM sends you to fleet-specific manuals for the route verification procedure.

> **FOM 5.2.11.5 Non-Defined or Unnamed Waypoints**
> *“"waypoints that are either not in the aircraft database, unnamed (a latitude/longitude, such as “26N170W” or “26N70”), or a waypoint that the Fight Crew manually enters."”*
> Oceanic tie-in: FOM 22.3.4 Unnamed Waypoint Verification carries the NAT-specific check. The (AS) note warns the flight planning system condenses lat/long waypoints with minutes into 5-character labels that will NOT load into the FMC; reference the full lat/long from the Flight Plan and enter a user-defined waypoint.
> *Workbook wording: What is the difference between a named vs non-defined/unnamed waypoint?*

DISPATCH AND RELEASE

What is in the dispatch release package?

Release, flight plan, TAF, METAR, SIGMETs, Class D and FDC NOTAMs

FOM 8.5.1 lists the package as: Dispatch Release including MEL/CDL listing, (AS) Flight Plan, Aerodrome Forecast (TAF or equivalent), Aviation Routine Weather Report (METAR or equivalent), appropriate special weather reports/amended TAFs/SIGMETs and turbulence information, Class D NOTAMs, FDC NOTAMs, (AS) F&Fs, and (HA) Company NOTAMs. A note adds that the Load Manifest, normally the Load Closeout weight and balance loadsheet, is also required prior to departure. It normally arrives electronically, (AS) in JetPack or (HA) by email. The PIC's signature on the release means all of it has been reviewed (FOM 5.2.11).

> **FOM 8.5.1 Flight Information Packet/Dispatch Release Package**
>
> *"• Dispatch Release (including MEL/CDL listing) • (AS) Flight Plan • Aerodrome Forecast (TAF or equivalent) • Aviation Routine Weather Report (METAR or equivalent) • Appropriate Special Weather Reports, Amended Aerodrome Forecasts, SIGMETs, and"*
>
> The full FOM list also includes (AS) F&Fs and (HA) Company NOTAMs, and the Dispatch Release itself includes the MEL/CDL listing. The Load Manifest/Load Closeout is separately required prior to departure.
>
> *Workbook wording: What documents are included in the Dispatch/Release package? ("Lucky 7"; in order or printing)*

FUEL PLANNING

B043 vs B044 flight plan?

B043 international fuel reserve, B044 planned redispatch

B043 is the Ops Spec authorizing special fuel reserves in international operations, where the 10 percent applies only to planned oceanic/remote continental time and the final reserve is 45 minutes at normal cruise consumption. Ops Spec B044 is one of the authorizations cited for planned redispatch under FOM 8.5.9, where the flight is dispatched to an intermediate destination and redispatched enroute to the intended destination to minimize the 10 percent fuel requirement.

> **FOM 8.3.2.3 and FOM 8.5.9 Redispatch Planned**
>
> *""Ops Spec B043""*
>
> Both reduce carried fuel but by different mechanisms: B043 narrows the window the 10 percent applies to; redispatch (B044) shortens the leg the 10 percent is computed against. 8.5.9 lists Ops Specs A002, A004, A008, B044 and B050.
>
> *Workbook wording: What is the difference between a BO44 vs B043 flight plan?*

FUEL PLANNING

Standard release vs redispatch?

Redispatch runs two plans, so you carry less fuel

A standard release dispatches you to the destination on one plan. Planned redispatch is an FAA-authorized flag procedure that minimizes the fuel needed to meet the 10 percent requirement. You operate with two plans: the original dispatch to the intermediate destination, and a second plan from a planned redispatch waypoint to the intended destination. The terms of the original release govern until the Captain and Dispatcher concur on the redispatch, and once redispatched those terms are operative and cannot be changed except by Captain and Dispatcher concurrence.

> **FOM 8.5.9 Redispatch Planned**
>
> *""The terms of original Dispatch Release must be adhered to until the Captain and the Dispatcher concur on the terms of the redispatch.""*
>
> Terminology to have cold: Intended Destination is the ultimate destination from the redispatch waypoint; Intermediate Destination is the one on the original release; the Redispatch Point is the common point on both flight plans where the continue decision is made. Regulatory basis 14 CFR 121.631 plus Ops Specs A002/A004/A008/B044/B050. Procedures at 8.5.9.1.
>
> *Workbook wording: Discuss the differences between a standard dispatch release vs a Re-Dispatch/Re-Release*

INTERNATIONAL THEATERS

Where is country-specific guidance?

FOM theater chapters plus Jeppesen State Rules and Procedures

FOM chapters 19 through 24 are organized by theater and carry country-specific airspace, navigation, communication and procedural rules. Where the FOM does not carry the detail it names the exact Jeppesen FD Pro path, for example FOM 19.3.1.2 sends you to Pubs > North America > North America Airway Manual > State Rules and Procedures - Canada for holding, and FOM 5.5.15.4 does the same for international holding speeds generally. Airport-specific guidance, including things like Starlink restrictions (FOM 5.1.15), is in the Jeppesen 10-7 for that airport. International documents required on board are listed at FOM 5.2.1.2.

> **FOM chapters 19-24 Theater chapters; example FOM 19.3.1.2 Holding Airspeeds**
>
> *“For holding procedures and airspeeds, see Jeppesen FD Pro app at the following location: Pubs > North America > North America Airway Manual > State Rules and Procedures – Canada > Instrument Flight Rules – Holding Procedures.”*
>
> The six theater chapters are 19 North America, 20 Pacific, 21 Caribbean, 22 Atlantic, 23 Europe, 24 Asia. They repeatedly hand off to Jeppesen FD Pro State Rules and Procedures, and airport-specific items live in the Jeppesen 10-7 pages.
>
> *Workbook wording: International Planning: Where can you find country-specific guidance?*

1.1.2.9 Know Weather Minima & Limitations for Dispatch, Takeoff, Approach, and Landing

Dispatch Weather Minimums

WEATHER AND MINIMUMS

Destination alternate weather required to depart?

At or above alternate minimums for the ETA

A flight may not be dispatched unless the weather at the destination alternate is forecast at or above the alternate minimums for the time the aircraft would arrive there. Those minimums come from the Derived Alternate Airport IFR Weather Minimums table in 8.2.2.1 under Ops Spec C055. For a single straight-in non-precision approach the derivation is MDA(H) plus 400 ft and visibility plus 1 sm or 1600 m. Exemption 20108 allows dispatch with alternates not meeting this criteria under specific conditional-language limits.

> **FOM 8.2.2.1 Destination Alternate Airport Weather Requirements**
>
> *“"A flight may not be dispatched to the destination airport unless the weather conditions at a destination alternate airport are forecast to be at or above the alternate minimums"”*
>
> Second alternate rules live in 8.2.2.2 and Exemption 20108 in 8.2.2.3. Rev 125.1 added a DOM 3.000 cross-reference and the Ops Spec A010(e) rule that a second alternate is required for flag or supplemental flights to an international airport with information missing from the METAR, and both destination alternates must have a complete METAR and TAF.
>
> *Workbook wording: Weather required to be at our destination/destination alternate (Flag) to depart?*

ETOPS

ETOPS alternate weather before departure?

C055 minima, earliest to latest landing time

The Dispatcher will not list an airport as an ETOPS alternate unless reports or forecasts indicate weather will be at or above the ETOPS alternate minima in Ops Spec C055 (see 8.2.2) for the whole window it might be used, earliest to latest possible landing time, and field condition reports indicate a safe landing can be made. The earliest and latest arrival times are printed on the Dispatch Release, so if the flight is significantly delayed the crew should reassess whether those times still hold.

> **FOM 6.2.2 ETOPS Alternate Airport**
>
> *""Field condition reports indicate that a safe landing can be made.""*
>
> Each designated ETOPS alternate must have RFFS equivalent to ICAO Category 4 or higher, with a 30-minute response time, which means equipment available on arrival of the diverting airplane, not equipment parked there in advance. Rev 125 keyed the adequate airport and ETOPS alternate definitions to 8.1.3 Authorized Airports rather than Ops Spec B342/C070.
>
> *Workbook wording: Weather required for ETOPS alternate airport(s) prior to departure?*

Enroute Weather Minimums

ETOPS

ETOPS alternate weather in flight?

Crew operating minima

From after takeoff until approaching the EEP, alternate weather must be at or above the crew's operating minima, that is what the operating crew needs to fly the approach given their minima and any applicable MEL, and the alternate must remain suitable to conduct an IAP. A turnback is required if an ETOPS alternate is no longer available. After the EEP a turnback is NOT required if the weather is below crew minima or the runway is unusable; instead you attempt to establish a new ETOPS alternate with Dispatch, which may lead to a turnback, reroute or continuation.

> **FOM 6.3.5 In-Flight ETOPS Alternate Airport Weather Requirements**
>
> *""Note that these weather requirements are less restrictive than the criteria used by Dispatch in planning.""*
>
> This is a favourite oral question because the answer changes at the EEP. If a METAR for an ETOPS alternate cannot be obtained after the EEP, a new ETOPS alternate must be designated. The FOM closes with joint PIC and Dispatcher judgment on the safest course of action.
>
> *Workbook wording: Weather required for ETOPS alternate airport(s) in flight?*

WEATHER AND MINIMUMS

Destination alternate weather in flight?

Crew operating minima

Dispatch planning uses the derived alternate minimums table at 8.2.2.1. In flight the standard shifts to the crew's operating minima, meaning what the operating crew needs to conduct the approach given their minima and any applicable MEL. FOM 6.3.5 states plainly that the in-flight weather requirements are less restrictive than the criteria used by Dispatch in planning.

> **FOM 6.3.5 / 8.2.2.1**
>
> *""Note that these weather requirements are less restrictive than the criteria used by Dispatch in planning.""*
>
> The dispatch-versus-inflight distinction is the whole question. Do not apply the 400-and-1 derived alternate minimums to an in-flight divert decision.
>
> *Workbook wording: Weather required at destination alternate airport in flight.*

Takeoff Weather Minimums

WEATHER AND MINIMUMS

Takeoff weather minimums at the departure airport?

1 sm or RVR 5000; lower than standard per table

Standard takeoff minimums are 1 sm visibility or RVR 5000 for aircraft of two engines or less (FOM 5.4.1.1). Under Ops Spec C078, FOM 5.4.4 allows lower than standard whenever a procedure allows standard: 1600 RVR or 1/4 sm with HIRL, centerline lights, centerline markings or adequate visual reference; 1200, 1000 and 500 RVR steps with progressively more lighting and two operative RVR readouts controlling; below 500 RVR (150 m) see the HGS/HUD Takeoff Briefing Card. If published minimums are higher than standard and no special procedure exists, you cannot depart below them (FOM 5.4.1.2), and 'NA' means no IFR departure. Captain must take the runway when RVR is 1600 or less or visibility 1/4 sm or less (FOM 5.4.3), and a takeoff alternate is required when departure weather is below landing minimums for the runway of intended use, within 393 nm for the 787 (FOM 5.4.2).

> **FOM 5.4.1.1 Standard Takeoff Minimums**
>
> *“Defined as 1 sm visibility or RVR 5000 for aircraft having 2 engines or less.”*
>
> FOM 5.4.4 authorizes lower than standard down to 500 RVR with the required lighting, and below 500 RVR (787) via the HGS/HUD Takeoff Briefing Card in the PRC. Charted SID/obstacle minimums may be more restrictive than the 10-9A.
>
> *Workbook wording: Weather required at departure airport for takeoff.*

WEATHER AND MINIMUMS

Who can fly a low visibility takeoff?

Below 500 RVR, work off the PRC low visibility card

Ops Spec C078 authorizes lower than standard takeoff minimums wherever a procedure allows standard minimums. For the 787, takeoffs below 500 RVR (150 m) require the Takeoff Low Visibility Card in the PRC. At foreign airports published 10-9A minimums may not conform to Ops Specs, and crews are restricted to the higher of the 10-9A or Ops Specs. If takeoff minimums for a runway are NA, IFR departure is not permitted.

> **FOM 5.4.4 IFR Lower than Standard Takeoff Minimum Requirements**
>
> *“"(737/787) For takeoffs less than 500 RVR (150 m), see Takeoff Low Visibility Card in PRC."”*
>
> NEW in Rev 125: 5.4.4 added 787 bannering and the HGS/HUD Takeoff Briefing Card, which is why this now applies to your fleet. The visibility table runs 5000/VIS 1 sm standard down through 1600, 1200 and 1000, with lighting and multi-RVR control requirements at each step. DESKTOP: the PRC card itself is not in this environment. Related: 5.3.11 Taxi Operations Low Visibility, and a takeoff alternate is required per 5.4.2 when departure weather is below landing minimums for the runway of intended use.
>
> *Workbook wording: Any limitations on who can perform a low visibility takeoff?*

Approach/Landing Weather Minimums

WEATHER AND MINIMUMS

Visibility drops on approach. Can you continue?

Yes, once the final approach segment has begun

Do not begin the final approach segment unless the latest reported RVR, visibility, or ceiling if required, is at or above the authorized minimums. However, if a weather report is obtained AFTER the final approach segment has been initiated and it indicates below minimum conditions, the pilot may continue the approach to DA/DH or MDA.

> **FOM 5.6.11 Visibility Requirements Approach**
>
> *“"the pilot may continue the approach to DA/DH or MDA."”*
>
> The trigger is the start of the FINAL APPROACH SEGMENT, not the FAF crossing or the clearance. For non-precision and CAT I: TDZ RVR is controlling, Mid and Rollout are advisory, and Mid may substitute if TDZ is unavailable. RVR above 6000 is advisory only. Visibility below 1/2 sm is not authorized and shall not be used for approach minimums.
>
> *Workbook wording: If the visibility drops while on the approach, can we continue the approach?*

WEATHER AND MINIMUMS

Different visibility rules at international airports?

Yes. Some ICAO states allow Converted Meteorological Visibility

Some ICAO countries allow use of Converted Meteorological Visibility (CMV) when RVR is not reported, for approaches that are not low-visibility. The FOM directs you to the Aerodrome Operating Minimums and Converted Meteorological Visibility references in the Jeppesen Publications for further guidance.

FOM 5.6.11 Visibility Requirements Approach

> *""Some ICAO countries allow for use of Converted Meteorological Visibility (CMV) when RVR is not reported for non low-vis approaches.""*

Separately, at foreign airports the published 10-9A takeoff minimums may not conform to Ops Specs, and crews are restricted to the HIGHER of the 10-9A or Ops Specs (FOM 5.4.4). Theatre chapters 19 through 24 carry the country-specific items.

Workbook wording: Are there different requirements at some international airports?

1.1.2.4 Know Operational Control (Dispatch) Requirements [CRITICAL OBSERVABLE ITEM]

DISPATCH AND RELEASE

Who has operational control of a flight?

The PIC and the Dispatcher, jointly

The PIC and the Aircraft Dispatcher are jointly responsible for preflight planning, delay, and the Dispatch Release, and jointly determine the suitability of weather, field and traffic conditions, airway facilities, and the filed route. Neither can dispatch alone; both must agree.

FOM 2.1.1 Joint Responsibility

> *""jointly responsible for the preflight planning, delay, and Dispatch Release of a flight""*

Regulatory basis 14 CFR 121.533/.535/.537/.551/.557/.627 and Ops Spec A008. Captain-side point: joint responsibility is not shared authority in flight. Once airborne the PIC retains final authority under 2.1.2.

DISPATCH AND RELEASE

What requires Dispatch notification in flight?

Any abnormality or irregularity

Flights experiencing abnormalities or irregularities, including but not limited to those listed in 7.2.3 Pilot Reports Quick Reference, shall contact Dispatch and report the details. Notification is mandatory; the timing flexes to safety of flight. If time does not permit contact in flight, contact shall be made after gate arrival. The 7.2.3 table also marks which events additionally need a FODO debrief.

FOM 11.1.2 Abnormal Operations Dispatch Notification

> *"""Contact Dispatch" implies that while notification is mandatory, such communications should take place as safety of flight considerations permit.""*

Rev 125.1 revised 7.2.3 to state the table is not all-inclusive and that the Company may request safety reports for items not listed when additional information is needed. Separately 11.1.3 covers Cabin notification, and 11.2.11 gives the NTSB briefing format for the Flight Attendants.

Workbook wording: What items require Dispatch notification during a flight?

1.2.2 Perform ACARS and FMS Initialization and Verification

COMMUNICATIONS AND PA

Where are the FMS/ACARS uplink and setup procedures?

FCOM Communications chapter, Supplementary Procedures, and Performance Inflight AeroData

FCOM SP.5 gives the verification procedures for messages uplinked in COMPANY format: pre-departure clearance and oceanic clearance must be manually compared against the filed flight plan with voice contact if anything disagrees, D-ATIS altimeter numeric and alpha values must match, and weight and balance and takeoff data are verified per the FOM. FCOM Chapter 5 Section 40 documents the MFD communications menus, ATC and Flight Information menus, the Company menu, and the ACARS, VHF, HF, SATCOM, AOIP and ADS managers (FCOM 5.40.72-5.40.85). ACARS takeoff and landing performance requests through the aFDP tool are FCOM PI.19.1 and PI.19.8. The FMS initialization flow, IDENT, POS INIT, route, performance and takeoff pages, is the CDU Preflight Procedure at FCOM NP.21.2.

FCOM SP.5 Communications - Flight Deck Communications System (Datalink)

> *"The following procedures are one means which may be used to verify Pre-Departure Clearance, Digital-Automatic Terminal Information Service (D-ATIS), Oceanic Clearances, Weight and Balance and Takeoff Data messages transmitted via the COMPANY format."*

Three places to know: FCOM SP.5 for uplink verification policy, FCOM 5.40 MFD Communications Functions for the datalink/ACARS pages and managers, and FCOM PI.19 AeroData for the ACARS takeoff and landing performance requests. FMS setup itself is FCOM NP.21.2 CDU Preflight Procedure.

Workbook wording: Where are the FMS/ACARS/Uplink and Setup Procedures located?

EFB AND FD PRO

How do you know the nav database is current?

CDU IDENT page, ACTIVE date range

During Initial Data set on the preflight flow, on the CDU IDENT page you verify the MODEL is correct, verify the ENG RATING is correct, cycle the active NAV database to erase all previously pilot-entered data, and verify that the navigation database ACTIVE date range is current.

FCOM NP.21.2 Preflight Procedure, Initial Data

> *""Verify that the navigation database ACTIVE date range is current.""*

Note the order: cycling the active database first is what erases previously pilot-entered data, so it is a data-hygiene step as much as a currency check. The IDENT page also confirms MODEL and ENG RATING, which is a real trap on a mixed fleet.

Workbook wording: How can you tell if the Nav database is current?

EFB AND FD PRO

Can you change the nav database?

Yes, on the CDU NAV DATA page

The FMC carries two navigation database cycles, active and inactive. NAV DATA is selectable from the CDU and is where the database cycle is managed. The operational rule is that you only make a cycle active when its date range covers the flight, which is exactly what the IDENT page ACTIVE date range check confirms.

FCOM NP.21.2 / FMC NAV DATA

> *""Select NAV DATA.""*

DESKTOP: the full switchover procedure and any company restriction on changing the database in flight should be confirmed in the FMS Pilot Guide and FCOM Chapter 11, which are not in this environment. Do not perform a cycle change with a flight plan loaded without verifying the effect on the active route.

Workbook wording: Can you change the Nav database? How?

EFB AND FD PRO

How do you know the AMM database is current?

On the EFB and in the FCOM

AMM database currency is checked on the EFB. Distinct from the NAV database (FCOM NP.21.2 IDENT page) and the ECL AIRLINE DATABASE revision (FCOM NP.21.18). Verify the AMM check on the aircraft or in the EFB user guide.

> **EFB / FCOM**
> *Workbook wording: 787 – How can you tell the AMM database is current?*

COMMUNICATIONS AND PA

How do you verify the correct ECL is loaded?

Database revision must match the QRH front page

During the preflight flow you push the CHECKLIST display switch, select RESETS, then Verify/Set AIRLINE DATABASE. Select the AIRLINE DATABASE that has a REVISION matching the ECL Revision listed on the front page of the current approved QRH, then select RESET ALL.

> **FCOM NP.21.18 Preflight Procedure**
> *""Select the AIRLINE DATABASE that has a REVISION that matches the "ECL Revision" listed on the front page of the current approved QRH.""*
> The QRH is the authority here, not the airplane. If the airplane database revision does not match the QRH front page, the ECL is not the approved checklist. RESET ALL after setting it clears any prior crew state.
> *Workbook wording: 787 – How do you verify the correct ECL is loaded?*

Load Closeout/TPR Update and review procedure

DISPATCH AND RELEASE

Who reads the closeout data?

In the FOM and the fleet performance manuals

Load closeout reading and verification, and the takeoff/landing performance products. Takeoff performance hierarchy is Final TPR, then Preliminary TPR, then TLR. Using a TLR for takeoff demands attention to the temperature and altimeter assumptions baked into it. FOM 5.2 covers closeout and weight and balance; the SABLE Load Sheet is the official FAA weight and balance record. Landing analysis designators and request methods are in the performance system documentation.

> **FOM 5.2 / fleet performance manuals**

DISPATCH AND RELEASE

Who verifies the closeout, and when?

In the FOM and the fleet performance manuals

Load closeout and weight and balance are FOM 5.2, with the SABLE Load Sheet as the official FAA weight and balance record (17.2.4 for freighter). Takeoff performance hierarchy is Final TPR, then Preliminary TPR, then TLR. Using a TLR for takeoff requires attention to temperature and altimeter assumptions, since the TLR is built on different input conditions.

> **FOM 5.2 / fleet performance manuals**
> *Workbook wording: Who verifies it? When?*

DISPATCH AND RELEASE

What closeout items must be verified?

In the FOM and the fleet performance manuals

Load closeout reading and verification, and the takeoff/landing performance products. Takeoff performance hierarchy is Final TPR, then Preliminary TPR, then TLR. Using a TLR for takeoff demands attention to the temperature and altimeter assumptions baked into it. FOM 5.2 covers closeout and weight and balance; the SABLE Load Sheet is the official FAA weight and balance record. Landing analysis designators and request methods are in the performance system documentation.

> **FOM 5.2 / fleet performance manuals**
>
> *Workbook wording: List several items that are required to be verified.*

DISPATCH AND RELEASE

No TPR. How do you get takeoff performance?

In the FOM and the fleet performance manuals

Load closeout reading and verification, and the takeoff/landing performance products. Takeoff performance hierarchy is Final TPR, then Preliminary TPR, then TLR. Using a TLR for takeoff demands attention to the temperature and altimeter assumptions baked into it. FOM 5.2 covers closeout and weight and balance; the SABLE Load Sheet is the official FAA weight and balance record. Landing analysis designators and request methods are in the performance system documentation.

> **FOM 5.2 / fleet performance manuals**
>
> *Workbook wording: How do you obtain takeoff performance if a TPR is unavailable?*

DISPATCH AND RELEASE

Can you use the TLR for takeoff data?

Yes. It is the takeoff data on the Dispatch Release

FOM 5.4.5 requires takeoff data be available and checked for each takeoff, and names where it comes from: the Flight Information Packet carries the TPR, the Dispatch Release carries the TLR. So the TLR is a legitimate source of takeoff data, not a workaround. Validation criteria live in the fleet manuals, and invalid data is replaced by ACARS or Dispatch.

> **FOM 5.4.5 Takeoff Data**
>
> > *“Flight Information Packet (TPR)/Dispatch Release (TLR)”*
>
> Takeoff data must be available and checked for every takeoff. If the data is invalid against the fleet validation criteria, get new data by ACARS or from Dispatch, who has the online performance tool. FOM 8.5.2.2 points at the fleet Aerodata Performance Handbook on the EFB for the TLR description.
>
> *Workbook wording: Can we use the TLR for takeoff information? When would we use it?*

DISPATCH AND RELEASE

TLR concerns with temp and altimeter?

In the FOM and the fleet performance manuals

Load closeout reading and verification, and the takeoff/landing performance products. Takeoff performance hierarchy is Final TPR, then Preliminary TPR, then TLR. Using a TLR for takeoff demands attention to the temperature and altimeter assumptions baked into it. FOM 5.2 covers closeout and weight and balance; the SABLE Load Sheet is the official FAA weight and balance record. Landing analysis designators and request methods are in the performance system documentation.

> **FOM 5.2 / fleet performance manuals**
>
> *Workbook wording: Any concerns when using TLR for takeoff information? Temp and altimeter.*

HAZMAT

Dangerous goods on board. What governs?

Review and acknowledge the eNOTOC, then retain it

Dangerous goods on board means the PIC must receive a NOTOC (FOM 14.1.1). On HA the Captain reviews the eNOTOC or HAL902 for correct departure airport, flight number and tail number with no hand edits, acknowledges it by selecting ACCEPT or ACKNOWLEDGE NOTOC on the Comm page, retains the printed copy readily available in flight to pass HAZMAT type, quantity and location to ARFF, and makes a logbook entry if damaged or leaking HAZMAT is removed (FOM 14.1.6). If digital acknowledgment fails, use the hard-copy procedures at FOM 14.1.23. The PIC does not handle DG and carries no civil or criminal liability for it (FOM 14.1.2); damaged or leaking shipments are FOM 14.1.26, and no hazardous materials may be carried in the flight deck.

> **FOM 14.1.6 Captain Responsibilities (HA)**
>
> *"• Acknowledging the eNOTOC by selecting either "ACCEPT*" on the MCDU or "ACKNOWLEDGE NOTOC" on the Comm page (depending on fleet type)."*
>
> HA rules differ from AS: hand edits to the eNOTOC are not permissible (AS allows Ramp Lead edits to tail number and DG location on the paper NOTOC, FOM 14.1.5). eNOTOC arrives at departure minus 30 min international, minus 15 min interisland (FOM 14.1.4).
>
> *Workbook wording: Describe the Dangerous goods procedures if you have any on your flight*

12.1.3.16 Know FOM PA policies and Perform Effective Communication from the flight deck

COMMUNICATIONS AND PA

Where are the standard customer service PAs?

In the FOM customer service announcements section

Chapter 7 carries PA policy at 7.1.21. Pilots should identify themselves as Captain or First Officer, and should avoid aviation abbreviations, acronyms, jargon and detailed explanations of procedures or duties. Examples given of what to avoid are knots, broken clouds, chop and abort. Use of the PA should be limited during periods when passengers are asleep.

> **FOM 7.1.21.1 Standard Customer Service Announcements, General**
>
> *""Avoid using aviation abbreviations, acronyms, jargon, and detailed explanations of procedures or duties""*
>
> Company-culture item worth knowing: the FOM specifically asks crews not to use the "we know you have a choice of airlines" line to or from Southeast Alaska stations, because Alaska is the sole major carrier there and passengers may not have a choice.
>
> *Workbook wording: What section of the FOM are the Standard Customer Service Announcements (PA's) located?*

COMMUNICATIONS AND PA

Which PAs must the flight crew make?

The required PAs, starting with the seat belt sign

Flight Crews are required to make the PAs in 7.1.19 during each flight with passengers or persons on board, acknowledging that flight conditions and priorities may occasionally preclude timely execution in the interest of safety. These include 7.1.19.1 Seat Belt Advisory when turning the sign off the first time after takeoff, 7.1.19.2 Seat Belt Advisory when turning the sign on, and 7.1.19.3 Anticipated Turbulence. Crews may vary the wording using judgment EXCEPT where an announcement is noted as verbatim.

> **FOM 7.1.19 Required Announcements**
>
> *""Flight Crew may use judgment to vary the wording of these announcements except where the announcement is noted as verbatim.""*
>
> Workload permitting, pilots announce any change in Seat Belt Sign condition, and the F/As must also announce it regardless of whether the pilots do. The FOM encourages Aloha from the Flight Deck and Mahalo on Hawaii flights. DESKTOP: read the full 7.1.19.x list off the PDF for the complete set and to identify which are marked verbatim.
>
> *Workbook wording: Which announcements are required to be made by the Flight Crew?*

TAXI AND GROUND

Pushing back late. When is a PA required?

At scheduled departure time, then every 15 minutes

FOM 7.1.21.1 states that at the departure time the Captain will either authorize pushback/engine start or notify Operations and the passengers of the delay, giving the reason if known. Once on board, the crew owns passenger delay information: relay it to Station Operations immediately, advise the A Flight Attendant, then update passengers at 15 minute intervals or sooner as information arrives. The same 15 minute cadence applies to arrival ground delays. FOM 7.1.18 adds that for a ground delay the explanatory PA should be made when practical after the aircraft is stopped and the parking brake is set, and FOM 5.3.13.7 covers the separate tarmac delay announcements and the opportunity to deplane.

> **FOM 7.1.21.1 General - Departure Time Delay Announcement ('push or talk')**
>
> *"• Departure Time Delay Announcement ("push or talk") – At the departure time, the Captain will either authorize pushback/starting engines procedures or notify Operations and the passengers"*
>
> The 'push or talk' rule is the trigger: at departure time you either authorize pushback or you tell Operations and the passengers. FOM 7.1.21.1 requires delay updates at 15 minute intervals or sooner, and FOM 7.1.18 recommends the explanatory PA be made after the aircraft is stopped with the parking brake set.
>
> *Workbook wording: When should you need to make a PA if you are pushing off the gate late? Minutes?*

PUSHBACK AND START

Demonstrate a late pushback, go-around and engine failure PA

Use FOM sample PAs: delay, go-around, emergency equipment

Late pushback: at departure time either push or talk, give the reason and an approximate length, then update every 15 minutes (FOM 7.1.21.1). Go-around: FOM 7.1.21 says a short PA should be made when workload and ATC coordination permit, and below 10,000 ft, on an extended downwind or in holding, you may state the reason and your intentions. Engine failure: brief the A Flight Attendant first using the NTSB format (Nature, Time, Special instructions, Brace) per FOM 11.2.11, then use the FOM 7.1.22 sample for the passengers, which announces the situation, that emergency equipment is standing by as a precaution, and the time until landing. Identify yourself as Captain, avoid jargon such as knots, chop, or abort (FOM 7.1.21.1).

> **FOM 7.1.21 Standard Customer Service Announcements and FOM 7.1.22 Emergency Vehicles Requested Sample PA**
>
> *"• Missed Approach/Go-Around – Should a go-around be necessary, a short PA announcement should be made when workload and ATC coordination permits. The FAA has advised that during"*
>
> There is no verbatim engine failure PA in the FOM. The closest scripted text is FOM 7.1.22 (emergency equipment standing by) and the NTSB brief examples at FOM 11.2.11, one of which is an engine shutdown and return. Only FOM 7.1.20.1 'Flight Attendants should now be seated for takeoff' is verbatim-mandatory.
>
> *Workbook wording: Demonstrate a late pushback PA, a go around PA, and engine failure PA*

COMMUNICATIONS AND PA

How do you make a PA on the headset?

Push and hold the ACP PA mic switch

With a headset or boom mic, the PA is keyed from the audio control panel: push and hold the PA mic switch, which extinguishes the MIC light for any other transmitter and keys the announcement from that station to all PA areas (FCOM 5.10.2). You do not need to select a transmitter first; FCOM 5.30.2 says pushing a PA mic switch on an ACP provides direct access to all PA areas. Flight deck announcements from an ACP are the highest cabin PA priority. The PA can also be reached through the cabin interphone system or the flight deck handset, and PA sales mode lets announcements be made without interrupting IFE audio and video.

> **FCOM 5.10.2 Audio Control Panel (ACP) - PA Mic Switch**
>
> *"6 PA Mic Switch Push and hold – • the MIC light for any other transmitter extinguishes • keys passenger address announcement from this station to all PA areas"*
>
> The ACP PA mic switch is direct access to ALL PA areas and takes top cabin PA priority, above priority all-area announcements and above handset announcements (FCOM 5.30.2). Monitor the PA by pushing the PA receiver volume control on the ACP.
>
> *Workbook wording: How do you make a PA using a headset (which button/transmit switches)?*

GENERAL OPERATIONS

PA PRIO vs PA ALL?

PRIO is priority. ALL includes crew rest

The 787 audio control panel provides selectable PA transmit options. PA ALL keys the passenger address announcement from that station to all PA areas, which includes the crew rest facilities. The priority selection is used when the announcement must override lower-priority audio.

> **FCOM 5.30 Communications, Passenger Address System**
>
> *""keys passenger address announcement from this station to all PA areas""*
>
> DESKTOP: confirm the exact PRIO versus ALL behaviour and the crew-rest exclusion selection against FCOM 5.30.2 in the PDF. Practical point tied to FOM 7.1.21: if you do not want to wake resting pilots in the OFCR, the selection matters, and the FOM already tells you to limit PA use while passengers sleep.
>
> *Workbook wording: What is the difference between PA PRIO and PA ALL selections?*

COMMUNICATIONS AND PA

How do you make a PA on the flight deck handset?

Select the PA dial code, push handset PA push-to-talk

Take the handset off the cradle, select the PA area with the numeric keys, the step up/step down switches, or the dial code select switch, then push and hold the handset PA push-to-talk switch, which connects the handset microphone to the selected PA area and is only used in PA mode (FCOM 5.10.19). From the TCP, PA CALL on the CABIN INTERPHONE speed dial page selects the passenger address system (FCOM 5.30.4). Handset announcements are the lowest of the three cabin PA priorities, below ACP flight deck announcements and priority all-area announcements (FCOM 5.30.2). The reset switch cancels a call or a wrong code, and PUSH RESET appears if the handset sits off hook unused for more than 30 seconds.

> **FCOM 5.10.19 Handset (aft aisle stand panel)**
>
> *“2 Handset PA Push-To-Talk Switch Push – • connects the handset microphone to the selected PA area • only used in the PA mode”*
>
> Select PA CALL from the TCP speed dial list or enter/step to the PA area dial code on the handset, then key the handset PA push-to-talk. Dial codes entered on the handset are not shown on the TCP cabin interphone pages, and the PTT works only in PA mode.
>
> *Workbook wording: How do you make a PA using the flight deck handset?*

CREW REST

Which PA reaches the cabin but not crew rest?

Handset or TCP PA area selection, not the ACP switch

FCOM 5.30.6 shows the cabin interphone directory with separate PA AREAS and CREW REST entries, and the subdirectory offers PA call to all cabin areas, PA call to individual cabin areas, and PA priority call to all cabin areas. Choosing from PA AREAS on the TCP or handset is how you address the cabin without keying every PA area. By contrast FCOM 5.10.2 and 5.30.2 are explicit that the ACP PA mic switch reaches all PA areas, which is why it is the wrong tool during crew rest. Directory entries and dial codes are predefined by the airline (FCOM 5.30.6), so verify the exact label on the aircraft. FOM 5.1.23.8 gives the A330 method (PA+ALL with the flight deck handset PTT) as the model for keeping the crew rest quiet.

> **FCOM 5.30.6 Cabin Interphone Subdirectory Page (PA AREAS)**
>
> *“Typical stations or station groups are: • individual cabin station • two or more cabin stations for conference calls • PA call to all cabin areas • PA call to individual cabin areas”*
>
> Verified: the ACP PA mic switch always goes to ALL PA areas (FCOM 5.10.2), so it cannot exclude crew rest; the cabin interphone PA AREAS subdirectory is where a cabin-only PA area is selected. The specific dial code label is airline-programmed and is not printed in the FCOM, FOM or QRH, so confirm it on the TCP directory. The A330 equivalent, PA+ALL on the flight deck handset, is spelled out at FOM 5.1.23.8 but is not stated for the 787.
>
> *Workbook wording: What selection should you use to make a PA throughout the cabin, but NOT to the crew rest facilities?*

COMMUNICATIONS AND PA

Where is the list of cabin communication codes?

In the FOM PA section and the FCOM

PA and interphone policy is FOM 7.1.15 Interphone/Call System Procedures, 7.1.18 Public Address System, 7.1.19 Required Announcements and 7.1.21 Standard Customer Service Announcements. Handset and headset keying, PA selections and the 787 Cabin Communication Codes are FCOM Chapter 5 Communications. The late-pushback PA minute threshold is a company service standard.

> **FOM 7.1.15-7.1.21 / FCOM Ch 5**
>
> No list of cabin communication codes exists in the FOM, FCOM or QRH. The FCOM states the directory entries and dial codes are predefined by the airline, so the live list is on the aircraft TCP CABIN INTERPHONE directory and subdirectory pages; the printed list belongs to the Inflight/cabin crew documentation, not the flight manuals. Verify it there rather than guessing a code.
>
> *Workbook wording: (787) Where can you find a list of Cabin Communication Codes?*

1.3.2 Perform FCOM Procedures Prior to Engine Start

PUSHBACK AND START

What check is called to the flight deck before start?

Walkaround Check complete, verbally from the Ground Crew

FOM 5.2.20 requires, before movement, that the Flight Crew check all door indications to confirm closed and receive a verbal confirmation from the Ground Crew that the Walkaround Check is complete. The cabin side of the same paragraph is the A Flight Attendant/FFA checking with the Captain before closing the boarding door (expect 'Are you ready to close boarding door?'), then stating 'Cabin Secure' and closing the flight deck door. The Captain must confirm fueling is complete before the forward primary entry door is closed. Any change that makes the cabin unsafe for movement must be reported to the flight deck immediately.

> **FOM 5.2.20 Cabin Safety Check Prior to Taxi or Pushback**
>
> > *“to confirm they are closed and receive a verbal confirmation from the Ground Crew that the Walkaround Check is complete. If installed, the Installed Physical Secondary Barrier (IPSB) must”*
>
> Two separate confirmations happen here: the A Flight Attendant/FFA says 'Cabin Secure' and closes the flight deck door, and the Ground Crew verbally confirms the Walkaround Check. The crew must also check all door indications and verify the IPSB stowed and latched if installed.
>
> *Workbook wording: What “check” must be communicated to the Flt Deck prior to engine start?*

PUSHBACK AND START &MIDDOT; WALKTHROUGH ITEM, PERFORM RATHER THAN RECALL

Review the engine failure procedure

In the FCOM and the FCTM

Engine failure procedure review, RTO PM callouts, max angle climb speed and minimum autopilot engagement altitude are FCOM NP.21 and L.10 plus FCTM. Verify each figure against the PDF; do not carry them from memory into a Procedures Validation.

> **FCOM NP.21 / L.10 / FCTM R9**
>
> *Workbook wording: Review Engine failure procedure*

PUSHBACK AND START

Where do you find the EFP for a runway?

On the ACARS TPR; also AeroData PH engine out procedures

The engine failure procedure for the specific runway prints on the AeroData Takeoff Performance Report, item 28 of the TPR legend at FCOM PI.19.17, right after engine out flap retraction altitude at item 27. The TPR is obtained by ACARS through the aFDP tool from the Company PERFORMANCE menu (FCOM PI.19.1), in preliminary and final versions. The backup copy is the AeroData PH on the EFB, which FOM 2.4.2 requires to be carried with printed Special Engine Out Procedures for departure and destination. FOM 5.5.1 makes it binding: if an engine fails on takeoff from a runway that has an engine failure takeoff procedure, the published procedure must be flown to meet obstacle criteria.

> **FCOM PI.19.17 Takeoff Performance Report (TPR) legend, item 28**
>
> *“27. Engine out flap retraction altitude 28. Engine failure procedure”*
>
> Second source: FOM 2.4.2 requires the (HA) AeroData PH on the EFB with printed Special Engine Out Procedures for departure and destination. Also check the Jeppesen 10-7 pages for airport-specific engine out or obstacle guidance.
>
> *Workbook wording: Where can you find the EFP for a specific runway? Anywhere else?*

PUSHBACK AND START

Where is the departure briefing guidance?

Yes. FOM Threat-Based Departure Briefing/TPC

FOM 5.2.7 says the PF initiates the departure briefing prior to pushback and that it should be interactive, concise and big picture, with the PM identifying relevant threats first, using the Crew Briefing Reference Card threat list and 10-7 airport threats as needed, and the PF adding any others. Plan covers taxi and departure runway, takeoff performance and configuration, route (ATC clearance versus programmed route), and the return plan including emergency return and takeoff alternate. Considerations covers specific PM duties driven by the threats, any other issues, and a recap of critical threats. The FCOM side is NP.21.26, where taxi and takeoff briefings are completed before start, and FCOM NP.21.33 where the pilot taking off updates any changes.

> **FOM 5.2.7 Threat-Based Departure Briefing/TPC**
>
> *“Prior to pushback, the PF initiates the departure briefing. The departure briefing should be interactive, concise, and focused on the big picture. Pilots jointly identify any relevant threats,”*
>
> TPC = Threats (PM and PF), Plan (PF), Considerations (PF). The threat list is on the Crew Briefing Reference Card plus airport threats in the 10-7. The arrival mirror image is FOM 5.5.16 Threat-Based Approach Briefing/TPC.
>
> *Workbook wording: Do we have any guidance regarding the Departure briefing? Where can it be found?*

1.3.3 Perform Aircraft Pushback Procedure

PUSHBACK AND START

How do you talk to the ground crew before pushback?

Flight interphone headset, nose gear jack, standard phraseology

Normal means is the flight interphone. The ground crew plugs into the flight interphone jack on the nose landing gear (FCOM 5.30.1) and the flight deck keys INT on the ACP or control wheel. FOM 5.2.24 scripts the exact phraseology for brake release, cleared to start, and cleared to disconnect. Roughly 10 minutes before departure a ground agent does an operational interphone check with the flight deck (FOM 5.2.23), and all tug departures require at least two Wing Walkers.

> **FOM 5.2.23 Pushback / 5.2.24 Pushback Interphone Communication (717/737/787/A321/A330P)**
>
> *“Interphone communication between Flight Deck and pushback crew is normally required. Standard”*
>
> FCOM NP.21.30 Pushback or Towing Procedure makes it a Captain item: establish communications with ground handling personnel. The jack itself is described in FCOM 5.30.1.
>
> *Workbook wording: How do you communicate with your ground crew prior to pushback? (Cockpit/Ground Communications)*

PUSHBACK AND START

How do you alert the ground crew?

Flight interphone; CONNECT INTERPHONE hand signal if unavailable

Ground-to-flight-deck alerting is well documented: a ground crew call gives an aural chime, the GROUND CALL EICAS communications message, and a CALL light on the ACP transmitter select switch (FCOM 5.30.1). In the other direction the manuals only give the interphone itself plus the CONNECT INTERPHONE marshalling signal in FOM 5.3.12, and NEGATIVE combined with CONNECT INTERPHONE means communication has been lost. The wheel well crew call horn is driven by equipment cooling faults and the fire/overheat test, not by a crew call switch (FCOM 2.20.10 and 8.20.7).

> **FOM 5.3.12 Hand Signals**
>
> *“CONNECT INTERPHONE; indicates the need for direct contact”*
>
> The 787 FCOM shows no flight deck switch dedicated to calling the ground crew. The FLIGHT DECK CALL SW on the nose wheel well panel (FCOM 8.10.5) runs ground-to-flight-deck only.
>
> *Workbook wording: How do you alert/call the ground crew?*

PUSHBACK AND START

No interphone for pushback. What are the hand signals?

Set/release brakes, ready for pushback, emergency stop, all clear

The pushback-relevant signals in FOM 5.3.12 are SET BRAKES (arms out, palms to crew, fingers closing into a fist), RELEASE BRAKES (reverse of that), READY FOR PUSHBACK (hands at eye level, index fingers pointing aft, fore and aft movement), CLEARED TO PUSHBACK (Captain's hitchhiker thumb moving side to side), EMERGENCY STOP (arms and wands brought overhead and crossed), START ENGINES (right forearm up and rotating), and the ALL CLEAR salute. The Flight Crew responds with the same brake signal to confirm. Aircraft does not move until the Marshaller's salute is received and returned or the taxi light is flashed.

> **FOM 5.3.12 Hand Signals / FOM 5.2.23.1 Pushback - No Interphone**
>
> *“READY FOR PUSHBACK; indication that the Ground Crew is ready to pushback.”*
>
> Governing rule is FOM 5.2.23: if a No Interphone pushback must be made, the Captain follows the pushback lead's hand signals. FOM 5.2.23.1 requires the Captain to brief the ground crew first, including how to signal STOP.
>
> *Workbook wording: If pushing back without an interphone, what are the hand signals?*

PUSHBACK AND START

Can you be pulled forward with two engines running?

No. Prohibited with both engines running

FOM 5.2.23 states the Ground Crew policy currently prohibits forward tow operations with both engines running on the 787 and A330. Related cautions in the same section: never disconnect the towbar without confirming brakes are set, both pilots verify the parking brake is set before replying to the Ground Crew, and powerbacks are not authorized. FCOM NP.21.30 adds do not turn the nose wheel tiller and do not use airplane brakes to stop the airplane during pushback or towing.

> **FOM 5.2.23 Pushback**
>
> *“(787/A330) The Ground Crew policy currently prohibits forward tow operations with both engines”*
>
> This is a ground crew policy limit, not an aircraft limitation, and it is banner-tagged specifically to the 787 and A330.
>
> *Workbook wording: Can you have the aircraft pulled forward with two engines running? (787/330 only)*

1.3.4.1 Complete Normal engine Start FCOM Procedure

Non-Normal Start Operations

PUSHBACK AND START

Which quick action items apply during engine start?

Aborted Engine Start, ENG AUTOSTART, Fire Eng on Ground/Tailpipe

Quick Action Index items that can bite during ENG START are Aborted Engine Start L, R (7.1), ENG AUTOSTART L, R (7.3), ENG LIMIT EXCEED L, R (7.4), Fire Eng Tailpipe L, R (8.5), FIRE ENG L, R (8.2), and Fire Eng on Ground L, R (Back Cover.2). Aborted Engine Start is FUEL CONTROL switch CUTOFF, START selector START, motor 30 seconds, then START selector NORM. ENG AUTOSTART is FUEL CONTROL switch confirm CUTOFF then START selector NORM. Quick Action Index checklist titles print in bold and are listed first inside each system section (QRH CI.2.1).

> **QRH Quick Action Index QA.Index.1**
>
> *"*Aborted Engine Start L, R .... 7.1"*
>
> There is no separate 787 hot start or hung start checklist. Autostart manages start faults, and ENG AUTOSTART L, R (QRH 7.3) is the annunciated failure when autostart cannot start the engine.
>
> *Workbook wording: Which Quick Action Items (787) may be required during ENG START, e.g. Aborted Engine Start, Hot Start, Hung Start?*

PUSHBACK AND START

Is there a QRH procedure for a tailpipe fire?

Yes. FIRE ENG TAILPIPE L, R

Yes. The QRH carries FIRE ENG TAILPIPE L, R in Chapter 8 at 8.5. It appears in the QRH index in the engine fire and fire protection group.

> **QRH 8.5 Fire Eng Tailpipe L, R**
>
> *""Fire Eng Tailpipe L, R""*
>
> A tailpipe fire is distinct from an engine fire: it is a fire in the tailpipe with no engine fire indication, and the procedure differs accordingly. DESKTOP: read the full step sequence off QRH 8.5 before briefing it to an FO.
>
> *Workbook wording: Is there a QRH procedure for a TAILPIPE FIRE?*

PUSHBACK AND START

What is the hand signal for a tailpipe fire?

Horizontal figure eight; other hand points to fire

FOM 5.3.12 defines FIRE as a horizontal figure 8 made by hand or wand, with the other hand or wand pointing at the location of the fire. On the 787 a tailpipe fire on the ground with no engine fire warning drives QRH 8.5: FUEL CONTROL switch CUTOFF, verify the engine fire switch in, advise the cabin, START selector to START to motor the engine, advise ATC, then START selector NORM once out. Motoring only works with the APU or two external power sources on.

> **FOM 5.3.12 Hand Signals**
>
> *"FIRE; signifies a fire and it's location to the Flight Crew by indicating a horizontal"*
>
> Same FIRE signal is used for any fire location, not just a tailpipe fire. Pair it with QRH 8.5 Fire Eng Tailpipe L, R.
>
> *Workbook wording: What is the hand signal for Tailpipe Fire?*

PUSHBACK AND START

Is there a QRH procedure for a ground engine fire?

Yes. Fire Eng on Ground L, R

Fire Eng on Ground L, R applies when fire is observed or detected in the affected engine on the ground. Sequence: thrust levers both idle (C), PARKING BRAKE set (C), advise the cabin (C or F/O), FUEL CONTROL switch affected side CUTOFF (C), engine fire switch pull (F/O), rotate to the stop and hold 1 second (F/O), then choose evacuation (Back Cover.4) or advise cabin and ATC. Note this is distinct from FIRE ENG L, R at QRH 8.2, which is the airborne checklist, and from Fire Eng Tailpipe L, R at 8.5.

> **QRH Back Cover.2 Fire Eng on Ground L, R**
>
> *"Condition: One or more of these occur on the ground:"*
>
> It lives on the back cover of the QRH, not in the Fire chapter, and it is cross-referenced from QRH 8.4 and from the Quick Action Index.
>
> *Workbook wording: Is there a QRH procedure for an ENG FIRE on the ground?*

1.3.5 Perform FCOM Procedures Following Engine Start/Before Taxi

PUSHBACK AND START

What must you consider before pushback and taxi?

NOTAMs, ATIS, closures, construction, and the taxi route

FOM 5.3.4.2 requires reviewing NOTAMs and current ATIS for taxiway or runway closures, construction, and other airport risks affecting the taxi route, and requires both crewmembers to understand the expected taxi route and every clearance. If any confusion exists, stop the aircraft and query the controlling agency; complex taxi instructions shall be recorded. On the aircraft side FCOM NP.21.31 Before Taxi covers generator lights out, APU off, engine anti-ice, ground personnel clear (salute is sufficient), flaps, flight control check, recall, and the taxi briefing update.

> **FOM 5.3.4.2 Prior to Taxi**
>
> *"• Review NOTAMs and current ATIS for any taxiway or runway closures, construction activity, or"*
>
> FOM 5.3.4.1 adds the dual acknowledgment requirement for runway crossing, hold short, line up and wait, cleared for takeoff, and cleared to land.
>
> *Workbook wording: What items must be considered prior to pushback/taxi?*

1.3.1 Perform FOM Takeoff and Departure Briefing Including Identification of Operational Threats

Threat Based Briefing/TPC

TAKEOFF AND DEPARTURE

What is the main concept of the TPC briefing?

Threats jointly, then the PF plan, then considerations

TPC replaces a read-the-plate monologue with a threat-led, interactive exchange. The PM opens by naming relevant threats, the PF adds any others, then the PF briefs the plan and the considerations. It should be interactive, concise and focused on the big picture, with management strategies discussed as threats surface.

> **FOM 5.5.16 Threat-Based Approach Briefing/TPC**
>
> *""The approach briefing should be interactive, concise, and focused on the big picture.""*
>
> Captain-side: an interactive briefing is where you set flight-deck tone. If the FO never speaks during your brief, the T was skipped and you have lost the main safety value of the format.
>
> *Workbook wording: Discuss the main concept of the TPC briefing*

TAKEOFF AND DEPARTURE

Who starts the briefing?

The PF, but the PM speaks first on threats

Prior to beginning the descent, the PF initiates the approach briefing. Within the format the PM identifies relevant threats first, then the PF briefs any additional threats followed by the plan and considerations.

> **FOM 5.5.16 Threat-Based Approach Briefing/TPC**
>
> *""Prior to beginning the descent, the PF initiates the approach briefing.""*

TAKEOFF AND DEPARTURE

Where is the briefing guidance?

In the FOM threat-based approach briefing section

FOM 5.5.16 carries the TPC structure. The threat list on the Crew Briefing Reference Card supports the T element, and airport-specific threats come from the 10-7 pages.

> **FOM 5.5.16 Threat-Based Approach Briefing/TPC**
>
> *""Refer to the threat list on the Crew Briefing Reference Card, if desired""*
>
> *Workbook wording: Do we have any guidance regarding this briefing? Where can it be found?*

2.1.1 Perform Normal Taxi-Out (Taxi-In) [CRITICAL OBSERVABLE ITEM]

Taxiing

TAXI AND GROUND

Where is taxi guidance in the FOM?

In the FOM taxi section

Section 5.3 carries taxi policy: 5.3.1 Taxi Prohibited, 5.3.2 Taxi Thrust, 5.3.3 Taxi Guidance, 5.3.4 Runway Incursion Prevention, 5.3.9 Passengers Standing While Taxiing, 5.3.10 External Lights, 5.3.11 Taxi Operations Low Visibility, 5.3.12 Hand Signals and 5.3.13 Tarmac Delays. Fleet-specific technique is in the FCOM and FCTM.

> **FOM 5.3 Taxi**
>
> *""See fleet-specific manuals for further guidance.""*
>
> *Workbook wording: Where in the FOM can you find information on taxiing?*

EFB AND FD PRO &MIDDOT; WALKTHROUGH ITEM, PERFORM RATHER THAN RECALL

Must both pilots have the FD Pro taxi chart out?

Not stated in our manuals. FD Pro procedure

This is a Jeppesen FD Pro X or EFB workflow step, not FOM/FCOM content. Verify against the FD Pro Pubs Quick Reference guide in Comply365 and demonstrate on the actual device during OE. FOM 5.7.3.15 does require the ENTIRE cleared route be loaded into the approved charting app for ORCN segments.

> **FD Pro Pubs Quick Reference (Comply365)**
>
> *Workbook wording: Do both crewmembers must have the FD Pro TAXI chart out and available during surface movements?*

TAXI AND GROUND

Can the ND airport map be used for navigation?

Not stated in our manuals

The airport map on the ND is a situational awareness aid and is not approved as a primary navigation source; confirm the exact FCOM wording before stating it. Recommended speed prior to a 90 degree turn is FCTM Ground Operations. FOM 5.3.3 caps taxi speed at 30 kts straight and smooth, approximately 10 kts in congested areas.

> **FCOM Ch 1/11 / FCTM Ground Operations**
>
> *Workbook wording: (787) Can the Airport map displayed on the ND be used for navigation?*

PUSHBACK AND START

Max breakaway thrust N1?

40% N1 maximum for breakaway

FCOM NP.21.33 lists three engine warm up requirements before takeoff: run the engines at least 3 minutes, use no more than 40% N1 for breakaway, and engine oil temperature above the bottom of the temperature scale. FOM 5.3.2 Taxi Thrust only warns to stay aware of equipment, structures, and aircraft behind you when engines are above idle. Practically, break away then reduce to idle and let the aircraft accelerate; on the 787 the thrust levers are far more sensitive at low N1 than an A330 pilot expects.

> **FCOM NP.21.33 Before Takeoff Procedure - Engine warm up requirements**
>
> *“Use no more than 40% N1 for breakaway”*
>
> Same 40% N1 number appears in FCOM SP sand and dust guidance, paired with a 10 knot ground speed cap. FOM 5.3.2 Taxi Thrust carries no number, so the pointer to FOM is wrong.
>
> *Workbook wording: Thrust Use – Max breakaway thrust, N1?*

TAXI AND GROUND

Max taxi speed?

30 kts straight and smooth, 10 kts congested

Speed shall be limited to not more than 30 kts, achieved only on straight, smooth taxiways, and approximately 10 kts in congested areas. While taxiing in the ramp or gate area, any congested area, where taxi instructions are complex, or during reduced visibility, the Flight Crew shall devote full attention to taxiing the aircraft.

> **FOM 5.3.3 Taxi Guidance**
>
> *“"Speed shall be limited to not more than 30 kts, achieved only on straight, smooth taxiways, and approximately 10 kts in congested areas."”*
>
> Note shall, and note that 30 kts is conditional on straight and smooth, not a blanket allowance. When operationally feasible, single-engine taxi is the standard procedure for fuel efficiency. DESKTOP: carbon brake technique (one firm application rather than repeated light braking) is FCTM Ground Operations, not in this environment.
>
> *Workbook wording: Max Taxi Speed/s, and Braking technique?*

TAXI AND GROUND

Why keep groundspeed in your scan?

To hold 30 kts max, about 10 kts congested

FOM 5.3.3 caps taxi speed at 30 kts, only on straight smooth taxiways, and approximately 10 kts in congested areas, and requires full crew attention while taxiing in the ramp, congested areas, complex taxi instructions, or reduced visibility. FCOM SP adverse ground guidance repeats a 10 knot ground speed limit with thrust below 40% N1 where sand and dust ingestion is a concern. Ground speed is the only accurate cue at high gross weight because the 787 accelerates on idle thrust and decelerates slowly; single-engine taxi is the standard fuel-saving procedure when operationally feasible.

> **FOM 5.3.3 Taxi Guidance**
>
> *“Speed shall be limited to not more than 30 kts, achieved only on straight, smooth taxiways, and”*
>
> 787 has no taxi speed readout on the PFD in the same place an A330 pilot expects; ground speed on the ND is the reference. Turn speed discipline also protects tires and brakes.
>
> *Workbook wording: Emphasize importance of keeping groundspeed in scan*

TAXI AND GROUND

Recommended speed before a 90 degree turn?

In the FCOM and the FCTM

The airport map on the ND is a situational awareness aid and is not approved as a primary navigation source; confirm the exact FCOM wording before stating it. Recommended speed prior to a 90 degree turn is FCTM Ground Operations. FOM 5.3.3 caps taxi speed at 30 kts straight and smooth, approximately 10 kts in congested areas.

> **FCOM Ch 1/11 / FCTM Ground Operations**
> *Workbook wording: Review recommended speed prior to entering a 90n turn*

TAXI AND GROUND

When do you use differential braking for hot brakes?

In the FCTM ground operations chapter

Turning radii, oversteering technique, breakaway thrust, differential braking and carbon brake technique are FCTM Ground Operations content. FCTM R9 is not in the browser environment. Verify on desktop before quoting any figure.

> **FCTM Ground Operations**
> *Workbook wording: Demonstrate use of differential braking if brake temperature on one side is significantly hotter than the*

TAXI AND GROUND

Gear location and turning radius?

In the FCTM ground operations chapter

Turning radii, oversteering technique, breakaway thrust, differential braking and carbon brake technique are FCTM Ground Operations content. FCTM R9 is not in the browser environment. Verify on desktop before quoting any figure.

> **FCTM Ground Operations**
> *Workbook wording: Location of gear/turning Radius*

TAXI AND GROUND

Where is the guidance on oversteering in turns?

In the FCTM ground operations chapter

Turning radii, oversteering technique, breakaway thrust, differential braking and carbon brake technique are FCTM Ground Operations content. FCTM R9 is not in the browser environment. Verify on desktop before quoting any figure.

> **FCTM Ground Operations**
> *Workbook wording: Where can you find taxi guidance/technique regarding oversteering in turns?*

TAXI AND GROUND

Normal 180 degree turning radius?

In the FCTM ground operations chapter

Turning radii, oversteering technique, breakaway thrust, differential braking and carbon brake technique are FCTM Ground Operations content. FCTM R9 is not in the browser environment. Verify on desktop before quoting any figure.

> **FCTM Ground Operations**
> *Workbook wording: What is the normal 180 degree turning radius?*

TAXI AND GROUND

Pivot 180 degree turning radius?

In the FCTM ground operations chapter

Turning radii, oversteering technique, breakaway thrust, differential braking and carbon brake technique are FCTM Ground Operations content. FCTM R9 is not in the browser environment. Verify on desktop before quoting any figure.

> **FCTM Ground Operations**
> *Workbook wording: What is the non-normal (pivot) 180 degree turning radius?*

WEATHER AND MINIMUMS

When do SMGCS procedures apply?

At airports with a published low visibility taxi plan

Airports may develop a low visibility taxi plan such as SMGCS and may publish low visibility taxi procedures. If an airport is equipped with Surface Movement Guidance and Control System, use the AMM Low Vis feature in Jeppesen FD Pro or the Jeppesen Low Visibility Taxi Route charts when SMGCS is in effect. Canadian airports with low visibility taxi plans may produce SMGCS charts in Jeppesen format.

FOM 5.3.11 Taxi Operations Low Visibility

> *""If an airport is equipped with Surface Movement Guidance and Control System (SMGCS), see the AMM Low Vis feature in Jeppesen FD Pro or Jeppesen Low Visibility Taxi Route charts""*

DESKTOP: the specific RVR trigger value at which SMGCS becomes active is airport-published on the 10-9 series, not a single FOM number. Canada-specific low visibility taxi plans are at FOM 19.3.1.5.

Workbook wording: When do SMGCS procedures go into effect?

2.1.3 Perform FCOM Pre-Takeoff Procedures

Airport Ops

TAKEOFF AND DEPARTURE

Can you depart at night with runway edge lights out?

Only at 1/4 sm or RVR 1600, with FODO permission

If ALL runway lights are inoperative, takeoff is not authorized during darkness unless both conditions are met: reported visibility greater than or equal to 1/4 sm or RVR 1600, and permission of the Flight Operations Duty Officer. If only a PORTION of the runway lights are inoperative, visibility is at least 1/4 sm or RVR 1600, and the remaining lights or runway markings give adequate visual reference to continuously identify the takeoff surface and maintain directional control throughout the takeoff run, takeoff may proceed.

FOM 5.4.9 Runway Lights Inoperative

> *""Permission of the Flight Operations Duty Officer (FODO)""*

Two different tests depending on all versus partial. The FODO call is only required for ALL lights inoperative. This is a Captain decision item that people commonly get wrong by assuming any light outage grounds the departure.

Workbook wording: Can we depart at night from a runway where the edge lights are inoperative?

TAXI AND GROUND

Runway change during taxi. What do you do?

New takeoff data, reload FMC departure, re-brief

FOM 5.4.5 requires runway analysis data to be available and checked for each takeoff; data comes from the Flight Information Packet (TPR) or Dispatch Release (TLR), and if invalid, new takeoff data can be obtained via ACARS or from Dispatch. In the FMC, reselect the runway and SID on the DEPARTURES page, re-accept the takeoff uplink on TAKEOFF REF (the TAKEOFF DATA LOADED / UPLINK alerts flag a runway or thrust change), reset trim, and confirm V speeds. FCOM NP.21.33 has the PF update changes to the takeoff briefing as needed, and NP.21.34 requires verifying the runway and entry point before crossing the hold short line.

FOM 5.4.5 Runway Analysis

> *“new takeoff data can be obtained via ACARS or from Dispatch”*

On the 787, if the HUD TAKEOFF runway changes, FCOM NP.21.3 requires both flight director switches OFF, then back ON after the new runway is executed on the CDU.

Workbook wording: If given a change of departure runway during the taxi, how is this accomplished prior to takeoff and what

Passenger Ops

TAXI AND GROUND

Must you stop taxiing if a guest stands up?

No

There is no regulatory requirement that the aircraft stop if a passenger gets out of their seat. Depending on the circumstances, it may be safer to continue to taxi.

> **FOM 5.3.9 Passengers Standing While Taxiing**
>
> *""There is no regulatory requirement that the aircraft stop if a passenger gets out of their seat.""*
>
> Captain judgment call. Stopping on an active taxiway can create a bigger hazard than the standing passenger. Coordinate with the Cabin Crew rather than reflexively braking.
>
> *Workbook wording: Are we required to stop the aircraft during taxi if a guest stands up?*

Cold Weather Ops

EFB AND FD PRO &MIDDOT; WALKTHROUGH ITEM, PERFORM RATHER THAN RECALL

SureWx with active precipitation or a contaminated surface?

In the SureWx guide and the FOM

SureWx holdover-time workflow is vendor material. The governing policy is the pretakeoff contamination check requirement. Verify the app steps in the SureWx guide and the de-icing policy in FOM Chapter 9 and the fleet cold weather supplement.

> **SureWx guide / FOM Ch 9**
>
> *Workbook wording: While using SureWx app, if/when there is active precipitation or during contaminated surface operations,*

EFB AND FD PRO &MIDDOT; WALKTHROUGH ITEM, PERFORM RATHER THAN RECALL

SureWx holdover time expired. What check is required?

In the SureWx guide and the FOM

SureWx holdover-time workflow is vendor material. The governing policy is the pretakeoff contamination check requirement. Verify the app steps in the SureWx guide and the de-icing policy in FOM Chapter 9 and the fleet cold weather supplement.

> **SureWx guide / FOM Ch 9**
>
> *Workbook wording: While using SureWx app and the holdover time has expired, what check is required prior to takeoff?*

COLD WEATHER AND RUNWAY CONDITION

Which manual holds the cold weather and de-icing material?

The FODM, plus fleet-specific manuals

FOM 9.3.1 sends you to the fleet-specific manuals and the FODM for cold weather procedures, and FOM 9.3.1.1 sends you to the FODM again for Liquid Water Equivalent holdover times. FOM 16.2.9 names the ASA Deicing Procedures Manual alongside the FODM. FCOM SP.16 carries the 787 procedures: icing conditions defined as OAT or TAT 10 C or below with visible moisture or contaminated surfaces, the modified preliminary preflight, exterior inspection, engine start, taxi-out, de-icing/anti-icing, before takeoff, and fan ice removal.

> **FOM 9.3.1 Cold Weather Preflight Requirements**
>
> *“Comply with the cold weather procedures in the fleet-specific manuals and FODM.”*
>
> FOM does not spell out the FODM acronym in text. Aircraft-side cold weather procedures are in FCOM SP.16 Cold Weather Operations; FOM 9.3 is the policy layer only.
>
> *Workbook wording: In which manual is most of the cold weather/de-icing information found?*

COLD WEATHER AND RUNWAY CONDITION

What are your cold weather ops resources?

FODM, HOTs app, FCOM SP.16, Deicing Procedures Manual

Practical resource stack: FODM for policy and holdover tables, the HOTs app on the EFB for holdover times (with printed HOTs from the FODM as the backup), FCOM SP.16 for 787 cold weather flows and de-icing/anti-icing procedure, and the ASA Deicing Procedures Manual through Maintenance Control. FOM 9.3.1.1 makes LWE-derived holdover times FAA-authorized and mandatory when available, including with +SN or FZRA reported. The flight log de/anti-icing block (type fluid, mix, start time, start point, post check, holdover time) references FODM Section 5.4.

> **FOM 16.2.9 Deice - Charters / FOM 2.4.2 Required Manuals and Documents for Dispatch**
>
> *“All deice procedures are found in the ASA Deicing Procedures Manual and the FODM . The ASA”*
>
> FOM 2.4.2 lists FODM and the HOTs app as EFB apps not required for dispatch; the alternate for HOTs is printed Holdover Times from the FODM. The ASA Deicing Procedures Manual is obtained through Maintenance Control.
>
> *Workbook wording: What are your cold weather ops resources? (Manuals, cards etc.)*

Contaminated Runway Ops

COLD WEATHER AND RUNWAY CONDITION

Wet vs contaminated runway?

Wet is a water film. Contaminated is beyond that

The FOM carries the operational consequence rather than a bare definition: for a Landing Distance Available of less than 7000 ft, do not use WET data. The wet versus contaminated distinction drives which performance table you are allowed to use, which is why the LDA threshold matters more on the line than the textbook definition.

> **FOM 9.1.12.3 Wet vs. Contaminated Data**
>
> *“"For LDA less than 7000 ft, do not use WET data."”*
>
> DESKTOP: the full wet/contaminated definitions and the RCAM/Runway Condition Code tables sit in FCTM and the TALPA material, which are not in this environment. Verify the definition wording there. Related: FOM 9.1.13 Degraded Braking uses the Can U StoP mnemonic.
>
> *Workbook wording: What is the difference between a wet vs contaminated runway?*

COLD WEATHER AND RUNWAY CONDITION

Braking action report vs RCAM?

PIREP is pilot-reported deceleration; RCAM is an assessment tool

A braking action report is a pilot qualitative PIREP (GOOD, GOOD to MEDIUM, MEDIUM, MEDIUM to POOR, POOR, NIL) and airport personnel cannot issue one. The RCAM is a published FAA or ICAO matrix used primarily by airports to assess conditions and generate a Runway Condition Code, 0 (NIL) through 6 (dry), and correlate it with those same descriptive terms. For the 787, FOM 9.1.6 points you to the FCTM Landing section for the RCAM presentation and runway-condition-dependent wind limits. A PIREP from a similar size aircraft may be used when no FICON is available, or to downgrade, never to upgrade.

> **FOM 9.1.2 Braking Action Report / FOM 9.1.6 Runway Condition Assessment Matrix (RCAM)**
>
> *“A braking action report (PIREP) is a qualitative description of deceleration by pilots such as GOOD,”*
>
> Key operational rule in FOM 9.1.2: a PIREP may downgrade expected braking action from the RCAM or FICON, but must never be used to upgrade it.
>
> *Workbook wording: What is the difference between a braking action report vs RCAM?*

COLD WEATHER AND RUNWAY CONDITION

What is a FICON NOTAM?

NOTAM giving runway condition codes by runway third

A FICON reports condition for touchdown, midpoint, and rollout in the direction of takeoff and landing, listed as runway thirds such as 3/4/2, with an expected braking action expressed as a Runway Condition Code. Where a single report gives multiple conditions along the runway, assume the most restrictive for the whole runway. The RWYCC applies to stopping in a rejected takeoff and to landing. Mu values are generally no longer reported, though airports may use them internally to upgrade or downgrade the code. FOM 9.1.3.1 lists the NOTAM acronyms (IR, LSR, PSR, DRFT, FRZN, PLW, and so on).

> **FOM 9.1.3 Field Condition (FICON) NOTAM**
>
> *“A FICON is a specifically formatted NOTAM reporting runway condition for each third of the”*
>
> Absence of a FICON does not assure the runway is dry; airports are encouraged but not required to issue one for a wet runway.
>
> *Workbook wording: What is a FICON (Field Condition) NOTAM?*

COLD WEATHER AND RUNWAY CONDITION

What is a Runway Condition Code?

The TALPA number, 0 through 6

The Runway Condition Code is the TALPA output used to assess a contaminated runway. On the 787 it drives the landing crosswind limitation directly: RwyCC 6 gives 40 kts, 5 (Good) 40, 4 (Good to Medium) 35, 3 (Medium) 25, 2 (Medium to Poor) 17, 1 (Poor) 15, and 0 (Nil) is no operation.

> **FCOM L.10 Landing Crosswind Limitations TALPA**
>
> *“"The crosswind limitation is determined by entering the table below with Runway Condition Code or Control/Braking Action."”*
>
> A FICON NOTAM is how the runway condition code reaches you. Reduce the crosswind limit by 5 knots on wet or contaminated runways whenever asymmetric reverse thrust is used.
>
> *Workbook wording: What is a runway CC (Runway Condition Code)?*

3.1.1 Perform Takeoff [CRITICAL OBSERVABLE ITEM]

Crosswind Control

TAKEOFF AND DEPARTURE

Max takeoff crosswind?

Off the takeoff table, less 5 kts above 40 percent N1

Takeoff crosswind limits come from the FCOM L.10 takeoff table, entered using the columns labeled 25 percent MAC and 30 percent MAC, which is why the takeoff table is CG and weight sensitive while the landing TALPA table is not. Reduce all takeoff crosswind guidelines by 5 knots if thrust is above 40 percent N1 (GE engines) at brake release for the takeoff roll.

> **FCOM L.10 Takeoff Crosswind Guidelines**
>
> *“"Reduce all takeoff crosswind guidelines by 5 knots if thrust is above 40% N1 (GE engines) at brake release for takeoff roll."”*
>
> Demonstrated crosswinds are for certification only and do not represent operational limitations; the tables are the limits. Do not interchange the takeoff and landing tables, they serve different functions. DESKTOP: read the exact takeoff table values off the FCOM L.10 PDF rather than quoting from memory.
>
> *Workbook wording: Max takeoff crosswind limitations?*

COLD WEATHER AND RUNWAY CONDITION

What is TALPA?

Takeoff and Landing Performance Assessment

TALPA is the framework that translates reported runway surface condition into a Runway Condition Code and a braking action, which then drive performance and limits. On the 787 the LANDING crosswind limitation is determined by entering the TALPA table with Runway Condition Code or Control/Braking Action: RwyCC 6 is 40 kts, 5 (Good) 40, 4 (Good to Medium) 35, 3 (Medium) 25, 2 (Medium to Poor) 17, 1 (Poor) 15, and 0 (Nil) is no operation.

> **FCOM L.10 Landing Crosswind Limitations TALPA**
>
> *“"The crosswind limitation is determined by entering the table below with Runway Condition Code or Control/Braking Action."”*
>
> Reduce the crosswind limitation by 5 knots on wet or contaminated runways whenever asymmetric reverse thrust is used. Winds are measured at 33 ft (10 m) tower height and apply to runways 148 ft (45 m) or greater in width. Landing on untreated ice or snow only when no melting is present. Sideslip-only (zero crab) landings are not recommended at the higher crosswinds.
>
> *Workbook wording: (787) TALPA; What is TALPA?*

WEATHER AND MINIMUMS

Max crosswind for a HUD takeoff?

20 knots crosswind

FCOM L.10.9 gives HUD takeoff wind component limits of 25 knots headwind, 15 knots tailwind, and 20 knots crosswind, and restricts low weather minima takeoff to ILS guidance on U.S. Type II or Type III ILS facilities. Compare with the autoland limits on the same page: 25 headwind, 15 tailwind, 25 crosswind but only 15 knots for CAT II/III. Maximum demonstrated crosswind, certification only and not a limit, is 33 knots takeoff and 40 knots landing, gust included (FCOM L.10.3). Operational approval is required for low visibility operations using HUD TO/GA.

> **FCOM L.10.9 Autoflight - AFM Limitations - Low Visibility (HUD) Takeoff**
>
> *“on HUD takeoff operations: # Headwind 25 knots # Tailwind 15 knots # Crosswind 20 knots”*
>
> This limit applies only when takeoff weather minima are predicated on HUD takeoff operations. It is not the general takeoff crosswind limit, which comes from the TALPA and non-TALPA tables at FCOM L.10.4 through L.10.7.
>
> *Workbook wording: (787) Max Crosswind for a HUD takeoff?*

Gear/Flap Management

TAKEOFF AND DEPARTURE

Max altitude with flaps extended?

20,000 ft

The maximum altitude with flaps extended is 20,000 feet.

> **FCOM L.10 Limitations, Non-AFM Operational Information**
>
> *“"The maximum altitude with flaps extended is 20,000 feet."”*
>
> Adjacent limits worth carrying from the same page: prior to takeoff the maximum allowable difference between the Captain's or First Officer's altitude display and field elevation is 75 feet, and takeoff is permitted only in the normal flight control mode.
>
> *Workbook wording: Max altitude with flaps extended.*

Rotation Rate

TAKEOFF AND DEPARTURE

Normal rotation rate?

In the FCTM. Not a published limitation

FCOM refers to using the normal takeoff rotation rate without publishing a degrees-per-second figure in the limitations. The numeric rotation rate and target pitch attitudes are FCTM technique material.

> **FCOM (rotation rate not published as a limitation)**
>
> Not published as a number in the FOM, FCOM, or QRH. The FCOM only says at VR rotate toward 15 degrees pitch attitude (NP.21.34) and use the normal takeoff rotation rate after de-icing (SP.16). A rotation rate in degrees per second lives in the 787 FCTM, Takeoff chapter, which was not among the three source manuals for this batch.
>
> *Workbook wording: Normal rotation rate? (degrees/sec)*

Pitch Attitude

TAKEOFF AND DEPARTURE

Normal takeoff rotation target pitch?

In the FCTM

The normal takeoff rotation target pitch attitude is FCTM technique content and is not carried in FCOM L.10.

> **FCTM Chapter 3 Takeoff**
>
> *Workbook wording: What is the normal takeoff rotation target pitch?*

TAKEOFF AND DEPARTURE

Rotation target pitch after an engine failure at V1?

In the FCTM

The engine-failure rotation target pitch attitude is FCTM technique content, distinct from the all-engine target, and is not published in FCOM L.10.

> **FCTM Chapter 3 Takeoff**
>
> *Workbook wording: What is the takeoff rotation target pitch attitude after an engine failure at/after V1 and prior to liftoff?*

Callouts

TAKEOFF AND DEPARTURE

What are the PM callouts during an RTO?

In the FCOM and the FCTM

Engine failure procedure review, RTO PM callouts, max angle climb speed and minimum autopilot engagement altitude are FCOM NP.21 and L.10 plus FCTM. Verify each figure against the PDF; do not carry them from memory into a Procedures Validation.

> **FCOM NP.21 / L.10 / FCTM R9**

4.1.1 Perform Departure and Climb

Noise Abatement Departure Procedures (NADP)

TAKEOFF AND DEPARTURE

What is NADP1?

The close-in noise abatement profile

NADP procedures are governed by Ops Spec C068. Specific requirements, including gradients and altitudes for configuration changes, may be defined in aircraft-specific manuals, the 10-7, or the Jeppesen Airway Manual Flight Procedures Noise Abatement Procedures. Crews must comply with published noise abatement procedures at noise-sensitive airports unless severe weather, traffic conflicts or other safety considerations require deviation, in which case the Captain must request an appropriate ATC clearance. Normal takeoff procedures satisfy either the close-in (NADP1) or distant (NADP2) profile requirements.

> **FOM 5.4.7 Noise Abatement Departure Procedure (NADP)**
>
> *""Normal takeoff procedures satisfy either close-in or distant NADP profile requirements.""*
>
> That quote is (AS) bannered. This matters for the JCAB ramp-check question in section 1.2.3 about whether HA may perform NADP 1: the OpSpec is C068. DESKTOP: confirm the HA-specific position and the OpSpec paperwork the JCAB inspector wants to see.
>
> *Workbook wording: What is NADP1? Describe associated procedures*

TAKEOFF AND DEPARTURE

What is NADP2?

The distant noise abatement profile

NADP2 is the distant profile. Per FOM 5.4.7, normal takeoff procedures satisfy either close-in or distant NADP profile requirements, so no separate profile is flown unless a specific published procedure requires it. Gradients and configuration-change altitudes come from the aircraft-specific manuals, the 10-7, or the Jeppesen Airway Manual.

> **FOM 5.4.7 Noise Abatement Departure Procedure (NADP)**
>
> *""Flight Crews must comply with published noise abatement procedures when operating from noise-sensitive airports""*
>
> The distinction between the two is where thrust reduction and acceleration occur relative to the airport. DESKTOP: the 787 profile detail is FCTM/FCOM, not FOM.
>
> *Workbook wording: What is NADP2? Describe associated procedures*

Climb Airspeed / Mach Control

TAKEOFF AND DEPARTURE

How do you find max rate of climb speed?

In the FCOM performance section and the Jepps

Max rate and max angle climb speeds are FCOM performance content. Obstruction climb gradients, transition altitude and level, and CLIMB VIA behaviour come from the Jeppesen SID plate and FOM 5.5.2 Terminal Departures. A Climb and Maintain clearance does not by itself cancel published SID altitude restrictions unless ATC explicitly cancels them.

> **FCOM performance / Jeppesen / FOM 5.5.2**
>
> *Workbook wording: How do you determine max rate of climb speed?*

WEATHER AND MINIMUMS

Severe turbulence penetration speed?

290 KIAS below FL250, 310 or .84 above

Severe turbulent air penetration speed in severe turbulence is 290 KIAS below 25,000 feet, and 310 KIAS or .84 Mach, whichever is lower, at and above 25,000 feet.

> **FCOM L.10 Severe Turbulent Air Penetration Speed**
>
> *“"290 KIAS below 25,000 feet"”*
>
> Memory-marked limitation. Note the crossover is 25,000 ft, not FL250 by pressure altitude convention, and the upper figure is whichever is LOWER of 310 KIAS or .84 Mach. Additional guidance at FCOM SP Chapter 16 Severe Turbulence.
>
> *Workbook wording: What is severe turbulence penetration speed?*

TAKEOFF AND DEPARTURE

Where is max angle climb speed?

In the FCOM and the FCTM

Engine failure procedure review, RTO PM callouts, max angle climb speed and minimum autopilot engagement altitude are FCOM NP.21 and L.10 plus FCTM. Verify each figure against the PDF; do not carry them from memory into a Procedures Validation.

> **FCOM NP.21 / L.10 / FCTM R9**
>
> *Workbook wording: Where can you find max angle climb speed?*

SID Departure

TAKEOFF AND DEPARTURE

What else gets briefed on a SID beyond legs and restrictions?

On the Jeppesen SID plate

SID plate items to brief beyond the FMC legs (runway applicability, obstruction climb gradients, red boxed notes, top altitude for CLIMB VIA) come from the Jeppesen chart itself plus FOM 5.5.2 Terminal Departures (IFR/VFR) and 5.5.1 Turns After Takeoff. A CLIMB VIA clearance carries the SID top altitude; a Climb and Maintain does not cancel published altitude restrictions unless ATC says so.

> **Jeppesen SID plate / FOM 5.5.1-5.5.2**
>
> *Workbook wording: Though the SID legs/restrictions get briefed from the plan view vs the CDU/FMS, what other items should*

TAKEOFF AND DEPARTURE

Is a SID runway specific?

Not stated in our manuals. Read the SID plate

SID plate items to brief beyond the FMC legs (runway applicability, obstruction climb gradients, red boxed notes, top altitude for CLIMB VIA) come from the Jeppesen chart itself plus FOM 5.5.2 Terminal Departures (IFR/VFR) and 5.5.1 Turns After Takeoff. A CLIMB VIA clearance carries the SID top altitude; a Climb and Maintain does not cancel published altitude restrictions unless ATC says so.

> **Jeppesen SID plate / FOM 5.5.1-5.5.2**
>
> *Workbook wording: Is the SID for a specific runway(s) only?*

TAKEOFF AND DEPARTURE

Obstruction climb gradient restrictions?

In the FCOM performance section and the Jepps

Max rate and max angle climb speeds are FCOM performance content. Obstruction climb gradients, transition altitude and level, and CLIMB VIA behaviour come from the Jeppesen SID plate and FOM 5.5.2 Terminal Departures. A Climb and Maintain clearance does not by itself cancel published SID altitude restrictions unless ATC explicitly cancels them.

> **FCOM performance / Jeppesen / FOM 5.5.2**
> *Workbook wording: Obstruction climb gradient restrictions.*

TAKEOFF AND DEPARTURE

What are the red boxed notes on a SID?

On the Jeppesen SID plate

SID plate items to brief beyond the FMC legs (runway applicability, obstruction climb gradients, red boxed notes, top altitude for CLIMB VIA) come from the Jeppesen chart itself plus FOM 5.5.2 Terminal Departures (IFR/VFR) and 5.5.1 Turns After Takeoff. A CLIMB VIA clearance carries the SID top altitude; a Climb and Maintain does not cancel published altitude restrictions unless ATC says so.

> **Jeppesen SID plate / FOM 5.5.1-5.5.2**
> *Workbook wording: Red Boxed notes e.g., speed restriction until further advised, etc.*

TAKEOFF AND DEPARTURE

Cleared CLIMB VIA. What is the SID top altitude?

On the Jeppesen SID plate

SID plate items to brief beyond the FMC legs (runway applicability, obstruction climb gradients, red boxed notes, top altitude for CLIMB VIA) come from the Jeppesen chart itself plus FOM 5.5.2 Terminal Departures (IFR/VFR) and 5.5.1 Turns After Takeoff. A CLIMB VIA clearance carries the SID top altitude; a Climb and Maintain does not cancel published altitude restrictions unless ATC says so.

> **Jeppesen SID plate / FOM 5.5.1-5.5.2**
> *Workbook wording: SID top altitude (if given “CLIMB VIA” from ATC)?*

TAKEOFF AND DEPARTURE

ATC says climb and maintain. Do SID restrictions still apply?

In the FCOM performance section and the Jepps

Max rate and max angle climb speeds are FCOM performance content. Obstruction climb gradients, transition altitude and level, and CLIMB VIA behaviour come from the Jeppesen SID plate and FOM 5.5.2 Terminal Departures. A Climb and Maintain clearance does not by itself cancel published SID altitude restrictions unless ATC explicitly cancels them.

> **FCOM performance / Jeppesen / FOM 5.5.2**
> *Workbook wording: While on the SID, if ATC issues a “Climb and Maintain XXX” clearance, does the flight crew still need to*

TAKEOFF AND DEPARTURE

Transition altitude and transition level?

In the FCOM performance section and the Jepps

Max rate and max angle climb speeds are FCOM performance content. Obstruction climb gradients, transition altitude and level, and CLIMB VIA behaviour come from the Jeppesen SID plate and FOM 5.5.2 Terminal Departures. A Climb and Maintain clearance does not by itself cancel published SID altitude restrictions unless ATC explicitly cancels them.

> **FCOM performance / Jeppesen / FOM 5.5.2**
> *Workbook wording: Transition Altitude/LVL?*

Manual Aircraft Control

WEATHER AND MINIMUMS

Minimum altitude for autopilot engagement?

In the FCOM and the FCTM

Engine failure procedure review, RTO PM callouts, max angle climb speed and minimum autopilot engagement altitude are FCOM NP.21 and L.10 plus FCTM. Verify each figure against the PDF; do not carry them from memory into a Procedures Validation.

> **FCOM NP.21 / L.10 / FCTM R9**
>
> *Workbook wording: Minimum Altitude for autopilot engagement?*

TAKEOFF AND DEPARTURE

Is the autopilot required on an RNAV SID?

No

FOM 5.1.4.1 is the only autopilot requirement tied to RNAV/RNP procedures, and it is limited to RNP AR with RNP less than 0.3 or Radius to Fix legs, where autopilot or flight director driven by the RNAV system shall be used; RNP AR departure capability must be authorized by the fleet-specific manuals (FOM 5.1.4.3). LNAV with flight director satisfies a normal RNAV SID. Note the hard 787 limit at FCOM L.10.8: the autopilot must not be engaged below 200 feet AGL after takeoff, so it cannot be required from the start of any SID. FCOM NP.21.3 has you verify the correct RNP for the departure on POS REF page 2.

> **FOM 5.1.4.1 RNP Authorization Required (RNP AR) Procedures**
>
> *"RNP AR procedures with RNP values less than 0.3 or with Radius to Fix (RF) legs shall use autopilot"*
>
> Only RNP AR procedures with RNP below 0.3 or RF legs require autopilot or flight director driven by the RNAV system. Nothing in the FOM, FCOM, or QRH requires the autopilot on a standard RNAV SID.
>
> *Workbook wording: Is the autopilot required for an RNAV SID?*

16.1.1.7.6 ETOPS Verification Flight Procedure

ETOPS Verification Flight

ETOPS

Where are the ETOPS verification flight procedures?

In the FOM ETOPS verification flight section

FOM 6.5.10 carries the HA ETOPS Verification Flight procedure under 14 CFR 121.374. An EVF is a revenue or non-revenue flight during which the Flight Crew is asked to assist Maintenance to confirm an ETOPS Significant System is operating normally or that an ETOPS significant problem has been resolved. It may be conducted on a non-ETOPS flight or during an ETOPS revenue flight prior to the EEP. 6.5.9 is the parallel AS procedure.

> **FOM 6.5.10 ETOPS Verification Flight (HA)**
>
> *""An EVF may be conducted on a non-ETOPS flight or during an ETOPS revenue flight prior to the EEP.""*
>
> You are on the HAL list, so 6.5.10 is your procedure, not the AS 6.5.9 version. Know which one applies before an oral.
>
> *Workbook wording: Where can the ETOPS verification flight procedures be found?*

MEL AND MAINTENANCE

Must the verification request be in the logbook before departure?

Yes

The Flight Crew will be notified of an open item on the MIC sheet, and Maintenance places this entry in the Aircraft Maintenance Logbook: AIRCRAFT RELEASED TO ETOPS SERVICE IAW WITH HA ETOPS MAINTENANCE MANUAL FOR VERIFICATION FLIGHT DUE TO REPAIR OF ______.

> **FOM 6.5.10 ETOPS Verification Flight (HA)**
>
> *""AIRCRAFT RELEASED TO ETOPS SERVICE IAW WITH HA ETOPS MAINTENANCE MANUAL FOR VERIFICATION FLIGHT DUE TO REPAIR OF""*
>
> *Workbook wording: Does the verification flight item request have to be in the maintenance logbook prior to departure?*

MEL AND MAINTENANCE

Must Maintenance talk to the crew before an EVF?

Yes, verbally

Maintenance will verbally notify the Flight Crew and place the release entry in the Aircraft Maintenance Logbook. The notification is therefore threefold: the MIC sheet open item, the verbal notification, and the logbook entry.

> **FOM 6.5.10 ETOPS Verification Flight (HA)**
>
> *""Maintenance will also verbally notify the Flight Crew""*
>
> *Workbook wording: Is Maintenance required to communicate with the flight crew prior to departing on an ETOPS verification*

MEL AND MAINTENANCE

Must the PIC then call Dispatch?

Yes, a verbal acknowledgement

The Flight Crew must verbally acknowledge the EVF requirement with Dispatch. That closes the loop between Maintenance, the crew and Dispatch before departure.

> **FOM 6.5.10 ETOPS Verification Flight (HA)**
>
> *""The Flight Crew must verbally acknowledge the EVF requirement with Dispatch.""*
>
> *Workbook wording: Does the PIC have to communicate with dispatch after speaking with Maintenance?*

ETOPS

Any Dispatch call required after takeoff, before the EEP?

Yes. 30 to 60 minutes after takeoff

Between 30 and 60 minutes after takeoff and prior to reaching the EEP, the Flight Crew will call or send an ACARS message to Dispatch to report the status of the affected component or system. This report AND the return acknowledgment from Dispatch must be complete before reaching the EEP.

> **FOM 6.5.10 ETOPS Verification Flight (HA)**
>
> *""This report and the return acknowledgment from Dispatch must be complete before reaching the EEP.""*
>
> Two-way requirement. Your report alone is not enough; Dispatch's acknowledgment must also be received before the EEP.
>
> *Workbook wording: Is there a communication requirement to Dispatch between takeoff and prior to the EEP?*

ETOPS

How do you make that report?

Phone call or ACARS

The Flight Crew will call or send an ACARS message to Dispatch to report the status of the affected component or system, between 30 and 60 minutes after takeoff and prior to the EEP.

> **FOM 6.5.10 ETOPS Verification Flight (HA)**
>
> *""the Flight Crew will call or send an ACARS message to Dispatch to report the status of the affected component or system.""*
>
> *Workbook wording: How can/should you make that communication of system performance?*

ETOPS

Component fails before the EEP. Can you continue?

No

If the item being verified is not operating normally, the aircraft cannot proceed beyond the EEP and must land for repairs.

> **FOM 6.5.10 ETOPS Verification Flight (HA)**
>
> *""the aircraft cannot proceed beyond the EEP and must land for repairs.""*
>
> Unambiguous. The verification failing before the EEP removes ETOPS entry as an option.
>
> *Workbook wording: If the component being verified fails prior to EEP, can you continue?*

ETOPS

Component fails after the EEP. Can you continue?

Yes. The restriction is on proceeding beyond the EEP

FOM 6.5.10 states the aircraft cannot proceed BEYOND the EEP if the item is not operating normally. Once past the EEP the verification-flight restriction has already been satisfied or failed at that gate. A subsequent failure is handled as a normal in-flight system failure, using ETOPS contingency guidance at 6.4.4 and the PIC/Dispatcher judgment framework.

> **FOM 6.5.10 ETOPS Verification Flight (HA)**
>
> *""the aircraft cannot proceed beyond the EEP and must land for repairs.""*
>
> INFERRED from the wording of the EEP gate, not an explicit post-EEP statement. Confirm with a check airman. Treat a post-EEP failure on its own merits under 6.4.4 In-Flight System Failures and the diversion logic in 6.4.1.
>
> *Workbook wording: If the component being verified fails after EEP entry, can you continue?*

ETOPS APU Verification Flight

ETOPS

What is the APU verification procedure?

Cold soak 2 hours, start within 1 hour of top of descent

To complete the APU verification the Flight Crew must either remain outside ETOPS airspace; or on flights able to remain at cruising altitude two hours or more BEFORE entering ETOPS airspace, complete the verification before the ETOPS Entry Point; or on flights able to remain at cruising altitude two hours or more AFTER exiting ETOPS airspace, initiate the flight with the APU running, shut it down after exiting ETOPS airspace, allow at least a two-hour cold soak, and initiate the APU start within 1 hour of top of descent. If none of these conditions can be met, initiate the flight with the APU running and keep it running for the duration.

> **FOM 6.5.10 ETOPS APU Verification Flight**
>
> *""Initiate the APU start within 1 hour of top of descent.""*
>
> The cold soak requires the APU shut down at least 2 hours at a normal ETOPS cruising altitude, ideally between FL270 and FL350. That altitude band is the point of the test: it proves a cold start at altitude.
>
> *Workbook wording: What are the APU verification flight procedures?*

ETOPS

How does APU verification differ from a single component EVF?

A scheduled cold soak and start, not a status report

A single-component EVF is a monitor-and-report task: confirm the system works, report to Dispatch between 30 and 60 minutes after takeoff and before the EEP, and land if it is not normal. The APU verification is an active test requiring a minimum two-hour cold soak at altitude and a start within 1 hour of top of descent, and it is normally structured around staying clear of ETOPS airspace or completing it before entry or after exit.

> **FOM 6.5.10 ETOPS Verification Flight (HA) and ETOPS APU Verification Flight**
>
> *""Allow the APU to cold soak for at least two hours.""*
>
> If the APU verification is unsuccessful, the Dispatcher and PIC must confer to determine the best course of action, as ETOPS entry may not be possible. That is a joint decision, unlike the single-component case where the EEP rule is automatic.
>
> *Workbook wording: How do those procedures differ from an ETOPS Verification (single component) flight?*

ETOPS

How many APU start attempts?

Three attempts maximum

FOM 6.5.6 sets up the APU In-Flight Start Program: after engine start or takeoff the APU is shut down and cold soaked at least two hours at ETOPS cruise altitude, and the crew attempts a start during the 1-hour window before top of descent. If it fails on the first attempt, subsequent attempts may be made at lower altitude down to a minimum of FL270. A start within the maximum allowed three attempts is successful. The Captain logs FL, OAT, number of start attempts, successful or unsuccessful, and AVAIL AT seconds in the Aircraft Maintenance Logbook.

> **FOM 6.5.6 APU In-Flight Start Program (HA)**
>
> *"• If the APU starts within the maximum allowed three attempts, the start is deemed to be"*
>
> Subsequent attempts after a failed first attempt may be made at lower altitudes, minimum FL270. Do not confuse this with the ground start rules.
>
> *Workbook wording: How many start attempts?*

ETOPS

Must the APU be shut down before descent?

No. It may be shut down, not required

Under FOM 6.5.6 the test ends once the APU finishes its start sequence or after three failed attempts, and the APU may then be shut down; it is permissive, not required. Check for a conflicting MEL first: if an MEL requires the APU to run the entire flight, the MEL supersedes the In-Flight Start Program and you must not shut it down. APU bleed altitude limits still apply, and the program does not test the APU generator or bleed. For the ETOPS APU verification in FOM 6.5.10, initiate the flight with the APU running, shut down after exiting ETOPS airspace, cold soak at least two hours (ideally FL270 to FL350), then start within 1 hour of top of descent.

> **FOM 6.5.6 APU In-Flight Start Program (HA)**
>
> *"Once the APU finishes its start sequence or after three failed start attempts, the test is complete,"*
>
> FOM 6.5.10 ETOPS APU Verification Flight sets the same start window (within 1 hour of top of descent) and likewise imposes no shutdown requirement before descent.
>
> *Workbook wording: After APU start at altitude, is it required to shut the APU down before descent?*

ETOPS

APU shut down early, then fails to start after landing. What then?

Make a separate Maintenance Logbook entry

A post-landing APU start attempt should be made as workload permits to avoid delaying the next flight (FOM 6.5.5). If it will not start on the ground, that ground failure gets its own Maintenance Logbook entry, separate from the in-flight start or verification entry the Captain already owes per FOM 6.5.6 / 6.5.10. Per FOM 5.1.8.1 the APU is only 'failed to start' after three consecutive failed attempts, or when fleet procedures preclude further attempts. FOM 12.4.6 requires the entry to include start attempt altitudes and any APU caution light displayed.

> **FOM 6.5.5 APU Cold Soaked Start (CSS) (AS)**
>
> *"an APU start attempt should be made after landing as workload permits. If the APU fails to start on the ground, a separate Maintenance Logbook entry must be made."*
>
> FOM 5.1.8.1 defines a failed start as three consecutive failed attempts, and 12.4.6 lists APU failure to start as a required logbook entry. The (HA) 787 in-flight start program is 6.5.6; the ETOPS APU verification flight is 6.5.10.
>
> *Workbook wording: If APU is shutdown prior to top of descent, then subsequently the APU fails to start after landing,*

MEL AND MAINTENANCE

What logbook entry is required after the verification flight?

Verification flight completed satisfactorily or unsatisfactorily

The Captain must make an entry in the Aircraft Maintenance Logbook detailing the results: VERIFICATION FLIGHT FOR ____ COMPLETED SATISFACTORILY or UNSATISFACTORILY. If the test was unsuccessful the Captain must list any discrepancies. For the APU specifically, a successful entry must state: VERIFICATION FLIGHT FOR APU COMPLETED, APU OPERATES NORMALLY.

> **FOM 6.5.10 ETOPS Verification Flight (HA)**
>
> *“"VERIFICATION FLIGHT FOR ____ COMPLETED “SATISFACTORILY” or “UNSATISFACTORILY"”*
>
> Captain writes this, not the FO and not Maintenance. The APU wording is distinct from the generic component wording, so know both.
>
> *Workbook wording: What maintenance logbook entry (specific verbiage) must be entered after the verification flight?*

5.1.1.5 Perform Communications/Administrative Functions During Cruise [CRITICAL OBSERVABLE ITEM]

PUSHBACK AND START

How do you call the cabin on the headset or handset?

ACP CAB switch plus TCP call; handset dial code select

Headset method: push the CAB transmitter select switch on the audio control panel, select the station on the TCP CABIN INTERPHONE page (speed dial, directory, or two digit code plus MAKE CALL), then key the mic/interphone switch to MIC (FCOM 5.30.2). Handset method: enter the dial code with the dial code step switches or numeric keys and push the dial code select switch; the PA push-to-talk is not needed except for PAs. Pushing the ACP CAB switch twice within one second places a priority call to the airline-designated station and disconnects that station from any call in progress. End the call with END CALL on the TCP, by deselecting CAB, or by cradling the handset.

> **FCOM 5.30.2 Cabin Interphone System**
>
> *“Calls can also be answered or placed using the flight deck handset. The desired call location is entered using the dial code step switches or numeric keys. The call is then made by pushing the dial code select switch.”*
>
> Two ways in: TCP CABIN INTERPHONE page (SPEED DIAL, DIRECTORY, or a two digit dial code plus MAKE CALL) with CAB selected on the ACP, or the flight deck handset. A list of the two digit station codes is printed on the handset.
>
> *Workbook wording: How do you make an interphone call to the cabin using the headset? Using the flight deck handset?*

CRUISE AND DIVERSION

Review PA usage

No PA in critical phases except emergency announcements

FOM 7.1.18 keeps the PA out of critical air and ground phases except for emergencies, because scanning and flight deck duties come first; delay routine announcements until established in cruise. FOM 7.1.19 lists the announcements the crew is required to make, and crews may vary the wording except where the manual says verbatim. FOM 7.1.21 adds customer service guidance: identify yourself as Captain or First Officer, avoid jargon and acronyms, and limit PA use while passengers are asleep. If the PA is known inoperative, the Captain must brief the A Flight Attendant on alternate communication methods (FOM 7.1.18.1).

> **FOM 7.1.18 Public Address System**
>
> *“Flight Crewmembers should not use the Public Address System during any critical air or ground phase of flight for other than emergency announcements.”*
>
> Required PAs are listed in FOM 7.1.19 (seat belt sign on/off, anticipated turbulence, turbulence, stopping short of the jetway). Customer service PA guidance and sample scripts are in FOM 7.1.21.
>
> *Workbook wording: Review PA usage (see 12.1.3.16 Know FOM PA policies and Perform Effective Communication from the*

EFB AND FD PRO

Where is the AIREP form in Comply365?

In Comply365

Document location inside Comply365 changes with releases. Verify in the live app rather than trusting a written path.

> **Comply365**
>
> *Workbook wording: Where in Comply365 can the AIREP form be found?*

EFB AND FD PRO &MIDDOT; WALKTHROUGH ITEM, PERFORM RATHER THAN RECALL

Do you complete an AIREP every flight?

Not stated in our manuals. Comply365 form

AIREP completion and retention, CPDLC request types, and plot saving or submission are company workflow and aircraft menu content. FOM 5.7.3.15 requires the entire cleared route be loaded for ORCN segments but does not mandate saving plots after every flight. Verify in Comply365 and on the aircraft.

> **Comply365 / FCOM Ch 11 / FOM 5.7.3.15**
>
> *Workbook wording: Do you need to complete the AIREP form for each flight?*

EFB AND FD PRO &MIDDOT; WALKTHROUGH ITEM, PERFORM RATHER THAN RECALL

Do you save the AIREP after each flight?

Not stated in our manuals. Comply365 form

AIREP completion and retention, CPDLC request types, and plot saving or submission are company workflow and aircraft menu content. FOM 5.7.3.15 requires the entire cleared route be loaded for ORCN segments but does not mandate saving plots after every flight. Verify in Comply365 and on the aircraft.

> **Comply365 / FCOM Ch 11 / FOM 5.7.3.15**
>
> *Workbook wording: Do you need to save the AIREP form after each flight?*

Company Reporting

OCEANIC AND NAT

Are company position reports required?

Yes, and holding always requires one

FOM 7.1.12 Reports to Dispatch carries the requirement set. One that is explicit: through ACARS or radio the Captain will ensure Dispatch is contacted with the holding fix, Expected Further Clearance time, and fuel on board, and diversion plans should be discussed if applicable. Abnormalities require notification under 11.1.2.

> **FOM 7.1.12 Reports to Dispatch / 5.5.15.1 Dispatch Notification of Holding**
>
> > *""the Captain will ensure that Dispatch is contacted with the holding fix, Expected Further Clearance (EFC) time, and fuel on board.""*
>
> Three items for the holding report: fix, EFC, fuel. DESKTOP: read 7.1.12 in full for the complete routine position-reporting requirement.
>
> *Workbook wording: Are we required to make company position reports?*

MEL AND MAINTENANCE

How do you report a discrepancy before landing?

ACARS or radio to Dispatch, who relays to Maintenance

Report through Dispatch using ACARS or radio. FOM 11.1.2 requires contacting Dispatch for abnormalities as safety of flight permits, and if time does not permit in flight, after gate arrival. Reporting early lets Maintenance position parts and personnel at the gate rather than starting the troubleshooting clock on block-in.

> **FOM 11.1.2 Abnormal Operations Dispatch Notification**
>
> *""If time does not permit contact with Dispatch while in flight, contact shall be made after gate arrival.""*
>
> That is also the answer to the advantages question in the maintenance logbook section: enroute notification converts ground time from diagnosis into repair.
>
> *Workbook wording: How can you report a discrepancy to Maintenance in advance, prior to landing?*

RVSM Requirements

OCEANIC AND NAT

What altitudes are RVSM airspace?

FL290 through FL410

RVSM airspace is any airspace or route where aircraft are separated by 1000 ft vertically, between FL290 and FL410 inclusive.

> **FOM 5.5.6.5 Enroute RVSM Altitude Monitoring Requirements**
>
> *""between FL290 and FL410, inclusive""*
>
> At initial level off, pilots shall verify all altimeters are set to 29.92 and crosscheck them. Use of the standby altimeter as a primary is PROHIBITED in RVSM airspace.
>
> *Workbook wording: What altitudes define RVSM airspace?*

OCEANIC AND NAT

RVSM equipment requirements?

Both primary altimeters, one autopilot, altitude alerting

The following shall be operating normally prior to entry into RVSM airspace: both primary altimeters, one autopilot altitude-control system, and the Altitude Alerting System. The autopilot should be engaged during altitude capture and level cruise, except where circumstances such as the need to retrim or turbulence require disengagement. Report reaching assigned altitudes if not in radar contact.

> **FOM 5.5.6.3 Entry Into RVSM Airspace**
>
> *""Both primary altimeters""*
>
> Three items only. A common wrong answer adds transponder or TCAS to the list.
>
> *Workbook wording: Are there aircraft equipment requirements to operate in RVSM airspace?*

OCEANIC AND NAT

RVSM equipment fails in flight. What do you do?

Notify ATC and Dispatch

In the event of required RVSM equipment failure prior to or within RVSM airspace, notify ATC and Dispatch to coordinate a prudent course of action. ATC may not require any action; however, a reroute or diversion may be required.

> **FOM 5.5.6.4 System Failure Enroute**
>
> *""ATC may not require any action; however, a reroute or diversion may be required.""*
>
> Both ATC and Dispatch, not one or the other. The FOM deliberately leaves the outcome open rather than mandating an immediate descent out of RVSM.
>
> *Workbook wording: What are you expected to do if you have a failure of required equipment while in RVSM airspace?*

OCEANIC AND NAT

Diverting without a clearance. What altitude do you plan?

Below FL290

Under the oceanic in-flight contingency procedures (FOM 5.7.4.2), if a revised clearance cannot be obtained, turn at least 30 degrees left or right to establish a same-direction parallel track offset 5 nm, and be below FL290 before diverting from the assigned clearance. Once established on the parallel route, take a 500 ft vertical offset (FOM 5.7.4.3). Descending below FL290 gets you out of RVSM airspace and reduces conflict, TCAS RA, and delay risk on parallel airways. Squawk 7700 when able, turn on all exterior lights, run TCAS in TA/RA, and broadcast on the in-use frequency and 121.5 (backup 123.45).

> **FOM 5.7.4.2 General Procedures**
>
> *“The aircraft should be below FL290 prior to diverting from the assigned clearance unless a revised ATC clearance has been obtained.”*
>
> Same guidance is repeated in FOM 5.7.4.3. First turn at least 30 degrees off the cleared track to establish a parallel same-direction offset of 5 nm, then take a 500 ft vertical offset once established.
>
> *Workbook wording: When diverting to an alternate without a clearance, what altitude should you plan to be below when*

CPDLC Logon Procedures

OCEANIC AND NAT

How early do you log on to CPDLC leaving the contiguous 48?

About 30 minutes prior to departure

CPDLC DCL becomes available roughly 30 minutes before departure at DCL airports (FOM 5.2.14), so that is the window to log on and pull the clearance. Where DCL is available and the aircraft is equipped, CPDLC DCL is expected to be used for the preflight clearance; if the logon or the DCL fails, revert to Clearance Delivery on VHF. Both pilots must read and understand the complete clearance as received. The logon address for the lower 48 is KUSA (FOM 5.2.15).

> **FOM 5.2.14 CPDLC Clearances - ATC Data Link Communications**
>
> *“Controller Pilot Data Link Communications Departure Clearance (CPDLC DCL) is available approximately 30 minutes prior to departure at airports with DCL operational.”*
>
> FOM 5.2.15 gives the logon table: CONUS (Lower 48) is 'During preflight' with logon address KUSA. Contrast Hawaii, which is above 10k and 15-45 min prior to entry, logon KZAK.
>
> *Workbook wording: When departing the 48 Contiguous states, to assist ATC, how many minutes prior to departure is an*

OCEANIC AND NAT

Airborne CPDLC logon window before the FIR?

A time window before entry, not an altitude

If equipped with CPDLC, log on either on the ground during preflight or after takeoff above 10,000 ft MSL, or as required by the specific FIR/OCA Pilot Reference Card. Oceanic Control Areas typically have a TIME WINDOW prior to entry that defines the logon window, not a specific altitude. The logon location and address vary: CONUS lower 48 during preflight to KUSA; Alaska climbing through 10,000 to PAZA; American Samoa climbing through 10,000 to NZZO; Australia climbing through 10,000 to YBBB.

> **FOM 5.2.15 CPDLC Logon Policy**
>
> *“"Oceanic Control Areas typically have a time window prior to entry that defines the logon window, not a specific altitude."”*
>
> The direct answer to the second half is no, it is not the same for all FIRs. Use the FIR/OCA Pilot Reference Card. For KZAK Oakland the portal phase_flows page carries logon 45 to 15 minutes prior to entry and not below 10,000 ft.
>
> *Workbook wording: Once airborne, what is the CPDLC logon time window prior to the FIR (“Flight Information Region”)? Is it*

OCEANIC AND NAT

How do you log on for a departure clearance? Enroute?

MFD COMM: Departure Clearance Request page; enroute ATC LOGON/STATUS page

Both are done on the MFD communications pages. For enroute, select ATC then ATC LOGON/STATUS, enter LOGON TO (four letter ICAO center identifier), flight number, origin and destination, then SEND; status steps through SENDING then CONNECTING and the page exits five seconds later (FCOM 5.40.42). The page also shows ATC CONNECTION, ACTIVE CENTER, NEXT CENTER, MAX UPLINK DELAY and ADS STATUS. For the departure clearance, use FLIGHT INFORMATION then DEPARTURE CLEARANCE REQUEST and fill flight number, ATC facility, departure, destination, gate and ATIS, then SEND (FCOM 5.40.56). Note that entering a new flight number on that page terminates the current ATC connection.

> **FCOM 5.40.42 ATC Logon/Status**
>
> *"The ATC LOGON/STATUS page allows logon to the desired ATC facility for establishment of a datalink connection and displays ATC datalink connection status information."*
>
> Departure clearance request page is FCOM 5.40.56; it is reached under FLIGHT INFORMATION on the MFD communications menu. Logon addresses come from FOM 5.2.15 (KUSA for the lower 48, KZAK for Oakland Oceanic).
>
> *Workbook wording: How, specifically in your aircraft, do you logon for a departure clearance? Enroute?*

CPDLC Menus

EFB AND FD PRO

What can you request over CPDLC?

In Comply365 and the FCOM

AIREP completion and retention, CPDLC request types, and plot saving or submission are company workflow and aircraft menu content. FOM 5.7.3.15 requires the entire cleared route be loaded for ORCN segments but does not mandate saving plots after every flight. Verify in Comply365 and on the aircraft.

> **Comply365 / FCOM Ch 11 / FOM 5.7.3.15**
>
> *Workbook wording: What types of requests can be made through CPDLC?*

FOM Waypoint Transition Procedure

OCEANIC AND NAT

Two minutes before a waypoint, what do you do?

Confirm the next waypoint and the one after it

Approximately 2 minutes prior to reaching a waypoint, confirm that the upcoming waypoint and the subsequent waypoint (next plus one) agree with the Flight Plan and ATC clearance.

> **FOM 5.7.3.16 Waypoint Transition**
>
> *""confirm that the upcoming waypoint and the subsequent waypoint (next + 1) agrees with the Flight Plan/ATC clearance.""*
>
> The next-plus-one check is the whole point. Verifying only the active waypoint will not catch a gross navigation error that begins one leg downstream.
>
> *Workbook wording: Approximately 2 minutes prior to reaching a waypoint, what crew actions are required?*

OCEANIC AND NAT

What gets checked after waypoint transition?

Time, fuel, the difference, LNAV, slash the waypoint

After waypoint transition: record the time and fuel remaining from the FMC on the Flight Plan/Dispatch Release or the ACARS HOWGOZIT/Flight Plan Review and compare with the estimates, recording the difference where plus is ahead or underburn and minus is behind or overburn. Verify LNAV (or the required mode) is displayed in the FMA and mark the waypoint with a slash. Transmit the position report to ATC if required.

> **FOM 5.7.3.16 Waypoint Transition**
>
> *""Record the difference (+ is ahead/underburn and - is behind/overburn).""*
>
> Note the sign convention, it is a common oral trip-up. The slash mark is the physical evidence the check was actually done, which matters on a line check.
>
> *Workbook wording: What items are required to be checked on the flight plan after waypoint transition?*

OCEANIC AND NAT

Where is fuel remaining recorded at waypoint transition?

The flight plan, the release, or the ACARS HOWGOZIT

Time and fuel remaining from the FMC are recorded on the Flight Plan/Dispatch Release, or alternatively on the ACARS HOWGOZIT/Flight Plan Review, and compared against the estimates.

> **FOM 5.7.3.16 Waypoint Transition**
>
> *""Record the time and fuel remaining from the FMC/FMS on the Flight Plan/Dispatch Release or the ACARS HOWGOZIT/Flight Plan Review""*
>
> *Workbook wording: On which document is the fuel remaining (at waypoint transition) recorded?*

OCEANIC AND NAT

Ten minutes after waypoint transition, what do you do?

Crosscheck navigation performance and course

Approximately 10 minutes after waypoint passage, crosscheck navigational performance and course compliance. The FMC navigation crosscheck is: using the appropriate scale confirm the aircraft symbol is on the programmed route on the ND; check cross-track deviation; verify the TO waypoint is consistent with the current route clearance; verify flight guidance is in the appropriate mode to remain on intended track to the next waypoint; and investigate or correct any anomalies or unexpected deviations.

> **FOM 5.7.3.17 10 Minutes After Waypoint Passage**
>
> *""Take immediate corrective action for any anomalies or deviations.""*
>
> Cross-reference 5.7.4.11 Position Doubtful if the crosscheck does not resolve. If you are on a weather deviation, the FOM requires this process to be completed immediately rather than waiting for the 10-minute point.
>
> *Workbook wording: Approximately 10 minutes after waypoint transition, what crew actions are required?*

OCEANIC AND NAT

Why track fuel score?

It reveals fuel leaks and burn trends early

Scoring fuel at every waypoint is the primary in-flight leak detector: a growing negative trend that fuel flows do not explain is the signature of a leak or fuel system malfunction (FOM 5.5.3). The crew must crosscheck fuel and time at top of climb and at each Flight Plan waypoint, or hourly, whichever is longer, recording plus for ahead/underburn and minus for behind/overburn (FOM 5.5.3, 5.7.3.16). On ORCN segments the values must be recorded on the Flight Plan/Dispatch Release or HOWGOZIT and kept available to both pilots (FOM 5.7.3.11). Once the 787 is 3000 lbs off plan, contacting Dispatch is mandatory (FOM 7.1.12).

> **FOM 5.5.3 Flight Plan Time and Fuel Monitoring - Enroute**
>
> > *"The difference between the actual and the estimated fuel remaining is an important factor in identifying and troubleshooting fuel leaks or other fuel system malfunctions."*
>
> FOM 7.1.12 requires a Dispatch call when actual fuel differs from planned by 3000 lbs on the 787/A330 (2000 lbs on the 717/737/A321). Oceanic recording rules are FOM 5.7.3.11 Time and Fuel Scoring.
>
> *Workbook wording: Why is keeping track of the actual fuel remaining vs planned fuel remaining ("FUEL SCORE") important?*

OCEANIC AND NAT

Which fuel figure goes on the release, tank total or PROG?

In the FOM and the fleet performance manuals

Load closeout reading and verification, and the takeoff/landing performance products. Takeoff performance hierarchy is Final TPR, then Preliminary TPR, then TLR. Using a TLR for takeoff demands attention to the temperature and altimeter assumptions baked into it. FOM 5.2 covers closeout and weight and balance; the SABLE Load Sheet is the official FAA weight and balance record. Landing analysis designators and request methods are in the performance system documentation.

> **FOM 5.2 / fleet performance manuals**
>
> *Workbook wording: (787) Which fuel remaining is noted on the dispatch release, the EICAS fuel panel (tank total) or the PROG*

Cruise Altitude Considerations

TAKEOFF AND DEPARTURE

OPT vs MAX vs RECOMMENDED altitude?

OPT ignores winds; MAX is performance ceiling; RECMD uses forecast winds

OPT is computed from gross weight, cost index, cruise speed schedule and cruise CG only, with no forecast wind or temperature deviation input, so it is a still-air best-economy number. MAX is the performance ceiling built from gross weight less climb fuel, temperature, number of engines running, cruise thrust limit, speed mode and the airline residual rate of climb setting, and it caps every other altitude calculation. RECMD is the most economical altitude for the next 250 to 500 nm using entered forecast winds and temperatures, consistent with the step size (2,000 ft assumed if step size is zero), and it is set to CRZ ALT within 200 nm of T/D or 500 nm of destination (FCOM 11.42.23-24). With RTA cruise active, OPT, MAX and RECMD are not computed at all.

> **FCOM 11.31.27 Flight Management System Operation - Step Climb**
>
> > *"Optimum (OPT) altitude calculation does not use forecast or IRS winds and temperatures. Recommended (RECMD) altitude calculation is based on step size, current CRZ ALT, and entered forecast enroute winds and temperatures."*
>
> Expanded field definitions are at FCOM 11.42.23-24 (CRZ page). RECMD looks 250 to 500 nm ahead and evaluates from 9,000 ft below current CRZ ALT up to MAX.
>
> *Workbook wording: What is the difference between OPT/Max/Recommended altitudes?*

FUEL PLANNING

Why does the tropopause level matter for cruise?

It sets air temperature; descending below it warms fuel

FOM 9.3.2.3 is the only place the manuals tie tropopause height to cruise altitude selection, and it does so through fuel temperature: fuel drifts toward TAT at roughly 3 degrees C per hour, up to 12 degrees C per hour in extreme conditions, so on long cold-soak sectors the tropopause level tells you where warmer air is. A 4000 ft descent below the tropopause buys about 7 degrees C of TAT, and an increase of M 0.01 buys about 0.7 degrees C. Occasionally a temperature inversion sits at the tropopause, in which case climbing warms the fuel instead. Allow up to an hour for fuel temperature to stabilize and account for the fuel penalty.

> **FOM 9.3.2.3 Fuel Freeze Considerations**
>
> *"For instance, a 4000 ft descent below the tropopause can result in approximately a 7°C increase in TAT."*
>
> Tropopause height per waypoint (TROP) is printed on the Navigation Log. FOM 8.3.3 also notes ozone forms primarily above the tropopause, which is why ozone converters are fitted.
>
> *Workbook wording: Why is it important to consider the level of the tropopause regarding your cruise altitude?*

DISPATCH AND RELEASE

Is there a tropopause chart in the release package?

No chart. TROP is a Navigation Log column per waypoint

The Navigation Log columns on the flight plan include TROP, the tropopause level at each waypoint, alongside MORA, TTK, NM, IAS, Mach, FL and TEMP (FOM 8.5.7). That per-waypoint field, not a separate chart, is the released tropopause data. FOM 8.5.1 lists the required contents of the Flight Information Packet and does not include a tropopause chart. Vendor graphical products for turbulence and other hazards may be supplied with the weather package or relayed by ACARS (FOM 9.8.1), but the manuals do not promise a dedicated tropopause chart.

> **FOM 8.5.7 Dispatch Release Sample (HA) - Navigation Log Legend**
>
> *"MORA Minimum Off Route Altitude TROP Tropopause TTK True Track to WPT"*
>
> FOM 8.5.1 lists everything in the Flight Information Packet (release, flight plan, TAF, METAR, SIGMETs and turbulence products, NOTAMs); no tropopause chart is among them. Graphical hazard products from the weather vendor are covered in FOM 9.8.1.
>
> *Workbook wording: Do we have a chart in our dispatch release package that shows the level(s) of the tropopause with*

FUEL PLANNING

What do you expect crossing the tropopause?

Temperature stops falling; an inversion may warm you climbing

Near and through the tropopause the normal lapse rate breaks down: temperature stops decreasing with altitude and can invert, so a climb may actually warm TAT and help a cold fuel problem, while a 4000 ft descent below the tropopause is worth about 7 degrees C of TAT (FOM 9.3.2.3). Above the tropopause natural ozone concentration rises, which is why ozone converters are installed and why 14 CFR 121.578 ozone limits matter up there (FOM 8.3.3). The classic expectations of jet stream shear and clear air turbulence at the tropopause are not documented in these three manuals, so treat that as outside-source knowledge.

> **FOM 9.3.2.3 Fuel Freeze Considerations**
>
> *"Occasionally, a temperature inversion at the Tropopause may occur, and, in this case, climbing will result in warmer temperatures and be beneficial for fuel warming."*
>
> The FOM only addresses the tropopause in temperature/fuel-freeze terms (9.3.2.3) and ozone terms (8.3.3). Turbulence and wind shear expectations near the tropopause are not stated in the FOM, FCOM or QRH.
>
> *Workbook wording: What can expect when cruising near, or crossing through, the tropopause?*

FOM Electronic Plotting Chart Procedure

CRUISE AND DIVERSION

When is an electronic plot required?

On all ORCN segments

An electronic plotting chart should be used on all Oceanic and Remote Continental Navigation segments. The Flight Crew should load the entire cleared route into the approved charting services app, such as FD Pro X, on the EFB.

> **FOM 5.7.3.15 Electronic Plotting Chart**
>
> *""An electronic plotting chart should be used on all ORCN segments.""*
>
> Regulatory basis AC 91-70 and Ops Spec B036. Note the word is should, and the requirement is the ENTIRE cleared route, not just the oceanic portion.
>
> *Workbook wording: When and under what conditions are we required to do an electronic plot?*

CRUISE AND DIVERSION

How do you plot on the EFB?

Load the entire cleared route into the charting app

The Flight Crew should load the entire cleared route into the approved charting services app on the EFB. FD Pro X is the app named in the FOM.

> **FOM 5.7.3.15 Electronic Plotting Chart**
>
> *""load the entire cleared route into the approved charting services app (such as FD Pro X) on the EFB.""*
>
> DESKTOP: the FD Pro X screen-by-screen plotting workflow is vendor documentation, not FOM content. Confirm with the FD Pro Pubs Quick Reference guide in Comply365.
>
> *Workbook wording: How do you plot using the EFB?*

CRUISE AND DIVERSION

Must you save or submit plots after every flight?

No published requirement to save or submit

FOM 5.7.3.15, written against AC 91-70 and Ops Spec B036, requires an electronic plotting chart on all ORCN segments and requires the full cleared route be loaded into the approved charting services app on the EFB. It stops there. Nothing in the section, or in the surrounding oceanic material, requires the crew to save, export or submit the plot afterward.

> **FOM 5.7.3.15 Electronic Plotting Chart**
>
> *"An electronic plotting chart should be used on all ORCN segments."*
>
> 5.7.3.15 requires the plot be used and the entire cleared route loaded into the approved charting app such as FD Pro X, but it sets no retention, save or submission requirement. Screenshotting a plot is personal practice, not policy.
>
> *Workbook wording: Are you required to save/submit your plots and route after every flight?*

CRUISE AND DIVERSION

When is it prudent to screenshot your plotted route?

In the FOM oceanic and weather chapters

Fuel score is the running actual versus planned comparison recorded at each waypoint transition per FOM 5.7.3.16, where plus is ahead or underburn and minus is behind or overburn. It is the early warning for a fuel problem. Tropopause level affects OPT and MAX altitude and marks the jet core; FOM Chapter 9 and the performance chapter cover it. Plot retention is FOM 5.7.3.15 plus company practice.

> **FOM 5.7.3.15-5.7.3.17 / FOM Ch 9**
>
> Not in the FOM, FCOM or QRH. The FOM covers electronic plotting only at 5.7.3.15, which requires an electronic plotting chart on all ORCN segments and says to load the entire cleared route into the approved charting app (FD Pro X) on the EFB; it says nothing about screenshots. There is no occurrence of the word screenshot in the FOM. Guidance on capturing the plotted route would come from HA/AS EFB or Jeppesen FD Pro user guidance, or from the ORCN/plotting job aid, not from these manuals. Related manual hooks if you need them: FOM 5.7.4.13 (gross navigation error actions) and FOM 8.5.1.1 (document retention).
>
> *Workbook wording: When would it be prudent to save your plotted route (take screenshot postflight)?*

SATCOM / HF Procedures

DISPATCH AND RELEASE

What prompts a SATCOM call to Dispatch?

Pilot discretion; any required report to Dispatch

FOM 7.1.7 makes satellite voice to Dispatch or MedLink purely the pilot's call, so the circumstance is judgment driven: use it when the message is too complex, too urgent, or too consequential for ACARS text. What is not discretionary is the contact itself. FOM 7.1.12 makes notification mandatory for fuel off plan by 3000 lbs on the 787, significant Flight Plan deviations, turbulence, safety of flight concerns, diversions, mechanical irregularities and potential ETOPS reroutes, with the caveat that the communication happens as safety of flight permits. Routine satellite voice with ATC is a different matter and is not authorized unless HF or CPDLC are unavailable.

> **FOM 7.1.7 SATCOM (if installed)**
>
> *"Satellite voice communication with Dispatch or MedLink is at the pilot's discretion."*
>
> The mandatory triggers are in FOM 7.1.12 Reports to Dispatch: 3000 lbs fuel deviation (787), 100 nm lateral, 4000 ft altitude, 15 minute arrival delay, weather, moderate or greater turbulence, safety of flight, abnormalities, diversions, mechanical irregularities, ETOPS reroutes.
>
> *Workbook wording: What circumstance precipitates a SATCOM call to Dispatch?*

MEL AND MAINTENANCE

Who do you call on SATCOM for maintenance or a guest issue?

Dispatch

Inflight, the Dispatcher is the primary contact and shares operational control with the Captain (FOM 7.1.1.2). ACARS messages sent to Maintenance Control go automatically to the Dispatcher, and Maintenance does not normally communicate directly with an airborne aircraft because their diagnostic guidance differs from operational guidance for crews. When technical systems or reliability information is needed, the FODO provides the communication link to Maintenance and Engineering, and Maintenance/Engineering are kept out of direct operational control conversations. Satellite voice to Dispatch or MedLink is at the pilot's discretion (FOM 7.1.7); the FODO is reachable at the HA number in FOM 7.1.3 or through Dispatch.

> **FOM 7.1.1.2 In-Flight Operations**
>
> *"The Dispatcher is the primary contact for requests and notifications, both managing information flow and sharing operational control with the Captain."*
>
> Maintenance will not normally talk directly to an airborne aircraft; the FODO provides the link to Maintenance and Engineering. For a medical event, MedLink is the SATCOM call (FOM 7.1.7).
>
> *Workbook wording: When we make a SATCOM call for Maintenance or guest issue, who do we call?*

CRUISE AND DIVERSION

How do you make a SATCOM call?

Select SAT on TCP, pick number, set priority, MAKE CALL

Push SAT on the tuning and control panel to bring up the SATCOM main menu (FCOM 5.20.5). Either line select a number from DIRECTORY, or type the number into the scratchpad, then set priority with the priority key (EMG, HGH, LOW, PUB) and push MAKE CALL; the prompt changes to END CALL once connected and channel status steps DIALING, RINGING, ANSWERED (FCOM 5.10.9-5.10.10). Select SAT on your audio control panel and key the mic to talk. Incoming calls give a SELCAL chime and an ACP CALL light; pressing the transmitter select switch connects them. SAT-2 is SwiftBroadBand only and HA does not have that service, so that channel is inoperative.

> **FCOM 5.20.5 Satellite Communication (SATCOM) System**
>
> *"The SATCOM control pages are displayed by selecting SAT on the TCP. Directories of airline-defined numbers are line-selectable or numbers may be manually entered if function is enabled by the operator."*
>
> Audio is handled separately: push the SAT transmitter select switch on the audio control panel and key the mic. Using the captain's or first officer's ACP to establish a SATCOM/cabin conference is prohibited (FCOM L.10.9); use the observer ACP.
>
> *Workbook wording: How do you make a SATCOM (specifically which buttons, and in what order, must be pressed in/on which*

CRUISE AND DIVERSION

Where are the preprogrammed phone numbers on the 787?

SATCOM DIRECTORY key on the TCP SATCOM page

On the TCP SATCOM main menu, the DIRECTORY key displays the directory of available SATCOM phone numbers organized by geographic region (FCOM 5.10.10). On SATCOM page 2/2, the DIRectory Details key shows the list of phone number identifiers (FCOM 5.10.11). FCOM 5.20.5 confirms the directories hold airline-defined numbers that are line selectable, and that manual number entry is possible if the operator enables it. The FCOM does not publish the actual HA numbers because the directory is customer built.

> **FCOM 5.10.10 Communications Controls and Indicators - SATCOM Operation**
>
> *"Push – Displays directory of available SATCOM phone numbers by geographic region."*
>
> SATCOM page 2/2 also has a DIR DETAILS key that lists the phone number identifiers. The directory contents are airline defined, so they are not printed in the FCOM.
>
> *Workbook wording: In the ACARS/TCP (787) where can you find a list of telephone numbers that are preprogrammed to call?*

CRUISE AND DIVERSION

Total VHF and HF failure. Any other way to reach ATC?

Yes. SATCOM voice, and CPDLC datalink

Satellite voice is the fallback: it is not authorized for routine ATC use, but the FOM specifically permits it when HF or CPDLC are not available (FOM 7.1.7), and all areas of the route structure have satellite voice coverage. CPDLC over SATCOM datalink is the other path, and it is the primary means in oceanic airspace when paired with ADS-C. Beyond that, FOM 5.7.4.2 has you broadcast identification, condition, intentions, position and flight level on the in-use frequency and 121.5 MHz (backup 123.45 MHz) so another aircraft can relay. An enroute SATCOM datalink failure must be reported to ATC with a plan to use voice (FOM 7.1.7.1).

> **FOM 7.1.7 SATCOM (if installed)**
>
> *"Routine satellite voice communication with ATC is not authorized; however, it may be used when HF or CPDLC are not available."*
>
> If it is a datalink failure rather than voice, FOM 5.7.4.9 says revert to HF position reporting and tell ATC. For contingencies, FOM 5.7.4.2 also has you alert nearby aircraft on 121.5, backup 123.45, for relay.
>
> *Workbook wording: If you have a total VHF/HF radio failure, is there another way to contact ATC?*

CRUISE AND DIVERSION

Where are the current Pacific HF frequencies?

On the Jeppesen enroute pages and in the PRC

Current Pacific HF frequency sets and the position-report template come from the Jeppesen enroute pages, ARINC and the PRC, not the FOM. The day/night frequency rule of thumb is operating practice. FOM 7.1.6 covers HF policy and 5.7.3.19 the non-CPDLC report content.

> **Jeppesen / PRC / FOM 5.7.3.19**
>
> *Workbook wording: Where can you find the most current (pacific) HF FREQUENCIES in use for your flight?*

CRUISE AND DIVERSION

Rule of thumb for HF frequencies and the sun?

Operating practice, not published in the manuals

Current Pacific HF frequency sets and the position-report template come from the Jeppesen enroute pages, ARINC and the PRC, not the FOM. The day/night frequency rule of thumb is operating practice. FOM 7.1.6 covers HF policy and 5.7.3.19 the non-CPDLC report content.

> **Jeppesen / PRC / FOM 5.7.3.19**
>
> *Workbook wording: What is the rule of thumb with HF frequencies vs the Sun?*

OCEANIC AND NAT

Where is the HF position report template?

On the Jeppesen enroute pages and in the PRC

Current Pacific HF frequency sets and the position-report template come from the Jeppesen enroute pages, ARINC and the PRC, not the FOM. The day/night frequency rule of thumb is operating practice. FOM 7.1.6 covers HF policy and 5.7.3.19 the non-CPDLC report content.

> **Jeppesen / PRC / FOM 5.7.3.19**
> *Workbook wording: Where can you find a ‘template’ to make an HF Radio Position report?*

15.1.2.3.4 Analyze Weather Conditions Using Weather Radar System

FOM and FCTM guidance on Weather avoidance

OCEANIC AND NAT

Where are the ICAO oceanic weather deviation procedures?

In the FOM oceanic section, from ICAO Doc 4444

Oceanic weather deviation procedures are at FOM 5.7.4.4 Weather Deviation Procedures for Oceanic Airspace, with 5.7.4.5 covering the case where controller-pilot communications are established and 5.7.4.6 covering when a revised clearance cannot be obtained, including the Weather Deviation Table. The regulatory source is ICAO Doc 4444.

> **FOM 5.7.4.4 Weather Deviation Procedures for Oceanic Airspace**
> *“"ICAO Doc 4444"”*
> Sits inside 5.7.4 Special Procedures for In-Flight Contingencies, which also carries navigation systems failure (5.7.4.8), CPDLC/ADS-C failure (5.7.4.9), total LRNS or MCDU failure (5.7.4.10), Position Doubtful (5.7.4.11) and Gross Navigation Error (5.7.4.13).
> *Workbook wording: Where can you find the ICAO Weather Deviation procedures for Oceanic Controlled Airspace?*

OCEANIC AND NAT

How do you request a weather deviation?

Voice WEATHER DEVIATION REQUIRED, or CPDLC

Initiate communications with ATC via voice or CPDLC. A rapid response may be obtained by either stating WEATHER DEVIATION REQUIRED to indicate priority is desired, or requesting the deviation using a CPDLC lateral downlink message. Inform ATC when the deviation is no longer required or has been completed and you have returned to the cleared route.

> **FOM 5.7.4.4 Weather Deviation Procedures for Oceanic Airspace**
> *“"Requesting a weather deviation using a CPDLC lateral downlink message."”*
> If it escalates, the urgency call is PAN-PAN spoken three times, or a CPDLC urgency downlink message.
> *Workbook wording: How, specifically to your aircraft, do can you make a weather deviation request?*

OCEANIC AND NAT

How do you insert the offset in the CDU?

LATERAL OFFSET page, enter L or R and distance, execute

Open the LATERAL OFFSET page with the OFST function key, the OFFSET prompt on INIT REF INDEX, the OFFSET prompt on RTE page 1, or LOAD SLOP from the SLOP TRANSITION help window (FCOM 11.42.18). Enter the distance in 1L as L or R plus magnitude (0.1 to 99.0 nm), optionally a START WAYPOINT in 2L and END WAYPOINT in 3L, then execute. The offset shows white dashed on the ND until executed, then dashed magenta with the original route still solid magenta; with LNAV active and autopilot engaged the airplane turns to capture using a 30 degree entry and exit path. The offset propagates until an approach, end of route, a track change over 135 degrees, a hold, or a non-offsetable leg, and is removed by deleting it, entering zero, a direct-to or intercept, or a PPOS hold.

> **FCOM 11.42.18 Route Offset and Strategic Lateral Offset Procedure (SLOP)**
>
> *“The LATERAL OFFSET page can be accessed by: selecting the OFST function key, selecting the OFFSET prompt on the INIT REF INDEX page or on the RTE page 1”*
>
> Valid DIST entries are L or R with XX.X, or XX.X then L or R, from 0.1 to 99.0 nm in 0.1 nm steps. SLOP itself is right of course only, 0.0 to 2.0 nm (FOM 5.7.4.7).
>
> *Workbook wording: Once received, how (aircraft CDU/FMS specific) can you insert the offset distance?*

WEATHER AND MINIMUMS

How far do you stand off severe cells?

20 nm laterally; clear tops by 5000 ft

FOM 9.5.5 sets 20 nm minimum lateral avoidance for any thunderstorm identified as severe or giving an intense radar echo, and 5000 ft vertical clearance over the tops when in the clear on top; below 5000 ft of top clearance, revert to the 20 nm lateral standard. Do not fly under a cumulonimbus overhang, and if you must, use antenna up tilt to find hail. Keep corridors between echoes reasonably straight and check longer radar ranges frequently while maneuvering on short range. The radar is an aid to detouring severe weather, not a penetration tool (FOM 9.5.4), and steep rainfall gradients, one mile or less from green edge to red, mark the worst turbulence.

> **FOM 9.5.5 Thunderstorm Avoidance**
>
> *“Avoid by at least 20 miles any thunderstorm identified as severe or giving an intense radar echo. When “in the clear” on top, cloud tops should be cleared by 5000 ft.”*
>
> If you cannot clear the tops by 5000 ft, go back to 20 nm lateral avoidance. The FOM does not specify an upwind or downwind side for the deviation, so that preference is not manual sourced.
>
> *Workbook wording: How far (in nm’s) should you avoid Severe CB’s if possible? Which direction? Vertically?*

WEATHER AND MINIMUMS

No deviation clearance. What do you do?

Emergency authority, PAN-PAN three times, offset 300 ft beyond 5 nm

If you cannot continue per clearance due to dangerous weather and no prior clearance can be obtained, the Captain should exercise emergency authority and communicate the urgency signal PAN-PAN spoken three times, or a CPDLC urgency downlink, as early as possible. Until cleared: deviate away from an organized track or route system if possible; broadcast identification, flight level, position including route/track, and intentions at suitable intervals on the frequency in use and 121.5, with 123.45 as backup; watch for conflicting traffic visually and on TCAS; turn on all exterior lights. For deviations less than 5 nm maintain assigned flight level. For deviations of 5 nm or more, at approximately 5 nm from track initiate the prescribed altitude change.

> **FOM 5.7.4.6 Revised ATC Clearance Cannot be Obtained**
>
> *“"If possible, deviate away from an organized track or route system."”*
>
> Weather Deviation Table 5.7.4.6(1). Flying EAST (000 to 179 magnetic): deviating LEFT descend 300 ft, RIGHT climb 300 ft. Flying WEST (180 to 359 magnetic): LEFT climb 300 ft, RIGHT descend 300 ft. Returning to track, be at assigned flight level within approximately 5 nm of centerline. If you are cleared a specific distance and then denied more, apply the altitude offset immediately.
>
> *Workbook wording: If a weather deviation clearance cannot be attained, what is the procedure to deviate around weather?*

WEATHER AND MINIMUMS

What do you report to Dispatch when deviating?

Lateral 100 nm, altitude 4000 ft, 15 minute delay, weather

FOM 7.1.12 breaks a significant Flight Plan deviation into four reportable items: lateral deviation of 100 nm or more from the planned route, altitude deviation of 4000 ft or more from planned cruise altitude, any action likely to delay arrival by more than 15 minutes such as holding, and enroute adjustments due to weather. Contact Dispatch means notification is mandatory but timed as safety of flight permits. If the deviation is holding, FOM 5.5.15.1 additionally requires the holding fix, EFC time and fuel on board, with holding fuel recomputed every 15 minutes. Oceanic weather deviation communication with ATC is FOM 5.7.4.4.

> **FOM 7.1.12 Reports to Dispatch**
>
> *"– Lateral deviation: 100 nm or more from the planned route. – Altitude deviation: deviation of 4000 ft or more from the planned cruise altitude."*
>
> These are the four sub-items under 'A significant deviation from the Flight Plan.' Separately, a Dispatch call is required when 787 fuel is off plan by 3000 lbs, and any diversion must be coordinated with Dispatch.
>
> *Workbook wording: What are the four items that are required to be reported to dispatch when deviating?*

5.1.2.6 Know Contingency Plan (i.e., Alternates, WEATHER Monitoring, Fuel, etc.) [CRITICAL OBSERVABLE ITEM]

Analyze Fuel Factors for Diversion

FUEL PLANNING

What drives normal fuel planning?

Weather, delays, MEL penalties, turbulence, altitude, one approach and a miss

Per 14 CFR 121.647 the following must be considered: wind, severe weather and other forecast weather conditions; anticipated traffic delays; MEL/CDL limitations and penalties; turbulence and cruise altitude; one instrument approach and possible missed approach at destination; and any other conditions that may delay landing.

> **FOM 8.3.1.1 Fuel Planning Factors**
>
> *""One instrument approach and possible missed approach at destination""*
>
> Captain-side: if you and the Dispatcher cannot agree on extra fuel, the Captain has the final decision for fuel on board (8.3.1.2). Both parties must agree BEFORE fueling begins, and the crew should contact Dispatch at least 30 minutes prior to pushback with changes. No flight may begin takeoff roll with less than (HA) MIN TAKEOFF FUEL.
>
> *Workbook wording: What items must you keep in mind when doing fuel planning (normal flight)?*

FUEL PLANNING

Domestic dispatch fuel requirement?

Destination, alternate, plus 45 minutes

Flights must be dispatched with enough fuel to fly to the airport to which dispatched, and either fly to and land at the most distant alternate with enough fuel to fly 45 minutes at last cruise altitude consumption rate, or with no alternate required, fly for 55 minutes at last cruise altitude consumption rate.

> **FOM 8.3.2.1 Domestic Operations**
>
> *""With no alternate, fly for 55 minutes at last cruise altitude consumption rate.""*
>
> 14 CFR 121.639. Both reserves are at LAST CRUISE ALTITUDE consumption rate, not holding. Applicable areas at 8.1.1.1.
>
> *Workbook wording: What are the dispatch fuel requirements for domestic operations?*

FUEL PLANNING

Flag dispatch fuel requirement?

Destination, 10 percent, alternate, plus 30 minutes at 1500 ft

Flag flights must be dispatched with enough fuel to fly to and land at the destination; then fly for 10 percent of the total time required to fly from the departure airport to and land at the destination; then fly to and land at the most distant alternate if one is required; then fly for 30 minutes at holding speed at 1500 ft above the destination airport, or the alternate if one is required.

> **FOM 8.3.2.2 Flag Operations**
>
> *""Fly for 30 minutes at holding speed at 1500 ft above the destination airport""*
>
> 14 CFR 121.645. The 10 percent is of TOTAL trip time, which is what distinguishes Flag from the B043 special reserve where the 10 percent applies only to oceanic/remote continental time. Applicable areas at 8.1.1.2.
>
> *Workbook wording: What are the dispatch fuel requirements for Flag operations?*

FUEL PLANNING

Supplemental dispatch fuel requirement?

Domestic or Flag rules, whichever applies

Supplemental (non-scheduled) operations are authorized within the areas approved for enroute operations and are conducted under Domestic or Flag rules per Ops Spec A030 or Exemption 11047 (FOM 8.1.1.3). The Dispatcher must notate on the release whether the flight is being operated as Part 121 Domestic or Flag (FOM 8.5.2). Fuel then comes straight from FOM 8.3.2.1 or 8.3.2.2. On the 787 flying international/oceanic, also check FOM 8.3.2.3 Special Fuel Reserves (Ops Spec B043): destination, 10% of planned time in oceanic/remote continental airspace, most distant alternate, then 45 min at normal cruise consumption computed at top-of-descent weight and altitude.

> **FOM 8.1.1.3 Supplemental Operations / FOM 8.5.2 Dispatch Release**
>
> *“Supplemental (non-scheduled) flights shall be dispatched using either Domestic or Flag rules as”*
>
> There is no separate supplemental fuel rule. Go to FOM 8.3.2.1 (Domestic: destination + most distant alternate + 45 min, or 55 min with no alternate) or 8.3.2.2 (Flag: destination + 10% of total enroute time + most distant alternate + 30 min hold at 1500 ft). Exemption 11047 (FOM 8.1.1.4) allows Domestic/Flag rules to airports not listed in Ops Spec C070.
>
> *Workbook wording: What are the dispatch fuel requirements for Supplemental operations?*

FUEL PLANNING

B043 special fuel requirement?

Destination, 10 percent of oceanic time, alternate, plus 45 minutes

Under Ops Spec B043 flights must be dispatched with enough fuel to fly to and land at the destination; then fly for 10 percent of the PLANNED TIME IN OCEANIC/REMOTE CONTINENTAL AIRSPACE; then fly to and land at the most distant alternate if one is required; then fly for 45 minutes at normal cruising consumption, calculated at the flight's weight and altitude at top of descent.

> **FOM 8.3.2.3 Special Fuel Reserves in International Operations (737/787/A321/A330)**
>
> *""Fly for a period of 10% of the planned time in oceanic/remote continental airspace""*
>
> This is the key contrast with Flag: B043 applies the 10 percent only to oceanic/remote continental time, not the whole trip, and the final reserve is 45 minutes at normal cruise consumption rather than 30 minutes holding at 1500 ft. Explicitly banners 787. That is what makes B043 fuel-efficient on Pacific routes.
>
> *Workbook wording: What are the special fuel requirements for a B043 flight plan?*

FUEL PLANNING

What is final reserve fuel?

30 minutes holding at 1500 ft, or 45 minutes under B043

Final reserve is the fuel that must remain untouched at the end of the planned profile. For Flag operations it is 30 minutes at holding speed at 1500 ft above the destination or alternate. Under the B043 special international reserve it is 45 minutes at normal cruising consumption computed at the flight's weight and altitude at top of descent. For domestic it is 45 minutes at last cruise altitude consumption with an alternate, or 55 minutes with none.

> **FOM 8.3.2.2 / 8.3.2.3**
>
> *""Fly for 45 minutes at normal cruising consumption.""*
>
> Tie this to 11.2.6: if it becomes apparent you will land with less than 30 minutes of fuel, declare MAYDAY MAYDAY MAYDAY FUEL. Final reserve is a planning number; the 30-minute emergency-fuel trigger is an in-flight number.
>
> *Workbook wording: What is Final Reserve Fuel?*

FUEL PLANNING

When do you ask ATC about the delay?

When delay plus divert could land you under 30 minutes

The PIC should request ATC delay information if it is anticipated that an airborne delay followed by a diversion could result in landing with less than a 30-minute fuel supply. FOM 11.2.5 reinforces the same posture: crews should be proactive in assessing estimated arrival fuel and, if delays are anticipated that could lead to minimum fuel, should evaluate and inquire well in advance.

> **FOM 11.2.6 Emergency Fuel**
>
> *""The PIC should request ATC delay information if it is anticipated that an airborne delay followed by a diversion could result in landing with less than a 30-minute fuel supply.""*
>
> Inquire early. By the time you can declare minimum fuel you have already committed to an airport and given up the divert option.
>
> *Workbook wording: When should you plan to “inquire” from ATC the current delay situation when unanticipated*

FUEL PLANNING

When do you declare minimum fuel?

Once committed to one airport and able to accept no delay

Minimum fuel means the fuel supply has reached a state where the flight can accept little or no delay. Declare it only after committing to land at a specific airport. Do not declare it at the destination while a divert to the alternate is still available. It is an advisory, not an emergency, and grants no priority.

> **FOM 11.2.5 Minimum Fuel**
>
> *""should only be made when the flight has committed to land at a specific airport""*
>
> The FOM also directs crews to be proactive and inquire well in advance if delays could lead to minimum fuel.
>
> *Workbook wording: When should you declare Minimum Fuel?*

FUEL PLANNING

When do you declare MAYDAY FUEL?

When you will land with less than 30 minutes

The PIC shall declare MAYDAY MAYDAY MAYDAY FUEL when it becomes apparent that landing will occur with less than a 30-minute supply of fuel, to ensure priority handling. The 30-minute supply is based on holding fuel flow or higher, as required by configuration, at 1500 ft above field elevation.

> **FOM 11.2.6 Emergency Fuel**
>
> *""the PIC shall declare, “MAYDAY, MAYDAY, MAYDAY, FUEL” to ensure priority handling.""*
>
> Note shall, not should. Thirty minutes is a floor, not a target. Contrast with 11.2.5 minimum fuel, which is advisory and grants no priority.
>
> *Workbook wording: When should you declare “MAYDAY, MAYDAY, MAYDAY, FUEL”?*

Assess Diversion and Alternate Airport Weather Conditions and Minima

ETOPS

Airborne, what weather do you need at destination and alternates?

Crew operating minima

In flight, alternates must be at or above the crew's operating minima and remain suitable to conduct an IAP. To START the approach, do not begin the final approach segment unless the latest reported RVR, visibility, or ceiling if required, is at or above the authorized minimums. If a below-minimums report arrives after the final approach segment has begun, you may continue to DA/DH or MDA.

> **FOM 6.3.5 / FOM 5.6.11 Visibility Requirements Approach**
>
> *""Do not begin the final approach segment of an instrument approach procedure unless the latest reported RVR, visibility, or ceiling (if required) is at or above the authorized minimums.""*
>
> Three different standards get confused here: dispatch planning minima (8.2.2.1), in-flight suitability (crew operating minima, 6.3.5), and the approach-start gate (5.6.11). Keep them separate in an oral.
>
> *Workbook wording: Once airborne, what does the weather have to be at your destination, destination alternate, ETOPS*

ETOPS

How do you get current METAR and TAF oceanic?

ACARS, or SATCOM or HF to Dispatch

ACARS is the primary means of pulling current METAR and TAF in oceanic airspace, with SATCOM or HF contact with Dispatch as the alternative. FOM 6.3.5 makes clear that Dispatch monitors airports in the ETOPS area of operation and will inform the Flight Crew of significant changes at adequate airports and designated alternates.

> **FOM 6.3.5 In-Flight ETOPS Alternate Airport Weather Requirements**
>
> *""Dispatch will inform the Flight Crew of significant changes at adequate airports and designated alternates.""*
>
> DESKTOP: the specific ACARS weather-request menu path is aircraft/ACARS documentation. The obligation direction is worth noting for an oral: Dispatch pushes significant changes to you, you do not rely solely on pulling them.
>
> *Workbook wording: How can you get the most current METAR/TAF when in oceanic airspace?*

ETOPS

Who monitors destination and alternate weather?

You and Dispatch, in parallel

The FOM frames this as joint PIC and Dispatcher responsibility rather than a fixed interval. Dispatch must evaluate weather, landing distances, airport services and facilities for the earliest to latest ETA at designated ETOPS alternates, will inform the crew of significant changes, will update alternates if conditions preclude safe approach and landing, and will provide a revised Flight Plan if alternates change.

> **FOM 6.3.5 In-Flight ETOPS Alternate Airport Weather Requirements**
>
> *""Dispatch will update ETOPS alternates if any conditions preclude safe approach and landing at existing alternates.""*
>
> No fixed interval is published. The practical standard is that you should never be surprised by destination or alternate weather at top of descent. Tie to the descent gouge: ATIS, Build, Bug, Brakes, Brief, RNP/Recall.
>
> *Workbook wording: How often should you be checking the weather at your destination and alternates?*

FUEL PLANNING

How do you get updated enroute and descent winds?

WIND DATA REQUEST prompt on the RTE DATA page

On the RTE DATA page the WIND DATA REQUEST prompt downlinks a request and returns both enroute wind and descent forecast data (FCOM 11.42 FMC Cruise). Up to four altitudes entered on a WIND page qualify what comes back. The FMC applies the first entered wind to all waypoints, and a wind entered at a later waypoint changes data from that waypoint downtrack only, so load winds nearest the airplane first. Dispatch can also push uplinks to the PERF INIT, TAKEOFF REF, DESCENT FORECAST, RTE, ALTN, ALTN LIST and WIND pages, annunciated by the FMC EICAS communications alert and a hi-lo chime (FCOM 11.34).

> **FCOM 11.42 FMC Cruise - RTE DATA Page, WIND DATA REQUEST**
>
> *“Select – transmits a datalink request for wind and descent forecast data.”*
>
> You may enter up to four altitudes on any WIND page to qualify the request. Winds can also be uplinked directly to the DESCENT FORECAST and WIND pages from Dispatch (FCOM 11.34 Company Datalink, Uplinks).
>
> *Workbook wording: How (aircraft specific) can you get updated enroute/descent winds?*

13.1.1.1.1 Know Engine Failure in Cruise Procedure [CRITICAL OBSERVABLE ITEM]

TAXI AND GROUND

Engine failure over the western states. What do you do?

Driftdown on the ETCP route, land nearest suitable airport

FOM 11.2.7 requires landing at the nearest suitable airport based on time considerations, and the Captain must notify ATC and Dispatch as soon as possible and file a safety report. Suitability factors are listed in FOM 11.2.7.1; fuel sufficiency, passenger accommodation, and maintenance availability are explicitly NOT justification to fly past the nearest suitable airport with one engine out. For terrain, fly the routing shown on the Flight Plan's terrain clearance / driftdown solution: if the engine fails prior to a charted decision point, terrain clearance is assured along the listed route to the listed airport (FOM 8.3.7 and 8.5.5.3). Use QRH PI-QRH.12 for driftdown speed, level off altitude, and range capability.

> **FOM 11.2.7 Engine Shutdown in Flight / FOM 8.3.7 Enroute Terrain Clearance Program (ETCP)**
>
> *“If an engine fails or is shut down in flight, land the aircraft at the nearest suitable airport, based on”*
>
> Over the western US the driver is terrain. The Flight Plan carries the ETCP/driftdown solution with decision points and any Driftdown Alternates (FOM 8.3.7); weather for a named Driftdown Alternate must be in the Flight Information Packet. Driftdown speed and level off altitude tables are QRH PI-QRH.12.5 / FCOM PI.13.5.
>
> *Workbook wording: What are the Engine Failure and diversion procedures while flying over the western part of the 48*

Oceanic Contingency Procedures

OCEANIC AND NAT

Engine failure in oceanic airspace. What do you do?

Get a clearance; if not, turn 30 degrees, offset 5 nm

Always try for a revised ATC clearance first (FOM 5.7.4.2). Without one, turn at least 30 degrees left or right to establish and maintain a parallel same-direction track offset 5 nm, be below FL290 before diverting, squawk 7700 when able, turn on all exterior lights, run TCAS in TA/RA, and broadcast intentions on the in-use frequency and 121.5 (backup 123.45) with MAYDAY or PAN-PAN spoken three times. Once established on the parallel route, take a 500 ft vertical offset (FOM 5.7.4.3). Combine this with the engine checklist and FOM 11.2.7 nearest suitable airport requirement, and advise Dispatch.

> **FOM 5.7.4.2 General Procedures (Oceanic Emergency/Abnormal)**
>
> *“If prior clearance cannot be obtained, the following contingency procedures should be employed”*
>
> Be below FL290 prior to diverting from the assigned clearance unless a revised clearance is obtained, then establish a 500 ft vertical offset once on the parallel route (FOM 5.7.4.3). ETOPS: if the contingency follows an engine shutdown or ETOPS-critical system failure, tell ATC, remind them of the aircraft type, and request expeditious handling.
>
> *Workbook wording: What are the engine failure and diversion procedures while in oceanic airspace?*

OCEANIC AND NAT

Immediate action items for engine failure in cruise?

None; ENG FAIL L, R has no memory items

The QRH Quick Action Index marks memory-item checklists with an asterisk; ENG FAIL L, R (QRH 7.14) is not listed, and the checklist has no dashed memory-item line. The first reference actions are A/T ARM switch confirm OFF and thrust lever confirm idle on the affected side, then the engine automatically attempts a relight - N2 above 30% and steadily increasing means it is starting; after one minute failed, fuel control switch confirm CUTOFF. Route first to Dual Eng Fail/Stall (QRH 7.2) if thrust is lost on both, or to Eng Svr Damage/Sep L, R (QRH 7.10) if there is airframe vibration with abnormal engine indications or a separation. Only Dual Eng Fail/Stall, Eng Svr Damage/Sep, ENG SURGE, ENG LIMIT EXCEED, ENG AUTOSTART, FIRE ENG and Aborted Engine Start carry memory items in the engine group.

> **QRH Quick Action Index / QRH 7.14 ENG FAIL L, R**
>
> *“*indicates a checklist which includes memory items.”*
>
> ENG FAIL L, R does not appear on the QRH Quick Action Index at all, so there are no memory items to recite. Reference actions start with A/T ARM switch (affected side) confirm OFF and thrust lever (affected side) confirm idle. A Hawaiian-issued QRC card is not part of the QRH file, so verify the card itself.
>
> *Workbook wording: Review the immediate action items for engine failure in cruise (787, back of the QRC).*

OCEANIC AND NAT

Driftdown off the route. Which way do you turn?

At least 30 degrees, left or right per listed factors

FOM 5.7.4.2 has you leave the cleared track by initially turning at least 30 degrees left or right to establish a parallel, same-direction track offset 5 nm. Direction is a judgment call driven by where you sit in the track system, what is allocated on adjacent tracks, which way the alternate lies, the SLOP you are already flying, and terrain. If you are flying a 1 or 2 nm right offset, turning right adds to that spacing rather than driving you back through the centerline. Maintain the assigned level until established on the parallel route if possible, then take a 500 ft vertical offset, and get below FL290 before diverting unless recleared (FOM 5.7.4.3).

> **FOM 5.7.4.2 General Procedures (Oceanic Emergency/Abnormal)**
>
> *“The direction of the turn should be based on one or more of the following factors:”*
>
> The five factors are position relative to any organized track or ATS route system, direction of flights and levels on adjacent tracks, direction to an alternate airport, any strategic lateral offset being flown, and terrain clearance. Turning away from the offset you are already flying is what buys the full lateral spacing.
>
> *Workbook wording: When turning off the route during a drift-down, which direction will you turn? Why?*

8.3.4.26 Demonstrate Ability to Correctly Program Aircraft Guidance Systems Including Approach

SLOP

OCEANIC AND NAT

What is SLOP for?

Collision risk and wake turbulence

The Flight Crew should use SLOP as standard operating practice in normal oceanic operations to mitigate collision risk and wake turbulence. It addresses both wake vortex encounters and the heightened collision risk when non-normal events such as operational altitude deviation errors and turbulence-induced altitude deviations occur.

> **FOM 5.7.4.7 Strategic Lateral Offsets in Oceanic Airspace**
>
> *“"to mitigate collision risk and wake turbulence."”*
>
> SLOP is generally accepted but disallowed in specific regions; check the AIP. Any aircraft overtaking another should offset within the procedure to create the least wake turbulence for the aircraft being overtaken.
>
> *Workbook wording: What is the purpose of SLOP?*

OCEANIC AND NAT

How far can you offset with SLOP?

Up to 2 nm right of centerline. Never left

SLOP is applied in relation to a route or track. The aircraft may fly zero, or up to a maximum of 2 nm right of course only, from 0.0 to 2.0 nm right of centerline in 0.1 nm increments. SLOP left of course is not authorized. The Captain may apply an offset outbound at the oceanic entry point but must return to centerline at the oceanic exit point.

> **FOM 5.7.4.7 Strategic Lateral Offsets in Oceanic Airspace**
>
> *“"SLOP left of course is not authorized."”*
>
> Three legal positions in practice: centerline, 1 nm right, 2 nm right, though the FOM permits any 0.1 increment. No ATC clearance is needed. Must be back on centerline by the oceanic exit point.
>
> *Workbook wording: Up to what distance/direction from the route is SLOP used/acceptable?*

In Trail Procedures (OpSpec A354)

DESCENT AND APPROACH

What is ITP?

In OpSpec A354 and NAT training

ITP is authorized under OpSpec A354 and detailed in the oceanic/NAT training material and Jeppesen ATC pages. Confirm the request phraseology and the distance/speed criteria there before using it.

> **OpSpec A354 / NAT training**
>
> *Workbook wording: What is “ITP”?*

DESCENT AND APPROACH

When would you use ITP?

In OpSpec A354 and NAT training

ITP is authorized under OpSpec A354 and detailed in the oceanic/NAT training material and Jeppesen ATC pages. Confirm the request phraseology and the distance/speed criteria there before using it.

> **OpSpec A354 / NAT training**
>
> *Workbook wording: When might you consider using the ITP?*

TAKEOFF AND DEPARTURE

How do you request an ITP climb or descent?

In OpSpec A354 and NAT training

ITP is authorized under OpSpec A354 and detailed in the oceanic/NAT training material and Jeppesen ATC pages. Confirm the request phraseology and the distance/speed criteria there before using it.

> **OpSpec A354 / NAT training**
>
> *Workbook wording: How do you (specifically) request an ITP CLB/DES clearance?*

Diversion Procedures/Select New Destination

DESCENT AND APPROACH

How do you select a new destination?

ALTN page, select the alternate, then DIVERT NOW, execute

ALTN page 1/2 lists four alternates in ETA order airborne, distance order on the ground, each with its own XXXX ALTN page reached by the adjacent right LSK (FCOM 11.43). Manual entries are valid from the nav database and display in large font. Select the airport, review or modify the route on the XXXX ALTN page, then select DIVERT NOW and execute. ALTN REQUEST downlinks for a preferred list of up to four alternates, and ALTN INHIBIT keeps one or two airports off the list. The Emergency Report page also has a DIVERTING TO field that defaults to the active route destination (FCOM 5.40.46).

> **FCOM 11.43 FMC Descent and Approach - Alternate Airport Diversions**
>
> *“Selecting DIVERT NOW displays the route from the present position to the”*
>
> Executing the divert changes the route destination airport, folds the modification into the active flight plan, deletes the parts of the old route not in the diversion, and deletes all descent constraints if a descent path existed (DESCENT PATH DELETED message). Entering a new airport on ALTN page 1/2 does not select it - you must select it and then DIVERT NOW.
>
> *Workbook wording: How (aircraft specific) do you “Change” or “Select” a new destination?*

Alternate Navigation System

DESCENT AND APPROACH

What is alternate navigation?

TCP backup navigation after losing all displays or all FMCs

Alternate navigation is a backup capability in the Tuning and Control Panel used after unrecoverable loss of all displays or all three FMCs (FCOM 11.50). Under normal power the NAV key on any of the three TCPs brings it up; on standby power only the Captain's TCP works. Information is shown on the TCPs and the ISFD, enough to land at the nearest suitable airport. The TCPs have no performance or navigation database, so only a single manually entered waypoint is supported. Two pages exist: ALTN NAV and ALTN NAV RADIO.

> **FCOM 11.50 Alternate Navigation System Description**
>
> *“In the unlikely event that an unrecoverable loss of all displays or loss of all three”*
>
> LNAV and VNAV are not available in alternate navigation, but autopilot and autothrottle may be. On standby power only the left (Captain's) TCP is available.
>
> *Workbook wording: What is Alternate Navigation?*

DESCENT AND APPROACH

How do you fly alternate navigation to a point?

Push TCP NAV, enter waypoint lat/long, fly TRK TO

Push the NAV key on the TCP to get the ALTN NAV page, then enter the TO waypoint latitude and longitude at LSK 2 L/R using DDMM.M and DDDMM.M formats with the hemisphere selection keys (FCOM 11.50). The page then shows true track to the waypoint (TRK TO, magenta), current ground track, groundspeed, time to go accurate to 1 minute, and distance to go accurate to 10 nm. You must manually update the waypoint as the diversion progresses. ALTN NAV RADIO page 2/2 lets you tune an ILS frequency and course or a GLS channel into the left integrated navigation receiver so left-side deviations show on the Captain's PFD and the ISFD - check that both sides are tuned correctly, since the TCP ALTN NAV advisory means left and right can be on different frequencies.

> **FCOM 11.50 Alternate Navigation System Description - Alternate Navigation Page**
>
> *“capability. Once the waypoint’s latitude and longitude have been entered at LSK”*
>
> The waypoint must be manually updated as required - there is no database and no leg sequencing. TRK TO is displayed in magenta; GTK, GS, TTG and DTG are also shown. LNAV and VNAV are unavailable, so hand-fly or use heading select.
>
> *Workbook wording: How do you perform Alternate Navigation to a point or destination?*

6.1.1.4 Assess Operational and Environmental Factors for Descent and Arrival

Descent Preparation

DESCENT AND APPROACH

What goes into preparing for descent?

Alerts, landing performance, VREF, minimums, autobrake, FMC crosscheck, briefing, checklist

Start the Descent Procedure before descending below cruise altitude and finish it by 10,000 ft AFE (FCOM NP.21.38). Items: verify RNP as needed, recall and review all alert and memo messages and operational notes, check landing performance, verify/enter VREF on APPROACH REF, set RADIO/BARO minimums (BARO to MDA+50 for an MDA approach unless the chart authorizes DA in lieu of MDA, BARO to published DA, RADIO to DH for CAT II, blank for CAT III), set the NAV RADIO page, set the AUTOBRAKE, crosscheck FMC waypoints/speeds/altitudes/sequence against the chart if LNAV or VNAV will be used, do the approach briefing, then the DESCENT checklist. FOM 5.5.16 governs the briefing format itself.

> **FCOM NP.21.38 Descent Procedure (Amplified Procedures)**
>
> *“Complete the Descent Procedure by 10,000 feet AFE.”*
>
> Before top of descent the PM also modifies the active route for the arrival/approach and sets descent speed to CI mach/300 on the VNAV DES page unless ATC asks for something else. The approach briefing must include the reminder that go-arounds on or near the ground require go-around thrust set manually and can trigger a flap configuration warning.
>
> *Workbook wording: What items are included in preparing for descent?*

STAR/Airport/Arrival Review

TAXI AND GROUND

What do you brief off the STAR, approach and taxi charts?

Route, constraints, minimums, alternate, missed approach, runway, exit, taxi, autobrakes, flaps, speed

Under Plan the PF briefs route (STAR, approach, approach mode), relevant altitude and airspeed constraints and approach minimums, fuel/route to alternate, missed approach with fuel/route to alternate, landing runway and assessment plus exit and taxi, autobrakes, and flaps and approach speed (FOM 5.5.16). Considerations covers specific PM duties driven by the threats and plan, other issues, and a recap of critical threats. The briefing is initiated prior to beginning the descent. For contaminated runways the briefing also includes LTP and both anticipated and actual braking action (FOM 9.x). Add the FCOM NP.21.39 note about manual go-around thrust near the ground, and for CAT II/III add aircraft capability, airport facilities, crew qualification, weather minima, task sharing, go-around strategy, and runway turnoff and initial taxi plan (FCOM SP.4.8).

> **FOM 5.5.16 Threat-Based Approach Briefing/TPC**
>
> *“Review any relevant altitude and/or airspeed constraints and approach minimums”*
>
> Format is T-P-C: Threats (PM identifies, PF adds), Plan (PF), Considerations (PF). It is interactive and big-picture, not a script. Threat list is on the Crew Briefing Reference Card and airport-specific threats are in the Jeppesen 10-7.
>
> *Workbook wording: What items should be briefed on the STAR/Approach/Taxi charts?*

Descent Plan Execution

DESCENT AND APPROACH

How do you start the descent?

Set lower MCP altitude and push the altitude selector, or DES NOW

At T/D with a lower altitude set in the MCP window, VNAV starts the descent automatically. For an early descent, set the lower MCP altitude and push the altitude selector, or page to VNAV DES and line select DES NOW then execute (FCOM 11.42, 11.31). DES NOW gives roughly 1250 fpm at ECON speed until the planned path is intercepted, then VNAV commands pitch to hold the path at ECON speed with idle thrust. In an early descent the autothrottle annunciates THR then HOLD so thrust levers can be repositioned to adjust rate, and the pitch mode is VNAV SPD. Beyond 50 NM from T/D the MCP selector produces a cruise descent instead, levelling at the new cruise altitude.

> **FCOM 11.42 FMC Cruise - Early Descent / FCOM 11.31 Flight Management System Operation - Early Descent**
>
> *“Start an early descent by selecting the DES NOW prompt on the DES page or by”*
>
> Pushing the MCP altitude selector activates DES NOW only within 50 NM of T/D, or if the MCP altitude is below the highest descent altitude constraint. Beyond 50 NM from T/D it becomes a cruise descent to a new cruise altitude and VNAV may never capture the path.
>
> *Workbook wording: How do you get the aircraft to begin the descent?*

DESCENT AND APPROACH

Cleared the arrival vs cleared to descend via?

In the FCOM and the FMS Pilot Guide

FMC and CDU manipulation: alternate navigation, destination change, offset entry, wind uplink, descent initiation, and autopilot requirements on RNAV procedures. FCOM Chapter 11 and the FMS Pilot Guide. Related FOM policy: 5.7.4.7 SLOP for offsets and 5.7.4.8 Navigation Systems Failure.

> **FCOM Ch 11 / FMS Pilot Guide**
>
> Not in the FOM, FCOM or QRH. The US definitions of a route-only STAR clearance versus a 'descend via' clearance (lateral route plus published vertical profile and speeds) live in the AIM 5-4-1 and FAA JO 7110.65, and are also on the Jeppesen STAR chart. The FOM only covers foreign analogues: FOM 19.3.1.3 for Canada (STAR altitude restrictions stay mandatory unless specifically canceled by ATC), FOM 21.3.3.5 for Mexico phraseology, and FOM 23.16.2 for Europe.
>
> *Workbook wording: What is the difference between “Cleared XX arrival” vs “Cleared to descend via XX arrival”?*

Weather Considerations

WEATHER AND MINIMUMS

What weather do you need to begin an approach?

Reported RVR or visibility at or above minimums

The reported weather at the destination must be at or above the visibility minimums prescribed for the approach. Do not begin the final approach segment unless the latest reported RVR, visibility, or ceiling if required, is at or above the authorized minimums. For non-precision and CAT I, TDZ RVR is controlling, Mid and Rollout are advisory, and Mid may substitute for TDZ if TDZ is unavailable.

> **FOM 5.6.11 Visibility Requirements Approach**
>
> *""TDZ RVR reports are controlling.""*
>
> Visibility is the controlling parameter; ceiling applies only where the procedure requires it. RVR values above 6000 are advisory only. Visibility below 1/2 sm is not authorized and shall not be used for approach minimums.
>
> *Workbook wording: What must the Weather be in order to begin an approach? Ceiling and visibility?*

Analysis of Airport Conditions

WEATHER AND MINIMUMS

Lowest visibility for CAT I?

1800 RVR

CAT I precision minima as low as 1800 RVR are authorized to approved runways without TDZ or runway centerline lights, including runways where those lights are installed but inoperative, provided the FD or AP is engaged in approach mode and used to DA or missed approach initiation unless adequate visual references are established, and provided the published Jeppesen procedure carries an 1800 RVR minimum (FOM 5.6.16.1, Ops Spec C059). If the FD or AP fails or is disengaged you go around unless another FD/AP/HUD is available or you have the runway environment. Separately, FOM 5.6.11 prohibits visibility values below 1/2 sm as approach minimums and makes RVR above 6000 advisory only. High Minimums Captains have their own equivalents in FOM 3.17 (published RVR 1800 or less becomes RVR 4500).

> **FOM 5.6.16.1 Reduced Landing Minima - 1800 RVR and 200 ft DH**
>
> *"Authorized precision CAT I landing minima as low as 1800 RVR to approved runways without TDZ"*
>
> SA CAT I down to 1400 RVR / 150 ft DH is 737 only (FOM 5.6.16.2), so it is not available to you on the 787. TDZ RVR is controlling for CAT I; mid RVR may be substituted only if TDZ is unavailable (FOM 5.6.11).
>
> *Workbook wording: What is the lowest visibility for a CAT I approach?*

DISPATCH AND RELEASE

Are there high mins Captain restrictions?

Yes, relieved by Exemption 5549

FOM 5.6.17 cross-references 3.16 Exemption 5549 for high minimums Captain limitations. A newly qualified Captain operates to increased minimums until the experience requirement is met, and Exemption 5549 is the FAA relief that modifies how those high-minimums restrictions apply.

> **FOM 5.6.17 / FOM 3.16 Exemption 5549**
>
> *""See 3.16 – Exemption 5549 for high minimums Captain limitations.""*
>
> DESKTOP: read FOM 3.16 for the exact hour threshold and the precise relief Exemption 5549 grants. This is directly relevant to you as a new Captain and is a near-certain oral question, so get the numbers from the PDF rather than from memory.
>
> *Workbook wording: Are there restrictions for High Mins Captains?*

DISPATCH AND RELEASE

What is FAA Exemption 5549?

Relief allowing High Minimums Captains to fly published minima

Exemption 5549 lets a High Minimums Captain fly to published CAT I minima instead of the increased high-minimums values, provided the approach is flown by the Captain with an autopilot coupled to DH or missed approach (HA wording, FOM 3.16.1). If crosswind exceeds 15 kts combined with braking action less than good, visibility must be at least 4000 RVR (3/4 mile). FOM 3.16.2 extends it to SA CAT I / CAT II if the Captain has at least 300 hours as PIC or SIC in a turbojet and complies with FCOM CAT II/III and Autoland procedures. CAT III requires 100 hours PIC in type at the company, not reducible (FOM 3.15). Dispatch annotates the release 'FLT RELEASED UNDER EXEMPTION 5549' and does not have to amend it if weather deteriorates enroute.

> **FOM 3.16 Exemption 5549**
>
> *“High Minimums Captains may be dispatched using Exemption 5549 and must comply with the”*
>
> A High Minimums Captain has under 100 hours PIC in type; without the exemption they add 100 ft and 1/2 sm to published minimums, but no less than 300-1 (FOM 3.17). OE flight hours do not count toward the 100 hours.

WEATHER AND MINIMUMS

Lowest visibility for CAT II?

1000 RVR

For CAT II the final approach segment may not begin unless the latest controlling RVR reports are at or above authorized minimums and required airborne and ground equipment is operational, including LOC and GS, an outer marker or DME defining the FAF, HIRL, and an approach light system (FOM 5.6.17.2). For TDZ 1000 RVR approaches and all international approaches, TDZ and centerline lights are required. CAT II must be flown Autoland on the 787 (the HGS AIII option is 737 only), crosswind is limited to the lesser of the AFM limit or 15 kts, and landing plus subsequent ground operations at 1000 RVR must follow the airport low visibility operations plan using the Low Visibility Taxi Charts. Once established on final, you may continue if RVR drops below minimums. FCOM SP.4.7 sets the aircraft side: radio DH, RVR controlling, Autoland and autobrake mandatory, ASA LAND 2 or LAND 3, Captain flying.

> **FOM 5.6.17.2 CAT II/SA CAT II Operating Limitations**
>
> *“Landings and subsequent ground operations for CAT II operations to 1000 RVR must be”*
>
> The controlling RVR table lives on the approach briefing cards in the PRC, not in the FOM - FOM 5.6.11 says so explicitly and FOM 5.6.14 notes CAT II/III minima are NOT tailored on the Jeppesen charts. Verify the number on the PRC card before the ride.
>
> *Workbook wording: What is the lowest visibility for a CAT II approach? (C060)*

WEATHER AND MINIMUMS

Lowest visibility for CAT II SA?

On the PRC approach briefing card

Same source as CAT II. FOM 5.6.17 defers to the PRC/QRH approach briefing references and required equipment tables. SA CAT II also carries the 5.6.17.2 gate: the final approach segment may not begin unless controlling RVR is at or above authorized minimums and the required airborne and ground equipment is operational.

> **FOM 5.6.17.2 CAT II/SA CAT II Operating Limitations**
>
> Not stated numerically in the FOM, FCOM or QRH. FOM 5.6.11 sends you to the approach briefing cards in the Pilot Reference Cards (PRC): 'RVR requirements for (737) SA CAT I, SA CAT II, CAT II/III approaches are found on the approach briefing cards in the Pilot Reference Cards (PRC).' FOM 5.6.14 adds that CAT II/III minima are not tailored on the Jeppesen charts. Also note the FOM tags SA CAT I as 737 only (FOM 3.14 note).
>
> *Workbook wording: What is the lowest visibility for a CAT II SA approach? (C060)*

WEATHER AND MINIMUMS

Lowest visibility for CAT III?

On the PRC approach briefing card

FOM 5.6.17 defers CAT III minima to the PRC/QRH approach briefing references and required equipment tables. Autoland is the approved means on the 787 and is permitted on CAT II/III runways only, with the PIC required to be Autoland qualified.

FOM 5.6.17 / 5.6.18

Not stated numerically in the FOM, FCOM or QRH. FOM 5.6.11 points to the approach briefing cards in the Pilot Reference Cards (PRC) for CAT II/III RVR requirements, and FOM 5.6.14 says CAT II/III approach minima are not tailored on Jeppesen charts. Pull the CAT III card in the PRC.

Workbook wording: What is the lowest visibility for a CAT III approach? (C060)

DESCENT AND APPROACH

Does the FOM table CAT II and III limitations?

No. It points to the fleet manuals, PRC and QRH

FOM 5.6.17 directs Flight Crews to use the approach procedures in the fleet-specific manuals for all CAT II/III approaches. The approach briefing references and required equipment tables in the PRC/QRH contain the complete listings of Flight Crew procedures and required equipment. Missed approach criteria for CAT II/III are also in the fleet-specific manuals.

FOM 5.6.17 Category II/III Operational Requirements

> *""The approach briefing references and required equipment tables in the PRC/QRH contain complete listings of the Flight Crew procedures and required equipment.""*

Ops Spec C060. Also: Alaska may be restricted by variant from CAT II/III at certain runway/airport combinations, see the tailored Jeppesen chart notes (5.6.17.1).

Workbook wording: Is there a table in the FOM categorizing CAT II/III Approach and Landing Limitations?

DESCENT AND APPROACH

When can you autoland?

CAT II and III runways only

Autoland approaches and landings are permitted on CAT II/III runways ONLY. Pilots should advise the tower any time Autoland operations are conducted so the ILS Critical Zone is protected; that critical area is normally protected when weather is below 800 ft or 2 miles. On the 787 the PIC must be Autoland qualified. The procedures themselves are in the fleet-specific manuals.

FOM 5.6.18 Autoland Operations

> *""Autoland approaches and landings are permitted on CAT II/III runways only.""*

Ops Spec C061. Two traps: it is a CAUTION-level restriction, and the tower advisory is on you because in good weather the ILS critical area is not automatically protected. Note the portal previously carried a wrong claim that the Hawaiian 787 cannot autoland on an ILS; the FCOM note is conditional on detected glideslope interference, not fleet-wide.

Workbook wording: Under what conditions can you perform an autoland?

DESCENT AND APPROACH

What qualifies a 787 Captain for autoland?

One automatic landing in the aircraft

For CAT II/III, Autoland and (737) HGS AIII hand-flown approaches are approved. For HA the Captain must complete one automatic landing in the aircraft to be qualified for CAT II/III Autoland operations. On the 787 the PIC must be Autoland qualified per 5.6.18. Autoland is permitted on CAT II/III runways only.

> **FOM 5.6.17 / 5.6.18**
>
> *""(HA) The Captain must complete one automatic landing in the aircraft to be qualified for CAT II/III (Autoland) operations.""*
>
> This is a currency item that bites new Captains: without that one automatic landing logged in the aircraft, you are not CAT II/III qualified regardless of sim performance. Autoland reporting requirements are at FOM 7.1.14.
>
> *Workbook wording: Under what conditions must you perform an autoland?*

DESCENT AND APPROACH

How do you log an autoland?

On the ACARS Flight Summary page

Record all Autoland/HGS AIII landings on the ACARS Flight Summary page. For guidance on completing the A/C Approach Survey section of the Maintenance Logbook, HA crews use 5.6.21 Maintenance Request for Autoland.

> **FOM 7.1.14 Autoland and HGS Reporting**
>
> *""Record all Autoland/HGS AIII landings on the ACARS Flight Summary page.""*
>
> Two separate actions: the ACARS Flight Summary entry and the Maintenance Logbook A/C Approach Survey. As HA, 5.6.21 is your reference, not the AS reliability program at 5.6.20.
>
> *Workbook wording: What are the requirements to log/report an autoland?*

Diversion Considerations

TAXI AND GROUND

What drives a diversion?

Anything degrading a safe landing at destination

If an engine fails or is shut down in flight, land at the nearest suitable airport based on time considerations where a safe landing can be made. Suitability factors include aircraft configuration, weight, systems status and fuel remaining, and wind and weather conditions enroute at the diversion altitude. As soon as possible after a shutdown the Captain shall notify ATC and Dispatch and continue to provide timely progress information, and a safety report shall be completed.

> **FOM 11.2.7 / 11.2.7.1 Engine Shutdown in Flight, Nearest Suitable Airport**
>
> *""land the aircraft at the nearest suitable airport, based on time considerations, where a safe landing can be made.""*
>
> Nearest Suitable Airport is FAA-approval-required text, so it is regulatory, not company preference. Nothing justifies flying past the nearest suitable airport after an engine shutdown.
>
> *Workbook wording: What factors may precipitate a diversion?*

DESCENT AND APPROACH

When do you start the divert?

Fuel against final reserve, weather trend, field, systems

Build the decision from the fuel rules and the suitability factors. Minimum fuel should only be declared once committed to a specific airport and not while a divert is still available (11.2.5), so the divert decision must be made before that point. Suitability factors from 11.2.7.1 are aircraft configuration, weight, systems status and fuel remaining, and wind and weather conditions enroute at the diversion altitude. Dispatch and the PIC exercise judgment jointly on the safest course of action.

> **FOM 11.2.5 / 11.2.7.1**
>
> *""it should not be made at the destination when the decision to divert to an alternate is still available.""*
>
> That quote is the timing test. If you are still holding at destination with the alternate available, you are not in minimum fuel yet, and that is exactly the window in which the divert decision belongs. Waiting until you can declare minimum fuel means you waited too long.
>
> *Workbook wording: What factors should you consider when deciding when to start diverting to your alternate?*

DESCENT AND APPROACH

Not using your listed alternate. What do you weigh?

Configuration, weight, systems, fuel, wind and weather enroute

Suitability factors listed in the FOM include aircraft configuration, weight, systems status and fuel remaining, and wind and weather conditions enroute at the diversion altitude. The list is explicitly non-exhaustive. After an engine shutdown you must land at the nearest suitable airport based on time considerations where a safe landing can be made.

> **FOM 11.2.7.1 Nearest Suitable Airport**
>
> *""The following factors and others may be relevant in determining whether or not an airport is suitable""*
>
> Nearest Suitable Airport is FAA-approval-required text. Note it says nearest suitable based on TIME, not distance. DESKTOP: read the full factor list off 11.2.7.1, I captured only the first two bullets this session.
>
> *Workbook wording: If you are not diverting to your listed alternate, what factors should you consider before selecting the*

Fuel Management

FUEL PLANNING

Where is fuel planning in the FOM?

In the FOM fuel requirements section

Section 8.3 carries fuel: 8.3.1 Fuel Planning, 8.3.1.1 Fuel Planning Factors, 8.3.1.2 Departure Fuel Coordination, 8.3.1.3 Tanker Fuel, 8.3.2 Fuel Requirements with 8.3.2.1 Domestic, 8.3.2.2 Flag and 8.3.2.3 Special Fuel Reserves in International Operations. In-flight fuel emergencies are separate, at 11.2.5 Minimum Fuel and 11.2.6 Emergency Fuel.

> **FOM 8.3 Fuel**
>
> *""No flight may begin its takeoff roll with less than (AS) REQ Fuel and (HA) MIN TAKEOFF FUEL.""*
>
> Tanker fuel labels are worth knowing: economic tankering shows as (HA) EXTRA, mandatory tankering as (HA) CONT.
>
> *Workbook wording: Where in the FOM is the fuel planning section?*

ETOPS

What is the ETOPS critical fuel scenario?

Worst-case diversion fuel required from the critical point

The critical fuel scenario is the most fuel-critical of three diversion cases from the critical point to the ETOPS alternate: immediate descent to 10,000 ft with both engines (depressurization), descent to best single-engine altitude with one engine out, and immediate descent to 10,000 ft with one engine out (simultaneous depressurization and engine failure) (FOM 6.2.5). To that add fuel to descend to 1500 ft above the alternate, hold 15 minutes, and fly an instrument approach and landing; a 5% increase in forecast wind speed; icing fuel taken as the greater of airframe icing drag plus engine and wing anti-ice for 10% of forecast icing time or engine and wing anti-ice for 100% of forecast icing time; engine deterioration; APU use; and MEL/CDL penalties. The complete critical fuel planning, departure to CP/ETP plus CP/ETP to the diversion airport, is compared against flag fuel requirements and ETOPS ADD fuel is loaded if needed.

> **FOM 6.2.5 Critical Fuel Planning**
>
> *“ETOPS (AS) FUEL REQD OVER CP (HA) CRITF is the most critical fuel scenario from the critical point”*
>
> It is a dispatch requirement, not an inflight limit (FOM 6.2.6 area, around FOM 6.4). If before the EEP it looks like actual FOB at the CP/ETP will be below the critical fuel requirement, coordinate with Dispatch. (HA) label on the Flight Plan is CRITF; (AS) is FUEL REQD OVER CP.
>
> *Workbook wording: What is the ETOPS Critical Fuel Scenario?*

LAHSO Policy

DESCENT AND APPROACH

What is LAHSO?

Land and Hold Short Operations

LAHSO involves landing and holding short of an intersecting runway, an intersecting taxiway, or some other designated point on a runway. The Available Landing Distance (ALD) is measured from the landing threshold to the hold short point, and the hold short point is marked by a painted holding position marking and red and white hold short position signs.

> **FOM 5.6.32.1 Definitions LAHSO**
>
> *“"A point on the runway beyond which a landing aircraft with a LAHSO clearance is not authorized to proceed."”*
>
> Revisions to this section require FAA approval, which signals it is regulatory text, not company preference.

DESCENT AND APPROACH

What must you check before accepting LAHSO?

Stopping within the ALD, using AFM distance plus 1000 ft

The Captain may accept a LAHSO clearance only if every condition in 5.6.32.2 is satisfied. The HA determination is that existing conditions allow the aircraft to safely land and stop within the Available Landing Distance, where calculated landing distance equals the FAA-approved AFM distance plus 1000 ft for the actual configuration, environment and weight.

> **FOM 5.6.32.2 Operational Information LAHSO**
>
> *“"The calculated landing distance will be the FAA-approved Aircraft Flight Manual (AFM) distance plus 1000 ft"”*
>
> RECONCILE: Rev 125 revised LAHSO guidance to prohibit LAHSO on wet runways and require tailwind to be calm, less than 3 kts. Confirm the current four-factor list against the Rev 125.1 PDF before quoting it on the line. The AS note also requires a published 10-7 LAHSO page for that airport and runway configuration.
>
> *Workbook wording: What are the 4 different factors that must be considered prior to accepting a LAHSO clearance?*

Complete Landing Performance Assessment

DESCENT AND APPROACH

What is a TLR?

Takeoff and Landing Report

The TLR is the performance product used for runway analysis. FOM 5.6.22.1 sets the surrounding policy: Captains shall ensure landing is allowed and executed within the appropriate performance data, and the landing data used for in-flight analysis is based on input conditions, which is what makes an accurate runway assessment critical.

> **FOM 5.6.22.1 Landing Policy**
>
> *""The landing data used for in-flight analysis is based on input conditions, this highlights the importance of an accurate runway assessment.""*
>
> Acceptance is not weight-only: environmental conditions matter equally. Do not interchange the takeoff and landing crosswind tables, they serve different functions and the takeoff table is CG and weight sensitive while the TALPA landing table is not.

DESCENT AND APPROACH

How do you request landing performance data?

In the FOM and the fleet performance manuals

Load closeout reading and verification, and the takeoff/landing performance products. Takeoff performance hierarchy is Final TPR, then Preliminary TPR, then TLR. Using a TLR for takeoff demands attention to the temperature and altimeter assumptions baked into it. FOM 5.2 covers closeout and weight and balance; the SABLE Load Sheet is the official FAA weight and balance record. Landing analysis designators and request methods are in the performance system documentation.

> **FOM 5.2 / fleet performance manuals**
>
> *Workbook wording: How can you request landing data performance?*

DESCENT AND APPROACH

What do HXX/TXX and LXX/RXX mean on the printout?

In the FOM and the fleet performance manuals

Load closeout reading and verification, and the takeoff/landing performance products. Takeoff performance hierarchy is Final TPR, then Preliminary TPR, then TLR. Using a TLR for takeoff demands attention to the temperature and altimeter assumptions baked into it. FOM 5.2 covers closeout and weight and balance; the SABLE Load Sheet is the official FAA weight and balance record. Landing analysis designators and request methods are in the performance system documentation.

> **FOM 5.2 / fleet performance manuals**
>
> *Workbook wording: On the landing data analysis printout, what do the designator/s “HXX/TXX” and “LXX/RXX” represent?*

Cold Temperature Altimeter Correction/VNAV Temperature Limitation

COLD WEATHER AND RUNWAY CONDITION

When do you make cold weather corrections on approach?

Airport temperature 0 degrees C or colder, when charted

Colder than ISA means true altitude is lower than indicated, and the error grows with height above the altimeter reference source and as surface temperature approaches -30 C or colder (FCOM SP.16.10, FOM 9.3.3.1). Within the US and its territories the Jeppesen briefing strip carries a note when cold temperature corrections must be applied. Apply corrections to all published minimum departure, enroute and approach altitudes including missed approach altitudes, and set MDA/DA at the corrected minimum. Enter the table with airport temperature and height above the altimeter reference source (published minimum altitude minus airport elevation), add the correction to the published altitude, round the MCP altitude up to the next 100 ft, and extrapolate linearly above 3000 ft. On the 787 you also add cold temperature corrections to FMC waypoint altitude constraints during the Approach Procedure, but do not manually build the approach (FCOM NP.21.40).

> **FCOM SP.16.10 Cold Temperature Altitude Corrections / FOM 9.3.3.1**
>
> *"Extremely low temperatures create significant altimeter errors and greater"*
>
> No correction is needed under ATC radar vectors, when maintaining an assigned flight level, or when reported airport temperature is above 0 C or at/above the procedure's minimum published temperature. Advise ATC of the corrections, and never correct the altimeter barometric reference setting.
>
> *Workbook wording: When/why do you make cold weather corrections while on approach?*

6.1.1 Perform Descent and Terminal Arrival

Altimeter procedures

DESCENT AND APPROACH

When do you set QNH in the descent?

In the FOM theater chapters and the Jeppesen pages

A level is flown on standard 29.92 (QNE); an altitude is flown on local QNH; QFE references height above field elevation. Transition altitude is charted and climbing; transition level is descending and often ATC-assigned. Regional specifics are in FOM chapters 19 to 24, for example the Oakland FIR 5500 ft crossover at 20.7.1.

> **FOM Ch 19-24 / Jeppesen**
>
> *Workbook wording: When, in the descent, do you change your altimeter setting to QNH (local Altimeter)?*

DESCENT AND APPROACH

Transition level vs transition altitude?

In the FOM theater chapters and the Jeppesen pages

A level is flown on standard 29.92 (QNE); an altitude is flown on local QNH; QFE references height above field elevation. Transition altitude is charted and climbing; transition level is descending and often ATC-assigned. Regional specifics are in FOM chapters 19 to 24, for example the Oakland FIR 5500 ft crossover at 20.7.1.

> **FOM Ch 19-24 / Jeppesen**
>
> *Workbook wording: What is the difference between a transition level vs transition altitude?*

DESCENT AND APPROACH

What altimeter setting defines a level?

In the FOM theater chapters and the Jeppesen pages

A level is flown on standard 29.92 (QNE); an altitude is flown on local QNH; QFE references height above field elevation. Transition altitude is charted and climbing; transition level is descending and often ATC-assigned. Regional specifics are in FOM chapters 19 to 24, for example the Oakland FIR 5500 ft crossover at 20.7.1.

> **FOM Ch 19-24 / Jeppesen**
>
> *Workbook wording: What (altimeter setting) defines a “level”?*

DESCENT AND APPROACH

What altimeter setting defines an altitude?

In the FOM theater chapters and the Jeppesen pages

A level is flown on standard 29.92 (QNE); an altitude is flown on local QNH; QFE references height above field elevation. Transition altitude is charted and climbing; transition level is descending and often ATC-assigned. Regional specifics are in FOM chapters 19 to 24, for example the Oakland FIR 5500 ft crossover at 20.7.1.

> **FOM Ch 19-24 / Jeppesen**
>
> *Workbook wording: What (altimeter setting) defines an “altitude”?*

DESCENT AND APPROACH

What is QFE?

In the FOM theater chapters and the Jeppesen pages

A level is flown on standard 29.92 (QNE); an altitude is flown on local QNH; QFE references height above field elevation. Transition altitude is charted and climbing; transition level is descending and often ATC-assigned. Regional specifics are in FOM chapters 19 to 24, for example the Oakland FIR 5500 ft crossover at 20.7.1.

> **FOM Ch 19-24 / Jeppesen**

Manually Perform Descent and Arrival (optional)

DESCENT AND APPROACH

Is the autopilot required on an RNAV1 STAR?

No, but recommended in high workload environments

FOM 2.3.9.2 makes autopilot use a recommendation, not a requirement, and FOM 2.3.9.3 encourages hand flying in lower workload terminal environments to keep proficiency. The one place the FOM mandates automation for PBN is 5.1.4.1: RNP AR procedures with RNP values less than 0.3 or with RF legs shall use autopilot or flight director driven by the RNAV system. FCOM NP.21 arrival and approach procedures list LNAV/VNAV modes but never require the autopilot be engaged on a STAR. Verify any airport-specific requirement in the FMS Pilot Guide or the 10-7 before assuming.

> **FOM 2.3.9.2 Autopilot Basic Operating Procedures**
>
> *“In high workload, high traffic environments, autopilot use is strongly recommended.”*
>
> INFERRED from absence: no autopilot requirement for an RNAV 1 STAR exists anywhere in FOM, FCOM or QRH. The only PBN autopilot mandate is FOM 5.1.4.1, and that applies to RNP AR only.
>
> *Workbook wording: Is the autopilot required to be engaged on an RNAV1 STAR?*

8.2.1.5 Conduct Approach Briefing

DESCENT AND APPROACH

What is in the arrival and approach TPC briefing?

Threats, Plan, Considerations. The PF starts it before descent

Prior to beginning the descent the PF initiates the approach briefing, which should be interactive, concise and focused on the big picture. T, Threats, shared: the PM identifies relevant threats using the threat list on the Crew Briefing Reference Card and any airport-specific threats in the 10-7, and the PF briefs additional threats as needed. P, Plan, PF: route including STAR, approach and approach mode; relevant altitude and airspeed constraints and approach minimums; fuel and route to alternate; missed approach; landing runway, assessment, exit and taxi; autobrakes; flaps and approach speed. C, Considerations, PF: specific PM duties based on threats and plan, any other issues, and a recap of critical threats as desired.

> **FOM 5.5.16 Threat-Based Approach Briefing/TPC**
>
> *“"Pilots jointly identify any relevant threats, followed by the PF presenting the arrival plan"”*
>
> Threats is the only shared element; Plan and Considerations are PF. Management strategies are discussed as threats are identified rather than saved to the end. The departure equivalent is briefed the same way, see FOM 1.3.1 in the workbook.
>
> *Workbook wording: What items compose the Arrival and Approach TPC Briefing?*

8.1.1.5 Configure Airplane for Approach/Landing [CRITICAL OBSERVABLE ITEM]

Flap and Gear Management

DESCENT AND APPROACH

Normal flap and gear schedule on approach?

In the FCTM

Handling technique: rotation rates and target pitch, bounced landing recovery, reverser use and stowing, carbon brake technique, flap and gear schedule, autothrottle policy on approach. All FCTM R9 plus FCOM NP.21. FCTM is not in the browser environment.

> **FCTM R9 / FCOM NP.21**
>
> *Workbook wording: What is the normal flap/gear approach configuration schedule?*

Stabilized Approach Criteria

DESCENT AND APPROACH

Where are the stabilized approach criteria?

Fleet-specific manuals; FOM only points you there

FOM 5.6.5 states the philosophy (speed, descent rate, lateral/vertical guidance, configuration), the WARNING not to land from an unstable approach, and the Alaska no-fault go-around policy, then routes 717/787/A321/A330 crews to fleet-specific manuals for the actual criteria. It adds that an approach is stabilized only if all criteria are met, that the aircraft must normally be aligned with the runway by 500 ft AFE, and that any planned deviation must be briefed. The gate parameters themselves are not in the FOM, FCOM or QRH.

> **FOM 5.6.5 Stabilized Approach**
>
> *“(717/787/321/330) Comply with the stabilized approach criteria and procedures contained in the fleet-specific manuals. An approach is stabilized only if all criteria are met.”*
>
> FOM 5.6.5 carries the policy and the warning but explicitly defers the numeric criteria to the fleet manual (787 FCTM/AOM). The one hard number the FOM does give is runway alignment by 500 ft AFE, with 300 ft AFE exceptions annotated on the airport 10-7.
>
> *Workbook wording: Where is the Stabilized Approach Criteria parameters located?*

TAXI AND GROUND

What is required at each stabilized approach gate?

In the FOM approach policy and the FCOM

FOM 5.6.5 Stabilized Approach sets the standard and the no-fault go-around policy: do not land from an unstable approach, and either crewmember may call for a go-around. 5.6.6 covers visual approaches, 5.6.1 approaches authorized, 5.5.16 the TPC briefing. Configuration schedules, callouts and FMA behaviour are FCOM NP.21 and FCTM.

> **FOM 5.6 / FCOM NP.21**
>
> Not in FOM, FCOM or QRH. The gate-by-gate parameters (1000 ft IMC / 500 ft VMC, speed, sink rate, thrust, configuration) live in the 787 fleet-specific manual (FCTM/AOM), per the deferral in FOM 5.6.5. Do not state numbers from memory.
>
> *Workbook wording: What are the Stabilized Approach requirements at each “gate”?*

DESCENT AND APPROACH

Why go around if not stabilized?

Unstable approaches must not be landed from

FOM 5.6.5 pairs the WARNING with the no-fault go-around policy: either crewmember may call for a go-around and the professional judgment of the flight crew is used to assess the probability of meeting the criteria. Captain-side, the leadership move is to state in the approach brief that you expect and will support an FO go-around call, which removes the hesitation that produces continued unstable approaches. Related: FOM 5.6.22.1 requires touchdown within the first 3000 ft or first third of the runway, which is the downstream consequence of pressing an unstable approach.

> **FOM 5.6.5 Stabilized Approach**
>
> *“Do not land from an unstable approach.”*
>
> This is a WARNING in FOM 5.6.5, not a recommendation. Either crewmember may call the go-around under the Alaska no-fault policy, so an FO go-around call is not a challenge to your authority as PIC.
>
> *Workbook wording: Emphasize the importance of going around if the stabilized approach criteria are not met and the no fault*

8.2.1 Perform CAT 1 ILS Approach

DESCENT AND APPROACH &MIDDOT; WALKTHROUGH ITEM, PERFORM RATHER THAN RECALL

Demonstrate the callouts and configuration steps

In the FCTM

Handling technique: rotation rates and target pitch, bounced landing recovery, reverser use and stowing, carbon brake technique, flap and gear schedule, autothrottle policy on approach. All FCTM R9 plus FCOM NP.21. FCTM is not in the browser environment.

> **FCTM R9 / FCOM NP.21**
>
> *Workbook wording: Discuss/demonstrate the callouts and configuration steps*

WEATHER AND MINIMUMS

Inside the FAF. Minimum visibility?

Once the segment has begun, continue to DA or MDA

For CAT II and SA CAT II, the Final Approach Segment may not begin unless the latest controlling RVR reports for the landing runway are at or above the authorized minimums AND the required airborne equipment is operational AND the ground equipment is operational: localizer and glide slope, an outer marker or DME defining the FAF (a published waypoint or minimum GS intercept altitude fix may substitute), and High Intensity Runway Lights or foreign equivalent. If a below-minimums report arrives after the final approach segment has begun, you may continue to DA/DH or MDA per 5.6.11.

> **FOM 5.6.17.2 CAT II/SA CAT II Operating Limitations**
>
> *“"The Final Approach Segment (FAS) of an IAP may not begin unless the latest controlling RVR reports for the landing runway are at or above the minimums authorized"”*
>
> Note visibility below 1/2 sm is not authorized and shall not be used for approach minimums (5.6.11).
>
> *Workbook wording: What if you are inside the FAF? Minimum visibility?*

DESCENT AND APPROACH

What shows on the FMA with the approach armed?

In the FCTM

Handling technique: rotation rates and target pitch, bounced landing recovery, reverser use and stowing, carbon brake technique, flap and gear schedule, autothrottle policy on approach. All FCTM R9 plus FCOM NP.21. FCTM is not in the browser environment.

> **FCTM R9 / FCOM NP.21**
>
> *Workbook wording: Discuss what is shown on FMA when the approach is armed*

8.3.1 Perform Managed Lateral/Managed Vertical Approach

DESCENT AND APPROACH

Which modes are eligible for a non-precision approach?

IAN or VNAV

IAN drives the APP mode with an ILS-look-alike FAC and G/P presentation, so it is armed with the APP switch; VNAV flies the FMC path with LNAV or LOC as the roll mode. FCOM NP.21.45 adds that IAN is not recommended when the approach has a visual maneuver segment not in the FMC database. FCOM NP.21.48 carries the WARNING that when using LNAV to intercept the localizer, LNAV can parallel the localizer without capturing it and the airplane can then descend on the VNAV path with the localizer not captured. RNAV (RNP) AR approaches use the separate SP.4 procedure, not IAN or VNAV.

> **FCOM SP.4 Instrument Approach - RNAV (RNP) AR (note); NP.21.45 and NP.21.48**
>
> *“Note: For RNAV (GPS) and RNAV (GNSS) procedures use the Landing Procedure - Instrument Approach using IAN or Landing Procedure - Instrument Approach using VNAV in Normal Procedures.”*
>
> FCOM NP.21.45 (IAN) and NP.21.48 (VNAV) are the two non-precision landing procedures. Both require a published GP angle on the LEGS page, an RWxx waypoint at the approach end, or a missed approach waypoint before the runway end. Neither is authorized using QFE.
>
> *Workbook wording: What different approach modes(s) are eligible to perform a non-precision approach?*

8.3.2 Perform Managed Lateral/Selected Vertical Approach (Airbus) (Optional)

DESCENT AND APPROACH

Which approaches are flown managed or selected?

LNAV for RNAV/GPS/VOR/NDB; LNAV or LOC for LOC/SDF/LDA

FCOM NP.21.48 lists the recommended final approach roll modes: LNAV for RNAV, GPS, VOR or NDB; LNAV or B/CRS for a back course; LNAV or LOC for LOC, SDF, LDA, ILS (G/S off) and IGS (G/S off). For intercepting, FCOM allows LNAV, HDG SEL, TRK SEL, HDG HOLD or TRK HOLD. Vertically, the 787 uses VNAV PTH or IAN G/P rather than a selected V/S on final; FOM 5.1.4.3 prohibits V/S on the final approach segment of an RNP AR procedure. A330 habit trap: there is no FINAL APP mode and no managed/selected speed target concept here.

FCOM NP.21.48 Landing Procedure - Instrument Approach Using VNAV

> *"The recommended roll modes for the final approach are: • For an RNAV, GPS, VOR or NDB approach use LNAV • For a B/CRS approach use LNAV or B/CRS"*

Managed/Selected is Airbus vocabulary and does not appear in the 787 FCOM. The equivalent split is FMC-driven modes (LNAV/VNAV) versus MCP-selected modes (HDG SEL, TRK SEL, V/S, FLCH). This is the closest 787 answer to the workbook question.

Workbook wording: What approaches would be flown in Managed/Selected?

8.1.1 Perform Visual Approach

DESCENT AND APPROACH

Can you fly a visual approach at night?

Yes, unless that airport restricts night visuals

FOM 5.6.6.1 permits visual approaches with 1000 ft ceiling and 3 sm visibility or better, VFR cloud clearance, within 35 nm of the destination, under ATC control, with the airport or preceding aircraft in sight. Terrain clearance must always be verified, by visual reference, by a published minimum altitude such as MSA or MEA, or by flying the lateral and vertical path of a published instrument procedure. FOM 5.6.6.3 requires you to tune, identify and monitor the approach aid and to stay at or above the ILS glide slope, VASI, PAPI or PLASI when available. Stations with a night visual restriction are flagged in the 10-7.

FOM 5.6.6.1 Visual Approach or RNAV Visual Flight Procedure (without CVFP)

> *"Airports that restrict night visual approaches require that an instrument approach procedure be flown to guarantee adequate terrain clearance."*

Night restrictions are published in the airport 10-7 (see FOM 5.6.6.3 note). FOM 5.6.6.1 also tells you to consider terrain a threat any time a night visual is planned or even possible.

Workbook wording: Can we fly a visual approach at night?

9.1.1 Perform Manual Landing [CRITICAL OBSERVABLE ITEM]

Airspeed Control

DESCENT AND APPROACH

Is the autothrottle required on approach and landing?

Not a published requirement. FCTM technique

Handling technique: rotation rates and target pitch, bounced landing recovery, reverser use and stowing, carbon brake technique, flap and gear schedule, autothrottle policy on approach. All FCTM R9 plus FCOM NP.21. FCTM is not in the browser environment.

> **FCTM R9 / FCOM NP.21**
>
> *Workbook wording: Is the auto-thrust/auto-throttle system required during approach and/or landing?*

LANDING AND ROLLOUT

What would make you turn the autothrottle off?

In the FCTM

Handling technique: rotation rates and target pitch, bounced landing recovery, reverser use and stowing, carbon brake technique, flap and gear schedule, autothrottle policy on approach. All FCTM R9 plus FCOM NP.21. FCTM is not in the browser environment.

> **FCTM R9 / FCOM NP.21**
>
> *Workbook wording: What factors should be considered if deciding to turn off the auto-thrust/auto-throttle system for the*

Centerline Drift / Crosswind Control

COLD WEATHER AND RUNWAY CONDITION

Where is crosswind landing guidance?

FCOM Limitations crosswind tables; FOM Wind Limits

FCOM L.10.7 gives 40 kt crosswind for RwyCC 6 and 5, 35 kt for 4, 25 kt for 3, 17 kt for 2, 15 kt for 1, and no landing at RwyCC 0, with a 5 kt reduction on wet or contaminated runways whenever asymmetric reverse thrust is used, and winds measured at 33 ft tower height for runways 148 ft or wider. The Non-TALPA table gives 40 kt dry and wet, 20 kt standing water or slush, 35 kt snow with no melting, 17 kt ice with no melting. FOM 8.4.4 adds the policy layer: 50 kt maximum steady-state wind, steady-state near the crosswind limit with higher gusts is unacceptable, and the Captain may reduce limits for MELs, gusts, runway width, braking action, weather and currency.

> **FCOM L.10.7 Landing Crosswind Limitations - TALPA**
>
> *"*** Sideslip only (zero crab) landings are not recommended with crosswind components in excess of 25 knots."*
>
> FCOM L.10.7 carries both the TALPA table (keyed to Runway Condition Code / braking action) and the Non-TALPA table. Landing technique itself (crab, decrab, sideslip) is in the 787 FCTM Landing chapter, which is not one of these three manuals.
>
> *Workbook wording: Where can you find guidance on landing the aircraft in a crosswind?*

Thrust Reverse Usage

LANDING AND ROLLOUT

Is max reverse required on landing?

Not a published requirement. FCTM technique

Handling technique: rotation rates and target pitch, bounced landing recovery, reverser use and stowing, carbon brake technique, flap and gear schedule, autothrottle policy on approach. All FCTM R9 plus FCOM NP.21. FCTM is not in the browser environment.

> **FCTM R9 / FCOM NP.21**
>
> *Workbook wording: Is max thrust reverser required upon landing?*

LANDING AND ROLLOUT

When do you stow the thrust reversers?

In the FCTM

Handling technique: rotation rates and target pitch, bounced landing recovery, reverser use and stowing, carbon brake technique, flap and gear schedule, autothrottle policy on approach. All FCTM R9 plus FCOM NP.21. FCTM is not in the browser environment.

> **FCTM R9 / FCOM NP.21**
> *Workbook wording: When should the thrust reversers be stowed?*

Brakes Usage

LANDING AND ROLLOUT

Braking technique with carbon brakes?

In the FCTM ground operations chapter

Turning radii, oversteering technique, breakaway thrust, differential braking and carbon brake technique are FCTM Ground Operations content. FCTM R9 is not in the browser environment. Verify on desktop before quoting any figure.

> **FCTM Ground Operations**
> *Workbook wording: What braking technique is used with carbon brakes?*

LANDING AND ROLLOUT

Factored vs unfactored landing data?

Factored adds margin. The 787 adds 15 percent

Unfactored landing distance is the raw computed distance. Factored distance applies a safety margin on top. Rev 125 moved the 787 to a TALPA landing model using a 7-second air/flare distance and a 15 percent margin on TOTAL landing distance including air distance, calculated to a computed rather than fixed touchdown point. The 737 LTP calculation shows the same principle: minimum landing distance includes air-run to touchdown plus ground roll plus a 15 percent safety factor for degraded braking action.

> **FOM 5.6.22.2 / 5.6.22.3 Landing Performance**
> *""this distance includes air-run to touchdown plus ground roll and a 15% safety factor for degraded braking action requests""*
> DESKTOP: read 5.6.22.3 in full for the 787-specific wording, since Rev 125 removed HA bannering there and added fleet-specific guidance. The quoted 15 percent text above is from the 737 LTP paragraph, so do not attribute it to the 787 model without checking.
> *Workbook wording: What is the difference between factored vs unfactored landing data?*

8.5.1 Perform Go-Around/Missed Approach

Bounced Landing Recovery

DESCENT AND APPROACH

Key to bounced landing recovery without a tail strike?

In the FCTM

Handling technique: rotation rates and target pitch, bounced landing recovery, reverser use and stowing, carbon brake technique, flap and gear schedule, autothrottle policy on approach. All FCTM R9 plus FCOM NP.21. FCTM is not in the browser environment.

> **FCTM R9 / FCOM NP.21**
> *Workbook wording: What is a key component during a bounced landing recovery to avoid a tail strike?*

9.1.1.14 Perform FCOM Procedure(s) Following Landing

LANDING AND ROLLOUT

What are the FCOM procedures after landing?

After Landing, Shutdown, then Secure Procedure

After Landing (NP.21.53): speedbrake down verified by the Captain, APU as needed, engine anti-ice as needed, exterior lights, weather radar off, autobrake OFF, flaps UP, transponder to XPDR, and the COMPANY ARRIVAL REPORT page ON/IN times and landing pilot. Shutdown (NP.21.54): parking brake set, electrical power set, FUEL CONTROL switches CUTOFF, seat belts OFF, hydraulic panel off, fuel pumps off, beacon off, flight directors off, transponder STANDBY, status messages checked (wait about 3 minutes after HYD PRESS SYS L+C+R before logging), then the SHUTDOWN checklist. Secure (NP.21.56): IRS OFF, battery off, FD power off, emergency lights OFF, packs OFF, EFB power off, HUD combiner stowed, then the SECURE checklist.

> **FCOM NP.21.53 After Landing Procedure**
>
> *"Start the After Landing Procedure when clear of the active runway."*
>
> After Landing starts when clear of the runway (FCOM NP.21.53), Shutdown starts after taxi is complete (NP.21.54), Secure is run any time the flight deck will be left unattended (NP.21.56). FOM 5.3.4.3 adds that at controlled airports nothing beyond flaps and speedbrake is done until clear of the runway and taxi instructions are received.

2.1.1.10 Know Single Engine Taxi Procedures

TAXI AND GROUND

Where are the single engine taxi procedures?

Fleet-specific manuals; FOM states the policy only

FOM 5.3.3 makes single-engine taxi the standard procedure when operationally feasible and points to fleet-specific manuals for the procedure. The identical green-policy sentence is repeated at FOM 5.6.22.1 for the taxi-in case. The two pieces the FCOM does supply are the engine cooldown recommendation in the After Landing Procedure (NP.21.53) and the FOM 5.6.33 requirement that the Captain ensure engine cool-down limitations will be met prior to arrival at the gate. Everything else, including which engine to shut down and the restart timing, is fleet-manual material.

> **FOM 5.3.3 Taxi Guidance**
>
> *"When operationally feasible, a single-engine taxi is the standard procedure to optimize fuel efficiency. See fleet-specific manuals for further guidance."*
>
> There is no single-engine taxi procedure anywhere in the FOM, FCOM or QRH. FOM 5.3.3 and FOM 5.6.22.1 both state the green policy and then defer to the fleet manual (787 FCTM/AOM). Pull the actual procedure from there before OE.
>
> *Workbook wording: Where can you find the single engine taxi procedures? Review any differences in SOP.*

TAXI AND GROUND

Engine cool requirement before shutdown?

At least 3 minutes at taxi thrust

FCOM NP.21.53 sets the cooldown at a minimum of 3 minutes at no more than normal all-engine taxi thrust, and this is the clock that governs a single-engine taxi-in shutdown as well as arrival at a short-taxi gate. FOM 5.6.33 puts the responsibility on the Captain to plan for it before arrival and offers the practical fix of slowing or stopping at the top of the lead-in line. Reversers and high thrust during the roll-out reset the clock in practice, so brief the taxi-in when the gate is close to the runway exit.

> **FCOM NP.21.53 After Landing Procedure**
>
> *"Engine cooldown recommendations: • Run the engines for at least 3 minutes • Use a thrust setting no higher than that normally used for all engine taxi operations"*
>
> It is worded as a recommendation, not a limitation, but it drives the taxi-in plan. FOM 5.6.33 makes the Captain responsible for ensuring cool-down will be met prior to arrival, and allows stopping or slowing at the top of the lead-in line so the engines can be shut down immediately at the gate.
>
> *Workbook wording: What is the engine cool requirements before shutting down an engine?*

TAXI AND GROUND

What do you weigh before a single engine taxi?

In the FCOM normal procedures and the FOM

After landing, parking, securing and single-engine taxi flows are FCOM NP.21. FOM policy that applies: 5.3.3 says single-engine taxi is standard when operationally feasible; 16.1.11 covers Pilot Duties With No Flight Attendants; fuel conservation is in Chapter 5 and 8.

> **FCOM NP.21 / FOM Ch 5**
>
> Not in FOM, FCOM or QRH. The decision factors (ramp slope, weight, surface condition, brake energy, hydraulic and electrical configuration, restart time before takeoff) live in the 787 fleet-specific manual, which FOM 5.3.3 defers to. Do not recite a factor list from memory on PV.
>
> *Workbook wording: What factors should you consider before conducting a single engine taxi?*

10.1.1.1 Complete Parking FCOM Procedure

TAXI AND GROUND

Gate arrival procedures

Marshaller or VDGS, two wing walkers, parking brake set

FOM 5.6.33 requires two wing walkers and one marshaller/VDGS operator for the 787. With manual marshalling the signals must be clearly visible to the Captain at all times (hand signals are in FOM 5.3.12). With VDGS, verify the correct aircraft type is displayed before entering the gate area, verify the centerline guidance screen before the nose passes the jet bridge, and stop for WAIT or STOP. If both a VDGS and a marshaller are giving guidance, follow the marshaller. Stop the aircraft any time equipment is inside the Circle of Safety, and once stopped set the parking brake with the FO verifying and confirming it to the Captain.

> **FOM 5.6.33 Parking Requirements (manual or VDGS)**
>
> *"• (787/A330) Two Wing Walkers and One Marshaller/VDGS Operator"*
>
> The 787 requires more people than the narrowbodies: two wing walkers plus one marshaller/VDGS operator. Station exceptions to the two wing walker requirement are published in the airport 10-7.
>
> *Workbook wording: Gate Arrival Procedures*

TAXI AND GROUND

What do you check before taxiing into the gate area?

Correct aircraft type displayed on the VDGS

FOM 5.6.33 gives the VDGS sequence: correct aircraft type displayed before entering the gate area, guidance screen with centerline indicator displayed before the nose passes the jet bridge, and stop on WAIT or STOP. If lead-in line visibility is poor at night or in weather, the crew may request manual marshalling from Station Ops. The Captain should ensure engine cool-down limitations will be met before arrival, slowing or stopping at the top of the lead-in line if needed. The crew shall stop the aircraft any time equipment is inside the safety lines, with narrow exceptions for fueling carts, portable PCA units in painted boxes, and nose gear chocks at the stop bar.

> **FOM 5.6.33 Parking Requirements (manual or VDGS)**
>
> *"Ensure the correct aircraft type is displayed prior to entering the gate area."*
>
> Also confirm required personnel are in place (two wing walkers plus marshaller/VDGS operator for the 787), that the Circle of Safety is clear, and that the engine cool-down will be satisfied. FOM 5.3.3 requires full attention on taxiing in the ramp area at about 10 kt.
>
> *Workbook wording: What should be checked prior to taxiing into the gate area?*

PUSHBACK AND START

No marshaller. Where do you stop?

Stop; marshalling signals must be visible throughout

FOM 5.6.33 lists one marshaller/VDGS operator and two wing walkers as required personnel for the 787, and requires marshalling signals be clearly visible to the Captain at all times under manual procedures. It also states the flight crew may stop the aircraft and request wing walkers at any point for safety or if clearance from an obstruction is in question, and shall stop when any equipment is inside the safety lines (Circle of Safety). The practical Captain answer is to hold short of the gate and get guidance before committing the nose past the jet bridge. Hand signals, including HOLD POSITION and NORMAL STOP, are in FOM 5.3.12.

> **FOM 5.6.33 Parking Requirements (manual or VDGS)**
>
> *"Manual – for standard manual procedures, the marshalling signals shall be clearly visible to the Captain at all times (see 5.3.12 – Hand Signals)."*
>
> The FOM does not name a specific stop point for a missing marshaller. What it gives you is the authority and the trigger: signals must be visible at all times, and the flight crew may stop and request wing walkers at any point for safety or if obstruction clearance is in question.
>
> *Workbook wording: If there is no marshaller or if they are not marshalling you in, where should you stop the aircraft?*

10.1.2 Perform Postflight

TAXI AND GROUND

Perform the securing procedure

Run it whenever the flight deck is unattended

FCOM NP.21.56 flow: IRS selectors OFF, BATTERY switch off, FD POWER switch OFF, EMERGENCY LIGHTS switch OFF, PACK switches OFF, EFB PWR switch off (both), HUD combiner stowed with the caution to ensure it is fully raised and securely latched, then the Captain calls for and the FO does the SECURE checklist. The CAUTION notes the Power Electronics Cooling System needs a cool-down of up to 22 minutes before power is removed to avoid damaging the units and degrading the coolant. Cold weather and other supplementary variants of the Secure Procedure are in FCOM SP.16.

> **FCOM NP.21.56 Secure Procedure**
>
> *"Run the Secure Procedure any time the flight deck will be left unattended."*
>
> Trigger is an unattended flight deck, not end of day. Watch the PECS caution: up to 22 minutes of cool-down may be required before removing power (FCOM SP.6 Electrical Power Down).
>
> *Workbook wording: Perform Securing Procedure (Optional/If not performed, discuss)*

DESCENT AND APPROACH

ACARS post-flight report pages?

All Autoland attempts and all RNP AR procedures

FOM 5.6.21.2 makes the Captain responsible for completing the ACARS Autoland Report page for every Autoland attempt, including the logbook page information; if ACARS/ATSU is inoperative the Captain makes a logbook entry stating Autoland successful or unsuccessful with airport, runway and category, and the reason for any failure. FOM 5.6.21.1 covers the logbook side: an Info to Maintenance entry for any UNSAT Autoland, and for a SAT Autoland only when required by an ACI. FOM 7.1.13.1 requires reporting the results of all RNP AR procedures and any RNP degradation, with a Maintenance Logbook entry when the cause is a system failure, and FOM 7.1.13.2 requires a postflight safety report for any unsatisfactory RNP approach (ANP exceeding RNP, excessive deviation, GPWS warning, autopilot disconnect, or termination due to malfunction). FOM 7.1.14 adds that Autoland and HGS AIII landings are recorded on the ACARS Flight Summary page.

> **FOM 5.6.21.2 Autoland Reporting - ACARS; FOM 7.1.13.1 RNP AR Required Reporting**
>
> *"For all Autoland attempts, the Captain is responsible for ensuring that all applicable fields on the ACARS Autoland Report page are filled in, including the logbook page information."*
>
> Autoland reporting is every attempt, satisfactory or not. RNP AR reporting covers results of all RNP AR procedures plus any RNP degradation, and excludes visual approaches even if an RNP AR approach was displayed as reference (FOM 7.1.13.1 note).
>
> *Workbook wording: ACARS Post-Flight Report (Autoland and RNP-AR Report pages)*

TAXI AND GROUND

Debrief

PF initiates it and asks PM for feedback

FOM 5.6.39 says the postflight debriefing should be interactive, concise and focused on relevant items in order to learn from experiences and facilitate mentoring, and that the PF initiates it by asking the PM for positive and/or constructive feedback, referencing the Crew Briefing Reference Card. Captain-side, this is your main structured mentoring opportunity on the line and it is a graded CRM behavior under AQP, so run it even after an uneventful leg. FOM 25 covers briefing/debriefing as a CRM element, including debriefing performance against SOPs, safety margins and CRM. Flight irregularities follow FOM 11.1.12 and the notification/debrief tables in FOM 18.

> **FOM 5.6.39 Debrief**
>
> *"After the completion of each flight, as workload allows, the PF should initiate the postflight debriefing."*
>
> The PF initiates regardless of seat, so as Captain you may be debriefed by your FO after their leg. Use the Crew Briefing Reference Card. Separate from FOM 11.1.12 Postflight Debriefing/Flight Irregularities, which is the irregularity reporting path.

1.2.1.2.5.3 Know Maintenance Logbook Procedures [CRITICAL OBSERVABLE ITEM]

MEL AND MAINTENANCE

Where are the logbook review procedures?

In the MEL and the FOM

Maintenance logbook review, PDSC validity, MEL placard colours, the four repair interval categories and (O)/(M) procedures are MEL R5 and FOM Chapter 12. Neither the MEL nor the logbook forms are in this environment.

> **MEL R5 / FOM Ch 12**
>
> *Workbook wording: Where in the FOM can you find the maintenance logbook review/checking procedures?*

MEL AND MAINTENANCE

How long is the PDSC good for?

In the MEL and the FOM

MEL placard colours, the four repair interval categories (A, B, C, D), (O) and (M) procedures, crew placarding and PDSC validity are MEL R5 and FOM Chapter 12 content. MEL R5 is not in this environment.

> **MEL R5 / FOM Ch 12**

MEL AND MAINTENANCE

White vs yellow MEL placard?

In the MEL and the FOM

MEL placard colours, the four repair interval categories (A, B, C, D), (O) and (M) procedures, crew placarding and PDSC validity are MEL R5 and FOM Chapter 12 content. MEL R5 is not in this environment.

> **MEL R5 / FOM Ch 12**
>
> *Workbook wording: What is the difference between a white (HAL 46-1333) vs yellow (HAL 46-1333b) MEL placard?*

MEL AND MAINTENANCE

What are the four repair intervals?

In the MEL and the FOM

MEL placard colours, the four repair interval categories (A, B, C, D), (O) and (M) procedures, crew placarding and PDSC validity are MEL R5 and FOM Chapter 12 content. MEL R5 is not in this environment.

> **MEL R5 / FOM Ch 12**
>
> *Workbook wording: What are the 4 different repair intervals*

MEL AND MAINTENANCE

What is an (O) procedure in the MEL?

In the MEL

An (O) beside an MEL item means an Operations procedure must be accomplished by the flight crew for the item to be deferred, as opposed to (M) which requires a Maintenance procedure. MEL R5 is not in this environment; confirm the exact definition and the placarding requirement there.

> **MEL R5**
>
> *Workbook wording: What is an "(O)" procedure listed in an MEL?*

MEL AND MAINTENANCE

Where is the crew placarding procedure?

In the MEL and the FOM

MEL placard colours, the four repair interval categories (A, B, C, D), (O) and (M) procedures, crew placarding and PDSC validity are MEL R5 and FOM Chapter 12 content. MEL R5 is not in this environment.

> **MEL R5 / FOM Ch 12**
>
> *Workbook wording: Where can you find the Crew Placarding Procedure?*

Maintenance Communication Procedure

MEL AND MAINTENANCE

EVF communication requirements with Maintenance?

Call or ACARS Dispatch 30-60 minutes after takeoff, before EEP

FOM 6.5.10 (HA): the crew is notified of an open item on the MIC sheet, Maintenance also verbally notifies the crew and enters AIRCRAFT RELEASED TO ETOPS SERVICE IAW WITH HA ETOPS MAINTENANCE MANUAL FOR VERIFICATION FLIGHT DUE TO REPAIR OF ___ in the logbook, and the crew must verbally acknowledge the EVF requirement with Dispatch. Between 30 and 60 minutes after takeoff and before the EEP, call or ACARS Dispatch with the status; the report and the return acknowledgment must be complete before the EEP. If the item is not operating normally the aircraft cannot proceed beyond the EEP and must land for repairs. The Captain then makes a logbook entry VERIFICATION FLIGHT FOR ___ COMPLETED SATISFACTORILY or UNSATISFACTORILY, listing discrepancies if unsuccessful. The Alaska-side equivalent, FOM 6.5.9, routes results through Dispatch to Maintenance Control and requires notifying Dispatch of satisfactory results prior to passing the EEP.

> **FOM 6.5.10 ETOPS Verification Flight (HA)**
>
> *“Between 30 and 60 minutes after takeoff and prior to reaching the EEP, the Flight Crew will call or send an ACARS message to Dispatch to report the status of the affected component or system.”*
>
> Gotcha: the in-flight report goes to Dispatch, not to Maintenance. Maintenance's role is the up-front verbal notification plus the logbook entry, and the crew verbally acknowledges the EVF requirement with Dispatch. The report AND Dispatch's acknowledgment must both be complete before the EEP.
>
> *Workbook wording: What are the ETOPS verification flight communication requirements regarding contacting Maintenance*

MEL AND MAINTENANCE

Why tell Maintenance enroute rather than at the gate?

In the MEL and the FOM

Maintenance logbook review, PDSC validity, MEL placard colours, the four repair interval categories and (O)/(M) procedures are MEL R5 and FOM Chapter 12. Neither the MEL nor the logbook forms are in this environment.

> **MEL R5 / FOM Ch 12**
>
> *Workbook wording: What are the advantages of notifying MX of a discrepancy enroute (time permitting) vs after blocking in?*

1.1.1.16.4 Know International Procedures (as applicable)

FD Pro ICAO Procedures

EFB AND FD PRO &MIDDOT; WALKTHROUGH ITEM, PERFORM RATHER THAN RECALL

Where in FD Pro are the ICAO country procedures?

In the FD Pro guide in Comply365

This is a Jeppesen FD Pro X or EFB workflow step, not FOM/FCOM content. Verify against the FD Pro Pubs Quick Reference guide in Comply365 and demonstrate on the actual device during OE. FOM 5.7.3.15 does require the ENTIRE cleared route be loaded into the approved charting app for ORCN segments.

> **FD Pro Pubs Quick Reference (Comply365)**
>
> *Workbook wording: Where in FD Pro can you find the ICAO (country) procedures for destinations that we serve?*

EFB AND FD PRO &MIDDOT; WALKTHROUGH ITEM, PERFORM RATHER THAN RECALL

Where in FD Pro are the ICAO emergency procedures?

In the FD Pro guide in Comply365

This is a Jeppesen FD Pro X or EFB workflow step, not FOM/FCOM content. Verify against the FD Pro Pubs Quick Reference guide in Comply365 and demonstrate on the actual device during OE. FOM 5.7.3.15 does require the ENTIRE cleared route be loaded into the approved charting app for ORCN segments.

> **FD Pro Pubs Quick Reference (Comply365)**
> *Workbook wording: Where in FD Pro can you find the ICAO (country) emergency procedures for destinations that we serve?*

EFB AND FD PRO &MIDDOT; WALKTHROUGH ITEM, PERFORM RATHER THAN RECALL

Review the FD Pro Pubs Quick Reference guide

In the FD Pro guide in Comply365

This is a Jeppesen FD Pro X or EFB workflow step, not FOM/FCOM content. Verify against the FD Pro Pubs Quick Reference guide in Comply365 and demonstrate on the actual device during OE. FOM 5.7.3.15 does require the ENTIRE cleared route be loaded into the approved charting app for ORCN segments.

> **FD Pro Pubs Quick Reference (Comply365)**
> *Workbook wording: Review FD Pro Pubs Quick Reference guide in Comply365*

16.1.2 Perform Long Range Communication Procedures

OCEANIC AND NAT

When is an HF position report required?

Only when you are not on CPDLC

Flight Crews should NOT make HF position reports when using CPDLC procedures. When using CPDLC you must contact ARINC after a successful CPDLC logon but prior to the initial Gateway Fix to obtain a SELCAL check on HF, establishing a backup if CPDLC fails. No other HF reports need be made while using CPDLC. Non-CPDLC position reporting varies by area and is specified on the enroute chart.

> **FOM 5.7.3.18 Position Reporting Oceanic (CPDLC)**
> *""Flight Crews should not make HF position reports when using CPDLC procedures.""*
> REV 125.1 ADDED: if SELCAL is not functioning, one member of the crew must listen to HF continuously. That is the new sentence in this revision and a likely oral question. Non-CPDLC reporting is 5.7.3.19; for the Oakland FIR the procedures are in FD Pro X under Pubs, Pacific, Oceanic Position Reporting Procedures.

OCEANIC AND NAT

Can you enter oceanic airspace with both HF radios out?

No. At least one HF radio is required

FOM 20.4.1 sets dispatch minimums of two HF radios or a single HF plus SATVOICE, and after pushback prior to entering oceanic airspace requires dual LRNS plus two HF radios, or a single HF with SATCOM data link, or a single HF with SATVOICE. FOM 7.1.8 shows the same pattern by region: Pacific ETOPS needs 2 HF or 1 HF and SATVOICE; NOPAC and NAT HLA need 2 HF and SATDATA/CPDLC or 1 HF with SATDATA/CPDLC and SATVOICE. FOM 22.4.3 states NAT operations outside VHF coverage require two independent LRCS, one of which must be HF. Narrow exceptions exist where an LRCS is not required at all because VHF coverage exists, and for a few specific routings such as ANC to/from BOS, DTW, JFK, MSP, ORD where SATCOM voice alone is listed. If both HF fail enroute, FOM 20.4.4.1 has you relay position reports on 123.45 through nearby flights, use 121.5 for initial contact if necessary, and contact ARINC directly if SATCOM is installed; ATC and Dispatch must be notified immediately.

> **FOM 20.4.1 Required Equipment for OAK, ANC, and FUK FIRs**
>
> *"After pushback, prior to entering oceanic airspace: • Dual Long Range Navigation Systems (to include 2 MCDUs) • Two HF radios, or – A single HF radio and SATCOM data link, or – A single HF radio and SATVOICE"*
>
> Every authorized combination for OAK/ANC/FUK includes at least one working HF. Same story in the NAT: FOM 22.4.3 requires two independent LRCS, one of which must be HF. FOM 20.4.4.1 covers what to do if both HF fail after you are already established oceanic.
>
> *Workbook wording: Can you enter Oceanic airspace with both HF radios not working?*

16.1.3.2 Analyze and Assess Terrain Critical Routes

CRUISE AND DIVERSION

Western US depressurization strategy?

Fly the charted decompression polygon escape procedure

FOM 11.2.9 covers the western high-terrain depressurization strategy: polygons show where a direct descent to 10,000 ft is not possible before passenger oxygen depletes. Unlike the 737/A330, which always use an initial descent altitude of 17,000 ft/FL170, the 787 takes its initial descent altitude(s) from the polygon detail drawer, and those vary by polygon because of the 787 passenger oxygen system design. Do not enter another polygon below the drawer altitude unless terrain clearance is assured, and never plan through a charted No Ops Area. The PIC is expected to apply judgment; weather, visibility, ATC, and grid MORAs also inform the decision (see 2.2.8.1).

> **FOM 11.2.9 Decompression Polygon Procedures (737/787/A330)**
>
> *"Decompression polygon procedures provide a calculated safe method for descending to 10,000 ft before passenger oxygen supplies are depleted following a rapid decompression over high terrain."*
>
> Polygons live on the Jeppesen FD Pro IFR High/Low charts; click the polygon label to open the detail drawer. The three escape types are Exit Any Dir, Exit Twd <fix>, and To Fix then Route.
>
> *Workbook wording: Western U.S. Depressurization Strategy*

16.1.1.7 Perform ETOPS Procedures [CRITICAL OBSERVABLE ITEM]

APU In-Flight Start Program (APU component currency)

MEL AND MAINTENANCE

What release items matter for the APU in-flight start program?

The APU Administrative Control Item and any conflicting MEL items

Per FOM 12.4.12.5, a Maintenance request for an APU in-flight start is issued as an Administrative Control Item (ACI) found in Section 05 of the fleet MEL, and that ACI is reflected on the Dispatch Release. The other release item that matters is the MEL section: FOM 6.5.6 requires the crew to verify no MEL item conflicts, such as one that requires the APU to run for the entire flight, in which case the MEL supersedes the program and the APU is not shut down. Note the ACI can reappear on a later Dispatch Release generated before the ACI was completed.

> **FOM 12.4.12.5 Maintenance Requests / FOM 6.5.6 APU In-Flight Start Program (HA)**
>
> *“These requests will be in the form of an Administrative Control Item (ACI) as found in Section 05 of the fleet MEL. The associated ACI will be applied to the aircraft and will be reflected on the Dispatch Release.”*
>
> The ACI comes from Section 05 of the fleet MEL. Crew is also notified by an open item on the MIC sheet and a placard card (form HAL-46-1333) in the AML per FOM 6.5.6.
>
> *Workbook wording: What items on the dispatch release must you take note of when planning an APU In-Flight Start Program*

ETOPS

How does that differ from the APU in-flight start program?

EVF confirms a specific repair; the start program is routine currency

FOM 6.5.10 defines an EVF as a flight where the crew assists Maintenance in confirming an ETOPS Significant System works normally or that an ETOPS significant problem has been fixed, and Maintenance makes an AML release entry the crew must verbally acknowledge with Dispatch. FOM 6.5.6, by contrast, is a routine operational test of cold-soaked APU start/run capability, triggered by an open MIC sheet item and a HAL-46-1333 placard card. The EVF version of the APU check also carries its own completion options (remain outside ETOPS airspace, complete before the EEP, or complete after the EXP), and if it is unsuccessful the Dispatcher and PIC must confer because ETOPS entry may not be possible. Logbook wording differs too: EVF uses SATISFACTORILY/UNSATISFACTORILY, the start program uses SUCCESSFUL/UNSUCCESSFUL.

> **FOM 6.5.10 ETOPS Verification Flight (HA) vs FOM 6.5.6 APU In-Flight Start Program (HA)**
>
> *“An ETOPS Verification Flight (EVF) is a revenue or non-revenue flight during which the Flight Crew is asked to assist Maintenance to confirm that an ETOPS Significant System is operating normally or that an ETOPS significant problem has been resolved.”*
>
> EVF adds a required report to Dispatch between 30 and 60 minutes after takeoff and before the EEP; if the item is not operating normally the flight may not proceed past the EEP. The APU In-Flight Start Program has no such Dispatch report requirement.
>
> *Workbook wording: How does this differ from the APU In-Flight Start Program?*

Verification flight vs APU in-flight start program

ETOPS

What do you log during the APU start?

FL, OAT, start attempts, successful/unsuccessful, seconds to available

FOM 6.5.6 gives the exact Aircraft Maintenance Logbook format: APU IN-FLIGHT START, FL ___ (flight level during start attempt), OAT ___ degrees C, START ATTEMPTS ___ , SUCCESSFUL or UNSUCCESSFUL (successful means a start within 3 attempts), and AVAIL AT ___ SECONDS, the time from the beginning of the start sequence until the APU is available. The Captain owns this entry and must list discrepancies on an unsuccessful test. Do not confuse this with the EVF entry in FOM 6.5.10, which reads VERIFICATION FLIGHT FOR ____ COMPLETED SATISFACTORILY or UNSATISFACTORILY, or for the APU specifically VERIFICATION FLIGHT FOR APU COMPLETED, APU OPERATES NORMALLY.

> **FOM 6.5.6 APU In-Flight Start Program (HA)**
>
> *"APU IN-FLIGHT START FL ___ (flight level during start attempt) OAT ___°C (Outside Air Temperature)"*
>
> The Captain makes the AML entry, and must list any discrepancies if the test was unsuccessful. The full entry adds START ATTEMPTS ___ , SUCCESSFUL or UNSUCCESSFUL, and AVAIL AT ___ SECONDS.
>
> *Workbook wording: What items must be logged/noted during the APU start?*

MEL AND MAINTENANCE

What goes in the logbook after the flight?

In the MEL and the FOM

Maintenance logbook review, PDSC validity, MEL placard colours, the four repair interval categories and (O)/(M) procedures are MEL R5 and FOM Chapter 12. Neither the MEL nor the logbook forms are in this environment.

> **MEL R5 / FOM Ch 12**
>
> *Workbook wording: What items must be entered into the maintenance logbook after flight completion?*

MEL AND MAINTENANCE

What aircraft currency items are the ACIs?

Max thrust takeoff, autoland, APU in-flight start, anti-ice check

FOM 12.4.12.5 lists the Maintenance requests that arrive as Administrative Control Items: max thrust takeoff, (787/A321/A330) autoland and APU in-flight start, and (717) anti-ice system functional check. The ACI is applied to the aircraft, appears on the Dispatch Release, and the crew complies by following the procedure written in the ACI in Section 05 of the fleet MEL. Related: FOM 7.1.14 covers autoland and HGS reporting, and the autoland logbook entry is required when the autoland was satisfactory and was required by an ACI.

> **FOM 12.4.12.5 Maintenance Requests**
>
> *"• Max thrust takeoff • (787/A321/A330) Autoland, APU in-flight start • (717) Anti-ice system functional check"*
>
> For the 787 the applicable ACIs are max thrust takeoff, autoland, and APU in-flight start; the anti-ice functional check is a 717 item. ACIs are found in Section 05 of the fleet MEL.
>
> *Workbook wording: What A/C Currency Items are listed in the "Administrative Control Items" (ACI's)?*

ETOPS

How many start attempts are allowed?

Three

FOM 6.5.6 allows a maximum of three start attempts; a start within three attempts is successful. The first attempt is made during the 1-hour period prior to top of descent after at least a two-hour cold soak at ETOPS cruise altitude, and follow-on attempts may be flown at lower altitude but no lower than FL270. After the start sequence completes or after three failed attempts the test is over and the APU may be shut down. FOM 5.1.8.1 mirrors this: the APU is considered failed to start after three consecutive failed attempts or when fleet procedures preclude further attempts.

> **FOM 6.5.6 APU In-Flight Start Program (HA)**
>
> *"• If the APU starts within the maximum allowed three attempts, the start is deemed to be successful."*
>
> If the first attempt fails, subsequent attempts may be made at lower altitudes, minimum FL270. FOM 5.1.8.1 uses the same three-attempt rule to define an APU start failure.
>
> *Workbook wording: How many start attempts can be made?*

Flight crew actions prior to the EEP

ETOPS

What steps are required?

ETOPS alternate forecast at crew minima; complete ETOPS verification checks

FOM 6.3.2 lists what is required before crossing the EEP. For the 787 the two that apply are forecast weather at the ETOPS alternate at or above the operating crew's minima (accounting for crew minima and any MEL), and completion of any ETOPS verification flight checks. These crew weather requirements are deliberately less restrictive than the criteria Dispatch used at planning. Also check FOM 20.4.1 for required equipment in the OAK, ANC, and FUK FIRs. After the EEP the crew monitors distance from an ETOPS alternate to stay inside the area of operation, at the EXP completes APU verification steps as required, and before top of descent performs the APU start if required.

> **FOM 6.3.2 Flight Crew Duties, Prior to Crossing ETOPS Entry Point (EEP)**
>
> *"• Forecast weather at the ETOPS alternate must be at or above crew's operating minima (this is the weather required for the operating crew to conduct the approach, given the crew's minima and any applicable MEL)."*
>
> The 737-only bullets (APU/generator status, two engine driven generators) do not apply to the 787. CAT II/III, RNAV/GPS/RNP and RNP AR approaches may be used for the alternate suitability determination.

ETOPS

How is the EEP determined and shown to the crew?

60 minutes from an adequate airport, 410 nm; listed on the release

Per FOM 6.3.3 the EEP is the point 60 minutes from an adequate airport at the approved one-engine-inoperative cruise speed in still air under standard conditions; for the 787 that is 410 nm (FOM 6.2.3 ETOPS Area of Operations table). The EXP is the same fixed distance on the way out. Depiction to the crew is threefold: the adequate airports defining EEP/EXP are listed on the Dispatch Release, ETOPS-specific waypoints are entered and monitored using fleet procedures and crosschecked (lat/longs on the route, range rings centered and spaced correctly), and Figure 6.3.3(1) shows the 60-minute green rings. Any change of the defining adequate airports must be coordinated with Dispatch, and the crew must identify arrival at EEP/ETP/EXP.

> **FOM 6.3.3 ETOPS Entry/Exit Point/Equal Time Point**
>
> *"When approaching the ETOPS area of operation, the aircraft reaches the ETOPS Entry Point (EEP) when it is 60 minutes from the adequate airport."*
>
> 787 60-minute distance is 410 nm at .84M/290 KIAS per the FOM 6.2.3 table. Adequate airports defining EEP/EXP must be listed on the Dispatch Release, and crew and Dispatch must use the same ones.
>
> *Workbook wording: How is the EEP determined and depicted to the flight crew?*

ETOPS

Max diversion distance for ETOPS 120 and 180?

806 nm at 120 minutes, 1207 nm at 180 minutes

Table 6.2.3(1) in FOM 6.2.3 lists the 787 at .84M/290 KIAS single engine speed, giving 410 nm for 60 minutes, 806 nm for 120 minutes, and 1207 nm for 180 minutes. FOM 6.2.9 explains these Maximum Diversion Distances come from an FAA-approved single-engine cruise speed strategy using a reference weight of 515,000 lb and reference FL350 for the 787; actual planned weights are used for dispatch planning. Note a reroute can push the flight out of ETOPS 180 airspace, so monitor distance from the alternates after the EEP (FOM 6.3.2).

> **FOM 6.2.3 ETOPS Area of Operation, Table 6.2.3(1)**
>
> *"787 .84M/290 KIAS 410 nm 806 nm 1207 nm"*
>
> Same table gives the 60-minute EEP/EXP distance of 410 nm. These are Maximum Diversion Distances derived from the approved one-engine-inoperative cruise speed (FOM 6.2.9).
>
> *Workbook wording: What is the maximum diversion distance, ETOPS 120? ETOPS 180?*

ETOPS

ETOPS diversion speed?

.84M/290 KIAS, the one-engine-inoperative cruise speed

FOM 6.4.5 ties the ETOPS diversion speed to the approved One-Engine-Inoperative Cruise Speed in FOM 6.2.3, which for the 787 is .84M/290 KIAS. That speed is what the critical fuel scenario, the EEP/EXP distance, and the Maximum Diversion Distance are all built on, so flying materially off it invalidates the fuel plan. The Captain retains authority to deviate from the planned profile after assessing the emergency and fuel remaining, which is the Captain-side judgment call worth rehearsing. For a decompression diversion with both engines running, use the flight plan one-engine diversion speed or LRC once level.

> **FOM 6.4.5 Diversion Speed Strategy / FOM 6.2.3 ETOPS Area of Operation**
>
> *"Engine inoperative ETOPS fuel requirements are based upon the approved One-Engine-Inoperative Cruise Speed (see 6.2.3 – ETOPS Area of Operation), which should be followed in case of a diversion following an actual engine failure."*
>
> The Captain may deviate from the planned speed profile after assessing the emergency and fuel remaining. For a two-engine decompression diversion the flight plan one-engine diversion speed or LRC is used when level (FOM 6.4.3).
>
> *Workbook wording: ETOPS Diversion speed?*

15.2.1 Demonstrate Dangerous Goods Policies and Procedures

HAZMAT

Where is the dangerous goods policy in the FOM?

The HAZMAT chapter, under Dangerous Goods

Dangerous goods policy lives in FOM Chapter 14, HAZMAT, section 14.1 Dangerous Goods, starting with 14.1.1 Regulatory Compliance (Ops Spec A055, 49 CFR Parts 171-175, IATA DGR). Key subsections for a 787 Captain: 14.1.4 eNOTOC (HA), 14.1.6 Captain Responsibilities (HA), 14.1.8 NOTOC acceptable versions (HA), 14.1.17 rejected shipments (HA), 14.1.20 ICAO ERG drill codes, and 14.1.26 damaged or leaking DG. Related material sits outside Chapter 14 as well: PED lithium batteries in FOM 13.4.1, the PED battery/fire containment bags in FOM 11.1.10, and PIC notification of onboard DG in FOM 17.2.7.1. No hazardous materials may be carried in the Flight Deck other than excepted items required for operation or personal HAZMAT allowances.

> **FOM 14.1 Dangerous Goods (Chapter 14 HAZMAT)**
>
> *"• The Company accepts Dangerous Goods (DG) in accordance with International Air Transport Association (IATA) Dangerous Goods Regulations and the domestic Code of Federal Regulations (CFR) Title 49."*
>
> FOM Chapter 14 is titled HAZMAT and section 14.1 is Dangerous Goods, running 14.1.1 through 14.1.28. eNOTOC and HA Captain responsibilities are 14.1.4 and 14.1.6.
>
> *Workbook wording: Where can the dangerous goods policy be found in the FOM?*

HAZMAT

What is an eNOTOC?

The electronic Notice to Captain of dangerous goods, sent by ACARS

Per FOM 14.1.4 the eNOTOC is the Hawaiian electronic Notice to Captain for dangerous goods, uplinked at departure minus 30 minutes for trans-Pacific and international flights and departure minus 15 minutes for inter-island, and it prints automatically. FOM 14.1.6 makes the Captain responsible for reviewing it for correct departure airport, flight number, and tail number, with no hand edits permitted (unlike the AS NOTOC, where the Ramp Lead may hand edit tail number and DG location per 14.1.5). Retain the printed eNOTOC and keep it readily available in flight to pass HAZMAT type, quantity, and location to ARFF. If digital acknowledgment is not possible, use the hard-copy procedures in FOM 14.1.23 or the manual NOTOC HAL 902 procedures in 14.1.23.1.

> **FOM 14.1.4 eNOTOC (HA)**
>
> *"The eNOTOC will be sent to the Flight Crew at departure time minus 30 minutes for trans-Pacific and international flights and at departure time minus 15 minutes for inter-island flights."*
>
> It prints automatically on all aircraft. Versions per FOM 14.1.8: eNOTOC (primary), hard-copy eNOTOC from the Load Master if ACARS/printer is inop but FM Amadeus works, and HAL Form 902 (freighter DG COMAT only).

HAZMAT

How do you acknowledge an eNOTOC?

Select ACKNOWLEDGE NOTOC on the Comm page

FOM 14.1.6 requires the HA Captain to acknowledge the eNOTOC by selecting ACCEPT* on the MCDU or ACKNOWLEDGE NOTOC on the Comm page, depending on fleet type, which on the 787 means the MFD COMPANY/communications page. Acknowledgment is a Captain duty alongside reviewing the eNOTOC for correct departure airport, flight number, and tail number, with no hand edits permitted, and retaining the printed copy for ARFF. If ACARS or the printer is inoperative but FM Amadeus is operative, the Load Master provides a printed eNOTOC and FOM 14.1.23 hard-copy procedures apply; if FM Amadeus is inoperative, use the manual NOTOC HAL 902 procedures in 14.1.23.1. Refusal procedures are 14.1.17.1 through 14.1.17.3.

> **FOM 14.1.6 Captain Responsibilities (HA)**
>
> *"• Acknowledging the eNOTOC by selecting either "ACCEPT*" on the MCDU or "ACKNOWLEDGE NOTOC" on the Comm page (depending on fleet type)."*
>
> ACCEPT* on the MCDU is the Airbus path; the Boeing 787 uses ACKNOWLEDGE NOTOC on the Comm page. If digital acknowledgment is impossible see FOM 14.1.23 and 14.1.23.1.

HAZMAT

How do you fight a lithium battery fire?

HALON, then water. Never ice

FIGHT THE FIRE with a HALON extinguisher. Water may be used to fight a lithium ION battery fire but should NOT be used on a lithium METAL battery fire. It is very important to prevent the fire spreading to adjacent flammable materials. Then COOL THE DEVICE using water preferably, or any non-alcoholic liquid if water is not available (milk, fruit juice, POG). AVOID USING ICE, which acts as an insulator; if the device is insulated before it is fully cooled, remaining cells that have not reached thermal runaway could ignite. If the fire is on the Flight Deck the device must immediately be removed by any means available (gloves, blanket).

> **FOM 11.1.10.2 PED Fire Containment Bag (Yellow Bag)**
>
> *""AVOID USING ICE to cool the device. Ice will act as an insulator.""*
>
> It may take 15 or more minutes to completely cool the device. REV 125.1: this is the YELLOW bag section, renumbered from 11.1.10. The new RED bag procedure at 11.1.10.1 is a containment procedure, not a firefighting one: gloves, device into the bag, pull the red tab, seal the Velcro, zip, and place it outside the Flight Deck Door. HA aircraft switch yellow to red from October 1.

HAZMAT

Where are the cabin lithium battery limits?

The Alaska Airlines onboard PED and prohibited-items web pages

FOM 13.4 sends you to the alaskaair.com onboard-policies portable electronic devices page and the prohibited-items page for the list of permitted and restricted devices and battery types/sizes. FOM 13.4.1 states the operative rules directly: checked bags may contain lithium batteries only when installed in a device, spare lithium batteries and external lithium-ion battery chargers may never go in checked baggage, and carry-on may contain lithium batteries whether installed or spare, with spare terminals covered or insulated in original retail packaging or non-conductive tape. Mobility device lithium-ion batteries may be carried in the cabin under FOM 13.6/13.6.1, with PIC verbal or SSR notification but no NOTOC. For a thermal event in the cabin see FOM 11.1.10 PED Lithium Battery Containment Bag (red) and PED Fire Containment Bag (yellow).

> **FOM 13.4 Portable Electronic Devices / FOM 13.4.1 Lithium Batteries in PEDs**
>
> *"See the list of portable electronic devices that are permitted and restricted on board at: https://www.alaskaair.com/content/travel-info/policies/onboard-policies-portable-electronic-devices"*
>
> The FOM itself does not publish watt-hour numbers; it points to the alaskaair.com PED policy and prohibited-items pages. FOM 13.4.1 gives the carry-on vs checked rules, and 11.1.10 covers the containment bags.
>
> *Workbook wording: Where can you find lithium battery type/size limitations to be carried in the cabin?*

16.1.2.1.3 Know ACARS Procedures

COMMUNICATIONS AND PA

Review the ACARS menus and functions

ATC, FLIGHT INFORMATION, COMPANY, REVIEW, MANAGER, NEW MESSAGES

FCOM 5.40.2 defines the six MFD communications top-level menus. ATC gives logon/status and downlinks to ATC. FLIGHT INFORMATION gives Departure Clearance, Oceanic Clearance, ATIS, Pushback Clearance, TWIP, and Expected Taxi Clearance requests. COMPANY gives airline downlinks: FLIGHT INITIALIZATION, REQUEST AUTO-INITIALIZATION, DELAY REPORTS, DIVERSION, ETA REPORT, FLIGHT TIMES, REMINDERS, DEPARTURE/ARRIVAL REPORT, MESSAGE TO GROUND, ATC DATALINK PROBLEM REPORT, WEATHER REQUESTS, CREW REQUESTS (flight release, NOTAMs, weight and balance, gate assignment), MAINTENANCE REPORT, MISCELLANEOUS CODES, SITUATION, SENSOR STATUS, and E-MAIL (FCOM 5.40.68). REVIEW lists transmitted and already-handled messages and is inhibited when empty. MANAGER holds ACARS, VHF, HF, SATCOM, AOIP, ADS, System Information, Automatic Messages, and Master managers plus the Communications Audit (FCOM 5.40.72 to 5.40.88). NEW MESSAGES lists undisplayed uplinks and is inhibited when there are none. AeroData performance is reached with PERFORMANCE on the Company menu (FCOM PI.19.1).

> **FCOM 5.40.2 Communications - MFD Communications Functions, Communications Menus**
>
> *"Selecting ATC, FLIGHT INFORMATION, COMPANY, REVIEW, MANAGER, or NEW MESSAGES selection: • places the appropriate title in the menu heading line • displays the subordinate menu selections for that function in the menu/data area"*
>
> Selectable items are white text on gray; inhibited items are cyan on black with a cyan border and cannot be selected. A selected top-level function turns green, and three dots (...) mean another subordinate menu follows.
>
> *Workbook wording: Review ACARS menus and quiz them on where to find specific functions to demonstrate proficiency.*

COMMUNICATIONS AND PA

Where is the directory of ACARS functions?

The FCOM Communications chapter, MFD Communications Functions

FCOM Chapter 5 Section 40, MFD Communications Functions, is where the ACARS/datalink function set is documented, and its table of contents at 5.TOC.2 to 5.TOC.4 lists each function by page: Communications Menus 5.40.2, Command Key Functions 5.40.5, Message List 5.40.16, ATC Datalink 5.40.23, the downlink pages 5.40.24 to 5.40.42, ATC Reports 5.40.46, Flight Information pages through 5.40.66, Company Menu 5.40.68, Review Menu 5.40.70, Manager Functions 5.40.72 to 5.40.88, and New Messages 5.40.90. Onboard, the Company menu itself is the practical directory of airline functions, and MISCELLANEOUS CODES on that menu carries the coded requests. Company functions are airline-customizable, so the FCOM illustrations are described as a typical installation and some items may be inhibited.

> **FCOM 5.40 Communications - MFD Communications Functions (see 5.TOC table of contents)**
>
> *“The MFD communications functions are used to control datalink features. Datalink messages not processed by the FMC are received, accepted, rejected, reviewed, composed, sent, and printed using communications functions on the MFD.”*
>
> The Communications chapter table of contents (5.TOC.2 through 5.TOC.4) is the closest thing to a directory: it indexes every menu and page from Communications Menus through Manager Functions and New Messages.
>
> *Workbook wording: Where can you find a list/directory of ACARS FUNCTIONS?*

COMMUNICATIONS AND PA

Where is the spare ACARS paper?

In the FCOM communications chapter and the FOM

ACARS menu structure, function directory, SATCOM keying sequence, preprogrammed numbers and printer paper are aircraft and ACARS documentation, best demonstrated on the flight deck during OE. FOM 7.1.7 covers SATCOM policy and 7.1.12 Reports to Dispatch.

> **FCOM Ch 5 Communications / FOM 7.1.7**
>
> Not in the FOM, FCOM, or QRH. The FOM only addresses replacing it: FOM 12.2.6 Consumables Replacement - Flight Deck allows the crew to replace ACARS printer paper and requires an Info to Maintenance logbook entry so spare paper is restocked. The physical stowage location is aircraft-specific and is not published in these three manuals; check the flight deck placarding, the AOM, or ask Maintenance.
>
> *Workbook wording: Where is the spare ACARS paper located on the flight deck?*

COMMUNICATIONS AND PA

Where are the instructions to change ACARS paper?

In the FCOM communications chapter and the FOM

ACARS menu structure, function directory, SATCOM keying sequence, preprogrammed numbers and printer paper are aircraft and ACARS documentation, best demonstrated on the flight deck during OE. FOM 7.1.7 covers SATCOM policy and 7.1.12 Reports to Dispatch.

> **FCOM Ch 5 Communications / FOM 7.1.7**
>
> No paper-loading procedure exists in the FOM, FCOM, or QRH. FCOM 5.10 Communications Controls and Indicators only describes the Printer Cover Release Lever (push to release the printer cover and handset cradle for access to the printer paper door) and the Printer Paper Door Lever (push to open the printer paper door). Step-by-step loading instructions are on the printer placard/decal at the unit or in the AOM.
>
> *Workbook wording: Where are the instructions on how to change the ACARS paper?*

8.3.5.23 [PM] Demonstrate Ability to Correctly Program Aircraft Guidance Systems Including Approach

Procedure selection, waypoint coding, navigation performance

OCEANIC AND NAT

Diversion. How do you select a new destination?

ALTN page, select the airport, then DIVERT NOW, then execute

Per FCOM 11.43.24 to 11.43.27, ALTN page 1/2 shows four alternate airports in ETA sequence, sourced by uplink, from the ALTN LIST page 2/2, automatically from the nav database (at least one runway over 6,500 ft), or by manual entry. Select the right LSK to open that airport's XXXX ALTN page, then DIVERT NOW builds an LNAV route modification from present position to that alternate using the route shown on the XXXX ALTN page and displays the MOD XXXX ALTN page. Confirm or modify, then execute. Execution changes the destination, deletes non-diversion legs, and deletes all descent constraints, so rebuild the arrival and approach and expect to re-run landing performance. Alternates show on the ND in cyan with an A in the airport symbol. Captain-side, coordinate with Dispatch before executing; FOM 6.4.4 requires an updated Flight Plan/Dispatch Release before the EEP if one is needed, and FOM 11.2.7.1 covers nearest suitable airport.

> **FCOM 11.43.27 Flight Management, Navigation - FMC Descent and Approach, Alternate Page, DIVERT NOW**
>
> *"Selecting DIVERT NOW displays the route from the present position to the selected alternate using the route displayed on the XXXX ALTN page for the diversion airport."*
>
> Execution changes the route destination, folds the modification into the active flight plan, deletes original route legs not part of the diversion, and deletes all descent constraints (CDU shows DESCENT PATH DELETED). DIVERT NOW is blank on the ground.
>
> *Workbook wording: Diversion Procedures/Select New Destination*

OCEANIC AND NAT

What is the alternate navigation system?

TCP backup navigation after loss of all displays or all FMCs

FCOM 11.50 describes the alternate navigation system as a backup capability hosted in the Tuning and Control Panels, reached by pushing the NAV key on any TCP; on standby power only the left (Captain's) TCP works. It runs two pages, ALTN NAV and ALTN NAV RADIO. The TCPs carry no performance or navigation database, so the crew manually enters a single waypoint latitude and longitude at LSK 2 L/R on the ALTN NAV page and then gets present position, TRK TO, GTK, GS, DTG, and TTG to that waypoint; the waypoint must be manually updated as the flight progresses. LNAV and VNAV are unavailable, though the autopilot and autothrottles may still be usable, and raw attitude/airspeed/altitude come off the ISFD. This is a real A330 difference: there is no equivalent to the Airbus backup nav on the MCDU, and the workload is single-waypoint dead-reckoning style navigation to the nearest suitable airport.

> **FCOM 11.50.1 Flight Management, Navigation - Alternate Navigation System Description**
>
> *"In the unlikely event that an unrecoverable loss of all displays or loss of all three FMCs occurs, a backup navigation capability is available through the Tuning and Control Panel (TCP)."*
>
> Push NAV on any of the three TCPs under normal power; on standby power only the Captain's left TCP is available. LNAV and VNAV are lost, but autopilot and autothrottles may still be available.
>
> *Workbook wording: Alternate Navigation System*

11.1.10.4 Know Airplane Shutdown and Securing Procedures

Fuel Conservation

FUEL PLANNING

Where are fuel conservation considerations?

Green Policy; green-symbol fuel efficiency items appear throughout

FOM 1.6.3 Green Policy establishes that green-symbol guidance supports Alaska Air Group's fuel efficiency commitment, with the explicit note that safety is always first and fuel saving measures must never compromise safety. The individual conservation items live where the phase of flight is covered: single-engine taxi as the standard procedure when operationally feasible, and 5.5.3 Flight Plan Time and Fuel Monitoring - Enroute which calls cruise the greatest opportunity to enhance efficiency by maintaining optimum altitude and adhering to cost index. 5.5.4 Flight Plan - Deviation/Optimization covers accepting or requesting deviations for fuel and time savings. Fuel planning numbers themselves are in 8.3.1 and 8.3.2, not under conservation.

> **FOM 1.6.3 Green Policy**
>
> *"operational decisions that align with Alaska Air Group's commitment to fuel efficiency"*
>
> There is no section titled 'Fuel Conservation' in the FOM. Fuel conservation guidance is flagged by the green symbol wherever it applies, e.g. single-engine taxi (5.2 taxi guidance), cruise/optimum altitude in 5.5.3, and deviation/optimization in 5.5.4.
>
> *Workbook wording: Where can you find the Fuel Conservation considerations in the FOM?*

Flight Deck Entry Procedures/Security

SECURITY AND DOORS

Flight deck door procedure with no flight attendants?

Yes. Captain covers cabin duties; door closed, critical phases

FOM 16.1.11 states that when Flight Attendants are not on board a Company-operated flight, the Captain must ensure the cabin duties normally done by F/As are accomplished, and refers to fleet-specific manuals for procedures. The WARNING requires galley carts to be properly secured and the Flight Deck Door closed during critical phases of flight. Normal Secure Flight Deck rules in 15.2.6.1 still apply on Part 121 legs (door locked from pushback until stopped at the gate with engines shut down), and Secure Flight Deck procedures are not required during Part 91 operations. Positioning and Special Flight Permit ferry sections (16.4.1, 16.4.2) repeat the same 'see fleet-specific manuals' direction when no F/As are aboard.

> **FOM 16.1.11 Pilot Duties With No Flight Attendants**
>
> *"properly secured and the Flight Deck Door is closed during critical phases of flight."*
>
> FOM 15.2.6.2 points you here: 'For further guidance regarding the Flight Deck Door, see 16.1.11 – Pilot Duties With No Flight Attendants.' The section also refers you to fleet-specific manuals for the actual procedures.
>
> *Workbook wording: Are there special Flight Deck Door Procedure on Flights with No Flight Attendants?*

Holding (Procedures, speeds)

TAXI AND GROUND

Where are the US holding speeds?

In the FOM holding section

US holding speeds: 6000 ft or below, 200 KIAS. Above 6000 through 14,000 ft, 230 KIAS (or 210 KIAS where restricted), inbound leg 1 minute. Above 14,000 ft, 265 KIAS, inbound leg 1 1/2 minutes. Standard holding patterns are right-hand turns. Begin speed reduction no earlier than three minutes prior to the fix.

> **FOM 5.5.15 Holding**
>
> *""Standard holding patterns are right-hand turns.""*
>
> Holding pattern protection is based on a maximum of 280 KIAS or Mach 0.8, whichever is lower. Civil aircraft at military or joint airports should expect max 230 KIAS. Alaska and New York are restricted to 210 KIAS above 6000 through 14,000 ft where charts show the standard holding symbol with 210K in the center; in Alaska that does not apply in the Anchorage or Fairbanks approach control areas. Advise ATC if a higher speed is needed for turbulence, icing or configuration.
>
> *Workbook wording: Where can you find the holding speeds (FAR) in USA?*

INTERNATIONAL THEATERS

Where are international holding speeds?

ICAO Doc 8168 and the theater chapters

FOM 5.5.15 cites AIM 5-3-8 and ICAO Doc 8168. Country-specific holding speeds and procedures appear in the theater chapters 19 through 24 and in the Jeppesen country pages, for example the Heathrow holding restriction in the Europe chapter.

> **FOM 5.5.15 Holding, ICAO DOC 8168**
>
> *""ICAO DOC 8168""*
>
> Rev 125.1 renumbering: the United Kingdom chapter is 23.17 with LHR subsections under 23.17.1, and Italy is 23.18.
>
> *Workbook wording: Where can you find international country-specific holding speeds?*

TCAS

TCAS AND TRAFFIC

Where are TCAS procedures in the FOM?

TCAS Policy, in the Operating - General chapter

FOM 2.3.17 TCAS Policy is the FOM home for TCAS. It requires compliance with an RA unless the pilot considers it unsafe, prohibits maneuvering opposite an RA unless visually determined to be the only means of safe separation, and reminds crews the aircraft in sight may not be the intruder. It also states TCAS does not reduce the see-and-avoid responsibility (see 2.3.16 See and Avoid). For the flying procedure use QRH Non-Normal Maneuvers, Traffic Avoidance, accomplished by recall for any TA or RA.

> **FOM 2.3.17 TCAS Policy**
>
> *“Compliance with a TCAS Resolution Advisory (RA) is required, unless the pilot considers it unsafe”*
>
> The FOM carries policy only. The actual maneuver is the QRH recall item Traffic Avoidance (QRH MAN.1.5), and the system description is FCOM 15.20.17. Oceanic TCAS guidance is at FOM 5.7 (TA/RA mode whenever possible).
>
> *Workbook wording: Where can you find TCAS procedures in the FOM?*

TCAS AND TRAFFIC

What is a TA?

Traffic advisory: traffic predicted to conflict in 20-48 seconds

Per FCOM 15.20.18, a Traffic Advisory predicts another airplane entering the conflict airspace in 20 to 48 seconds, and exists to help the crew get visual contact. Indications are the amber ND message TRAFFIC plus a single 'TRAFFIC, TRAFFIC' voice annunciation; there is no PFD/HUD vertical guidance for a TA. QRH MAN.1.5 Traffic Avoidance for a TA has the PF look for traffic using the traffic display as a guide and maneuver only if traffic is sighted, with the PM calling out conflicting traffic. FCOM L.10.12 non-AFM operational information warns evasive maneuvers must not be based solely on the traffic display or TA without visually sighting the traffic.

> **FCOM 15.20.18 Traffic Advisories (TA) and Display**
>
> *“A TA is a prediction that another airplane will enter the conflict airspace in 20 to 48 seconds.”*
>
> A TA gets an amber TRAFFIC message on the ND and one 'TRAFFIC, TRAFFIC' aural, no vertical guidance. TA symbol is a filled amber circle (amber circle with black chevron for ADS-B traffic).
>
> *Workbook wording: What is an “TA”?*

TCAS AND TRAFFIC

What is an RA?

Resolution advisory: conflict predicted in about 15-35 seconds

FCOM 15.20.18 defines an RA as a prediction that another airplane will enter the TCAS conflict airspace within approximately 15 to 35 seconds; without altitude data from the intruder, no RA can be generated. Indications are the red ND message TRAFFIC, a voice annunciation, and TCAS vertical guidance on both PFD and HUD, with OFFSCALE shown in red if the RA is beyond the selected ND range. Compliance is mandatory per FOM 2.3.17 and QRH MAN.1.5, and FCOM L.10.12 authorizes deviation from the ATC clearance to the extent necessary to comply. Fly to keep pitch and vertical speed outside the displayed red RA regions, and do not use flight director pitch commands until clear of conflict.

> **FCOM 15.20.18 Resolution Advisories (RA) and Display**
>
> *“An RA is a prediction that another airplane will enter the TCAS conflict airspace within approximately 15 to 35 seconds.”*
>
> No RA is possible if the intruder is not reporting altitude. RA gives a red ND TRAFFIC message, a voice annunciation, and PFD/HUD vertical guidance; the symbol is a filled red square.
>
> *Workbook wording: What is an “RA”?*

TCAS AND TRAFFIC

What is a phantom TA?

In the FOM and the FCOM warning systems chapter

A TA is a Traffic Advisory identifying proximate traffic with no commanded manoeuvre. An RA is a Resolution Advisory commanding a vertical manoeuvre, and it is followed even against an ATC clearance. Symbology and the Phantom TA definition are FCOM Warning Systems.

> **FOM Ch 5 / FCOM Warning Systems**
>
> The term 'phantom TA' does not appear anywhere in FOM Rev 125.1, FCOM R10, or QRH R7 (zero case-insensitive hits). It is informal line/training slang, typically a TA with no traffic visible or a false target from a bad transponder return or ground reflection. If a definition is required, look in the Alaska/Hawaiian ground school or CBT material, not the manuals.
>
> *Workbook wording: What is Phantom TA?*

TCAS AND TRAFFIC

Where are the TCAS symbology definitions?

FCOM Warning Systems chapter, TCAS traffic display pages

Symbol definitions are in FCOM Chapter 15 Warning Systems: the annotated display legend at 15.10.16-15.10.17 and the same set described in prose at 15.20.17-15.20.19. TCAS-only traffic uses a filled red square for an RA, a filled amber circle for a TA, a filled white diamond for proximate traffic, and an unfilled white diamond for other traffic. ADS-B traffic uses the same colors with a black directional chevron, and adds flight ID to the data tag. Proximate traffic is defined as within six miles and 1,200 feet vertically.

> **FCOM 15.10.16 TCAS Traffic and Alert Message TRAFFIC Display**
>
> *“filled red square indicates a resolution advisory (RA)”*
>
> Controls and Indicators at 15.10.13-15.10.20 has the annotated ND picture and symbol legend; the narrative repeat is in System Description at 15.20.17-15.20.19. ADS-B (In) symbology is separate, at FCOM 10.40.
>
> *Workbook wording: Where can you find TCAS symbology definitions?*

MedLink

MEDICAL

What is MedLink?

Contract ER physicians giving 24-hour inflight medical advice

FOM 11.1.9.1 describes MedLink as emergency room physicians available 24 hours a day with aviation-specific experience, providing recommendations on onboard medical issues, and notes contact with MedLink removes Company liability for passenger health. It is the primary resource for an informed divert decision, weighing severity against medical resources at the destination or diversion airport, and it arranges medical resources on arrival. F/As tell volunteer medical professionals onboard that Alaska uses the MedLink physician for medical advice and wants the onboard professional as an on-site consultant. Per 11.1.9.4 the Captain determines whether a medical diversion is necessary with advice from MedLink and Dispatch.

> **FOM 11.1.9.1 MedLink Overview**
>
> *“MedLink utilizes emergency room physicians 24 hours a day to provide crewmembers with”*
>
> Company policy is to follow MedLink's recommendation, but the Captain keeps final authority on operational requests. The one-way exception: if MedLink says a passenger is not fit to fly, the Captain will not override MedLink.

MEDICAL

When do you contact MedLink?

Any medical support need, aircraft oxygen use, fitness-to-fly doubt

FOM 11.1.9.2 makes MedLink contact mandatory, except during critical phases of flight, for any medical situation where the crew requires support or advice, any passenger or crew use of aircraft oxygen, and any uncertainty about whether a passenger is contagious or physically fit to fly (see 11.1.8.2 Communicable Diseases). If paramedics are called on the ground, MedLink must still be contacted to clear the passenger to fly. If MedLink cannot be reached and it is impractical for the Flight Crew, 11.1.9.5 puts the continue-or-divert decision on the Captain, requires ATC and Dispatch notification, and requires a safety report.

> **FOM 11.1.9.2 When to Contact MedLink**
>
> *“MedLink shall be contacted for the following, except during critical phases of flight:”*
>
> Note the carve-out: not during critical phases of flight (sterile flight deck per 2.3.12). Passenger or crew use of aircraft oxygen is a mandatory trigger even if the person seems fine.
>
> *Workbook wording: When should you contact MedLink?*

MEDICAL

Who contacts MedLink on the ground?

CSA at the gate; after door close, F/As or crew

FOM 11.1.9.3 On Ground has the Flight Crew, F/As, station personnel, and MedLink work together to decide fitness to fly, and requires MedLink contact to clear a passenger even if paramedics already responded. At the gate or prior to door closure the CSA should initiate contact, and may deplane the passenger for privacy while assessing fitness to fly. After door closure prior to takeoff, and again after landing, the F/As initiate with operational broadband or a SATCOM cabin device, otherwise the Flight Crew initiates. Contact Dispatch if a delay is anticipated, tell ATC if you need an expeditious return to the gate or expeditious taxi, and advise Operations if medical assistance is expected at the gate.

> **FOM 11.1.9.3 Responsibility for Contacting MedLink - On Ground**
>
> *“The CSA should initiate contact with MedLink.”*
>
> After door closure and after landing the split is by equipment: with operational broadband or a SATCOM cabin device the F/As initiate; without it the Flight Crew initiates. Advise Operations if medical assistance is expected at the gate.
>
> *Workbook wording: Who is responsible for contacting MedLink while on the ground?*

MEDICAL

Who contacts MedLink in flight?

F/As if cabin SATCOM works; otherwise the Flight Crew

Per FOM 11.1.9.3 In Flight, on HA aircraft with available and operational cabin SATCOM handsets the F/As advise the Flight Crew and then call MedLink directly; if the cabin device is unavailable, the Flight Crew initiates. On the 787 with SATCOM but no cabin handset, the Flight Crew calls MedLink Inflight Medical Assistance at (602) 282-4910 and conferences in the F/As with the cabin headset. With no operational SATCOM, the Flight Crew requests an ARINC patch to that same number. If the situation could require a diversion, either MedLink calls Dispatch and conferences them in, or the Flight Crew contacts Dispatch after finishing the MedLink call (see 7.1.12 Reports to Dispatch).

> **FOM 11.1.9.3 Responsibility for Contacting MedLink - In Flight**
>
> *“On aircraft with operational SATCOM handsets in the cabin, the F/As call MedLink directly.”*
>
> This is the (HA) 787/A330 rule. On the (AS) 737 the same logic runs off operational broadband and the F/A Inflight Mobile Device MedAire app. Either way the F/A advises the Flight Crew of the medical issue first.
>
> *Workbook wording: Who is responsible for contacting MedLink while in flight?*

MEDICAL

Do you call MedLink direct or Dispatch first?

Directly. Dispatch is conferenced in if diversion possible

FOM 11.1.9.3 has you go to MedLink first, not Dispatch. With operational cabin SATCOM the F/As call MedLink directly; with SATCOM but no cabin handset the 787 Flight Crew calls MedLink directly at (602) 282-4910 and conferences the F/As on the cabin headset. Without operational SATCOM, the Flight Crew contacts ARINC for a patch to MedLink, and only contacts Dispatch if unable to establish the patch. Dispatch involvement is driven by the operational consequence: if a diversion is possible, request MedLink conference Dispatch in, or call Dispatch immediately after the MedLink call. On the ground, 11.1.9.5 also requires a safety report and a Maintenance Logbook entry if you were unable to reach MedLink because of equipment.

> **FOM 11.1.9.3 Responsibility for Contacting MedLink - In Flight**
>
> *“The Flight Crew will call MedLink. MedLink Inflight Medical Assistance may be reached at”*
>
> Only with no operational SATCOM does the call route through a relay: ARINC patch, or Dispatch if ARINC fails. Dispatch always gets notified for a potential diversion, either by MedLink conferencing them in or by you calling after.
>
> *Workbook wording: Do we as flight crew contact MedLink directly, or do we call dispatch first?*

MEDICAL

How do you make a MedLink call from the aircraft?

SATCOM to (602) 282-4910, or ARINC phone patch

Per FOM 11.1.9.3, the 787 primary method is a SATCOM call to MedLink Inflight Medical Assistance at (602) 282-4910, with the Flight Crew conferencing in the F/As using the cabin headset. If SATCOM is not operational, contact ARINC (or Stockholm Radio, per 5.7 guidance on phone patches) and request a patch to that same number, contacting Dispatch only if the patch cannot be established. F/As may AirDrop a Situation Summary from the MedAire app, or use the MedLink Patch Checklist from the F/A Stationery Kit, to structure the report. If communication is degraded the Flight Crew relays the information the F/As provide; if MedLink still cannot be reached, 11.1.9.5 shifts the continue-or-divert call to the Captain and requires a safety report.

> **FOM 11.1.9.3 Responsibility for Contacting MedLink - In Flight**
>
> *"The Flight Crew will contact ARINC for a patch to MedLink Inflight Medical Assistance at (602)"*
>
> On the 787 conference the F/As in on the cabin headset so the physician talks to the person with eyes on the patient. Ground Medical Assistance is (602) 282-6613 or (800) 961-4847; the (AS) number listed elsewhere in Ch 11 is (602) 282-4967.

MEDICAL

Medical event on an international flight. Any special notifications?

Yes. General Declaration entry; Australia notified 30 minutes before landing

FOM 18.1.15 uses the US Public Health Service 'ill person' definition: temperature of 100F (38C) or greater persisting more than 48 hours; fever with rash, swollen glands, jaundice, persistent cough, vomiting, difficulty breathing, muscle pain, headache with stiff neck, decreased consciousness, or unexplained bleeding; or persistent diarrhea. For foreign destinations other than the US and Canada, any apparent death or illness of a crewmember or passenger enroute is entered on the General Declaration form. For flights to Australia, the Captain must report any apparent death or ill person to Australian authorities at least 30 minutes before landing. FOM 11.1.8.2 covers the communicable-disease reporting chain, including MedLink conferencing in the CDC and possible aircraft quarantine at one of the designated stations.

> **FOM 18.1.15 Death or Illness Reporting - International**
>
> *"For Flights to Australia: The Captain must report to the Australian authorities the presence on"*
>
> The General Declaration requirement applies to countries other than the US and Canada. For a suspected communicable disease inbound to the US, 11.1.8.2 adds the CDC/public health authority path, usually conferenced in by MedLink or set up through Dispatch.
>
> *Workbook wording: If on an international flight, are there special procedures/notifications that must be made by the flight*

De/Anti-Ice Procedures

COLD WEATHER AND RUNWAY CONDITION

What are icing conditions?

OAT or TAT 10C or below with visible moisture/contamination

FCOM SP.16.1 defines icing conditions as OAT on the ground or TAT in flight of 10C or below, plus either visible moisture (clouds, fog with visibility of one statute mile / 1600 m or less, rain, snow, sleet, ice crystals) or ice, snow, slush, or standing water on the ramps, taxiways, or runways. The companion CAUTION prohibits engine anti-ice above 10C OAT on the ground and engine or wing anti-ice above 10C displayed TAT in flight. A separate trigger sits at 3C or below with visible moisture, which requires the periodic engine run-up to shed fan ice. FOM 9.3.2.2 gives the separate airframe icing intensity table (trace, light, moderate, severe), which is about accumulation rate, not the icing-conditions definition.

> **FCOM SP.16.1 Cold Weather Operations**
>
> *"Icing conditions exist when OAT (on the ground) or TAT (in-flight) is 10°C"*
>
> Two-part test: temperature AND either visible moisture (including fog with visibility 1 SM / 1600 m or less) or ice, snow, slush, or standing water on ramps, taxiways, or runways. Note the ground/air split: OAT on the ground, TAT in flight.
>
> *Workbook wording: What is definition of icing conditions?*

COLD WEATHER AND RUNWAY CONDITION

Where are the aircraft de-ice and anti-ice procedures?

FCOM Supplementary Procedures, Adverse Weather; company deicing in FODM

FCOM SP.16 Cold Weather Operations carries the airplane procedures: Preliminary Preflight, Exterior Inspection, Engine Start, Engine Anti-ice Operation On the Ground (SP.16.4), De-icing / Anti-icing (SP.16.6), Fan Ice Removal (SP.16.8), Wing Anti-ice Operation In-flight (SP.16.8), Cold Temperature Altitude Corrections (SP.16.10), and the cold weather Secure Procedure. FCOM SP.3 Anti-Ice, Rain only covers windshield washer and wiper use and points you to SP.16. Company deicing policy, fluid types, holdover times, and post-deicing checks are in the FODM and the ASA Deicing Procedures Manual (FOM 9.3.1, 9.3.1.1, 16.2.9), with the HOTs app and FODM carried on the EFB. FOM 9.2.2 adds the heavy snow rule: undiluted Type IV and a successful contamination check within 5 minutes of takeoff.

> **FCOM SP.3.1 Anti-Ice Operation, referring to SP.16 Cold Weather Operations**
>
> *“Note: (Refer to Cold Weather Operations in Chapter SP, Section 16, for”*
>
> Split the question: airplane anti-ice system operation is FCOM SP.16 (SP.3 covers only windshield washer/wiper). Ground deicing/anti-icing fluid, holdover times, and the contamination check are FODM and the ASA Deicing Procedures Manual, per FOM 9.3.1 and 16.2.9.
>
> *Workbook wording: Where do you find the aircraft de/anti-Ice procedures?*

COLD WEATHER AND RUNWAY CONDITION

Where are airport specific deice procedures?

FODM and the ASA Deicing Procedures Manual

FOM 16.2.9 states plainly that all deice procedures are found in the ASA Deicing Procedures Manual and the FODM, with the Deicing Procedures Manual available through Maintenance Control. FOM 9.3.1 sends you to the fleet-specific manuals and the FODM for cold weather preflight requirements, and 9.3.1.1 sends you to the FODM for Liquid Water Equivalent, which is station-dependent (an operational LWES exists only at certain airports and its HOTs are FAA-authorized and shall be used). The FODM is EFB-only per the FOM required-documents table, with printed Holdover Times as the backup. Nothing in the FOM, FCOM, or QRH lists deice procedures by airport.

> **FOM 16.2.9 Deice - Charters**
>
> *“All deice procedures are found in the ASA Deicing Procedures Manual and the FODM”*
>
> Verified only to this level: the FOM contains no airport-specific deice pages, and states all deice procedures live in the FODM (on the EFB) and the ASA Deicing Procedures Manual, which is obtained or reviewed through Maintenance Control. Station-specific deice pad, remote deice, and LWES details are in the FODM, which is not one of the three manuals available here.
>
> *Workbook wording: Where do you find specific airport deice procedures?*

COLD WEATHER AND RUNWAY CONDITION

When must engine anti-ice be on?

Whenever icing conditions exist or are anticipated, above -40C

FCOM SP.16.7 requires engine anti-ice AUTO or ON during all flight operations when icing conditions exist or are anticipated, except below -40C SAT. SP.16.4 requires it selected ON and left on for all ground operations in the same conditions, except below -40C OAT, and confirms EAI shows on the primary engine display for both engines. The WARNING is the trap for an Airbus pilot: do not wait for airframe visual icing cues, use the temperature and visible moisture criteria, because late activation can allow excessive ice ingestion and engine damage or failure. Separately, at 3C or below with visible moisture, run up both engines one at a time to a minimum of 70% N1 for 10 to 30 seconds every 15 minutes to shed fan ice.

> **FCOM SP.16.7 Engine Anti-ice Operation - In-flight (and SP.16.4, On the Ground)**
>
> *“Engine anti-ice must be AUTO or ON during all flight operations when icing conditions exist or are anticipated, except when the temperature is”*
>
> Ground and air differ: on the ground it must be selected ON (manual) and remain on, in flight AUTO or ON satisfies the requirement. Cutoff is -40C OAT on the ground, -40C SAT in flight; do not use above 10C OAT/TAT.
>
> *Workbook wording: When are you required to turn the engine anti-ice on?*

COLD WEATHER AND RUNWAY CONDITION

When must wing anti-ice be on?

AUTO normally; select ON manually when wing icing possible

FCOM SP.16.8 says the wing anti-ice system is mainly used to prevent icing in flight, with the primary method being the automatic ice detection system controlling it throughout the flight envelope for the cleanest airfoil, least runback ice, and least thrust and fuel penalty. The secondary method is manual selection of WING ANTI-ICE ON when wing icing is possible; confirm WAI shows on the primary engine display for both engines, and if it does not, avoid icing conditions. When manual use is no longer needed, return the selector to AUTO if ice detection is available, OFF if ice detection is inoperative, and do not use wing anti-ice with displayed TAT above 10C. FCOM 30 adds that automatic operation is available in flight and on the ground above 75 knots, that wing anti-ice is inhibited on the ground below 75 knots, and that manual ON in flight powers the mats any time TAT is below 25C; engine or wing anti-ice with flaps out of up also raises stick shaker and minimum maneuvering speeds.

> **FCOM SP.16.8 Wing Anti-ice Operation - In-flight**
>
> *“The secondary method is to select the WING ANTI-ICE selector ON when wing icing is possible and use for anti-icing.”*
>
> Wing anti-ice is primarily an in-flight system and is inhibited on the ground below 75 knots (FCOM 30.20). Do not use it above 10C displayed TAT. Ice on window frames, windshield center post, wiper arm, or side windows is the cue for structural icing.
>
> *Workbook wording: When are you required to turn on the wing de/anti-ice*

EFB AND FD PRO &MIDDOT; WALKTHROUGH ITEM, PERFORM RATHER THAN RECALL

Discuss the SureWx app

In the SureWx guide and the FOM

SureWx holdover-time workflow is vendor material. The governing policy is the pretakeoff contamination check requirement. Verify the app steps in the SureWx guide and the de-icing policy in FOM Chapter 9 and the fleet cold weather supplement.

> **SureWx guide / FOM Ch 9**
>
> *Workbook wording: Discuss SureWx app*

COLD WEATHER AND RUNWAY CONDITION

Where are cold weather procedures?

FOM Adverse Weather chapter; also FCOM Supplementary Procedures and FODM

FOM Chapter 9 covers adverse weather, with 9.3 Cold Weather Operations as the FOM home for cold weather. 9.3.1 Cold Weather Preflight Requirements (14 CFR 121.629) simply directs you to comply with the cold weather procedures in the fleet-specific manuals and the FODM; 9.3.1.1 covers Liquid Water Equivalent HOTs and again refers to the FODM. 9.3.2 In Flight adds adverse weather restrictions, the airframe icing intensity table, and fuel freeze considerations at 9.3.2.3. The airplane procedures themselves are FCOM SP.16 Cold Weather Operations, including cold temperature altitude corrections at SP.16.10 and the cold weather Secure Procedure at SP.16.13.

> **FOM 9.3 Cold Weather Operations (9.3.1 Cold Weather Preflight Requirements)**
>
> *“Comply with the cold weather procedures in the fleet-specific manuals and FODM.”*
>
> The FOM section is short and delegates: it points to fleet-specific manuals and the FODM. Airplane procedures are FCOM SP.16; deicing fluid, HOTs, and LWE are FODM.
>
> *Workbook wording: Where can you find the cold weather procedures in the FOM? Any other manual?*

Freighter Operations

TAXI AND GROUND

Review QRH procedures without moving switches

Open it in the ECL, then select CHKL OVRD

The non-normal menu is used to access checklists for EICAS alert messages with rectangle icons specifically to allow review of the checklist (QRH CI.2, Electronic Checklist Operation). Nothing is moved because the ECL is read-only until the crew checks items off; closed-loop items only turn green when the action is actually taken. When done reviewing, select CHKL OVRD at the bottom of the checklist page before closing ECL or selecting another checklist, otherwise the checklist stays active and CHKL NON-NORMAL may post on EICAS. Memory items are never reviewed this way - they must be done from memory or the QRC before any reference items (QRH CI.2, Memory Items vs. Reference Items).

> **QRH CI.2 Checklist Instructions - Non-Normal Checklists (Electronic Checklist Operation / Using CHKL OVRD)**
>
> *"• Checklist is opened only for review without the intent to accomplish it (see section Using CHKL OVRD)"*
>
> Opening a checklist for review only leaves it ACTIVE; CHKL OVRD forces it back to inactive and also strips its Notes-page entries, inhibited checklists and deferred items (QRH CI.2). Do not use CHKL OVRD if you only want to pause and finish later.
>
> *Workbook wording: Review QRH procedures without moving any switch positions (when time permits). This reviews the*

ETOPS, ORCN and FIR cards

ETOPS

Terrain polygons with no own-ship position on the EFB?

In the FOM theater chapters and NAT training

North Atlantic and Europe operations. FOM chapter 22 covers NAT HLA, transition areas (22.2.1), OEP (22.2.2), Northern AMU (22.2.3), OTS (22.3.1) and RCL timing (22.7.4). Chapter 23 covers Europe, with United Kingdom at 23.17 and Italy at 23.18 in Rev 125.1. Also review the NAT HLA training deck, ETOPS/ORCN and FIR cards, and the Jeppesen UK/Europe pages.

> **FOM Ch 22-23 / NAT HLA training**
>
> Not in the FOM, FCOM or QRH. FOM 11.2.9 covers decompression (terrain) polygons but assumes Jeppesen FD Pro with the chart displayed; it gives no technique for losing EFB own-ship. The only own-ship-loss workaround the FOM publishes is for the AMU border in 22.2.3, where you read the border lat/long off the FD Pro IFR chart and enter it on the ND FIX INFO page. Applying that to polygons is a reasonable technique but is not written anywhere in these three manuals.
>
> *Workbook wording: Consider techniques for terrain polygons without own-ship position on the EFB*

ETOPS

Terrain polygon vs ETP considerations?

In the FOM theater chapters and NAT training

North Atlantic and Europe operations. FOM chapter 22 covers NAT HLA, transition areas (22.2.1), OEP (22.2.2), Northern AMU (22.2.3), OTS (22.3.1) and RCL timing (22.7.4). Chapter 23 covers Europe, with United Kingdom at 23.17 and Italy at 23.18 in Rev 125.1. Also review the NAT HLA training deck, ETOPS/ORCN and FIR cards, and the Jeppesen UK/Europe pages.

> **FOM Ch 22-23 / NAT HLA training**
>
> Not in the FOM, FCOM or QRH. Decompression polygons (FOM 11.2.9) and ETPs (FOM 6.3.7) are covered in separate chapters and the manuals never compare or sequence them. Raise it with Dispatch: FOM 6.4.1 makes diversion-airport selection a joint Flight Crew / Dispatch call.
>
> *Workbook wording: Terrain polygon procedure vs ETP considerations (example: ETP is usually west of Greenland and BIKF is*

ETOPS

When do you use the HDG REF switch?

When assigned a heading where true north is used

FCOM SP.10.1 is blunt: use TRUE when assigned a heading in a region where true north is used, and NORM in all other regions. Entering the polar region the autopilot goes true for all cruise roll modes regardless of switch position, annunciated by a white box around TRU on ND and PFD, with a T next to leg course on the RTE LEGS page (FCOM 11.31). The trap is the timing notes in SP.10.1: change the switch before the HUD TAKEOFF runway is selected and before APP mode is armed, or you degrade HUD TO/GA guidance and LOC/FAC capture. Also, changing it while in HDG SEL or TRK SEL drops the AFDS to HDG HOLD or TRK HOLD, so you must reselect.

> **FCOM SP.10.1 Supplementary Procedures - Flight Instruments, Displays (Heading Reference Switch Operation)**
>
> *"Use TRUE when assigned a heading in a region where true north is used. Use NORM in all other regions."*
>
> The 787 switches to true automatically in the polar regions with HEADING REF in NORM (FCOM 11.31, FMC Polar Operations), so TRUE is a manual selection you make for an ATC-assigned true heading, e.g. inside the AMU (FOM 22.2.3). FCOM 11.31 adds: when operating the autopilot in the polar region in other than LNAV, select TRUE.
>
> *Workbook wording: AMU, when do you use the HDG REF switch?*

ETOPS

How do you identify the OEP?

Named waypoint on or near the FIR boundary entering the OCA

FOM 22.2.2 defines the OEP as the waypoint from which RCL time estimates are calculated, generally a named waypoint on or near the FIR boundary where the aircraft enters an OCA. Concrete examples in the same section: OEPs sit along the W015 meridian on the western boundaries of NOTA and SOTA, AGORI is the OEP on the northern boundary of NOTA, and OMOKO, TAMEL and LASNO are OEPs on the southern boundary of SOTA. BOTA OEPs are on the W008 45' meridian. Watch the GOTA exception in 22.2.1 - it has OEPs in the middle of the airspace, which is why SLOP starts halfway through that transition area.

> **FOM 22.2.2 Oceanic Entry Point (OEP)**
>
> *"The OEP is the waypoint from which Request for Clearance (RCL) time estimates are typically calculated (e.g., "X" minutes prior to OEP via data link). It is generally a named waypoint located"*
>
> If the route has no waypoint sitting on the OCA boundary, the OEP backs up to the last named waypoint before it; on random routing it is the last named waypoint before the first LAT/LONG inside the OCA. FOM 5.7.2.16 gives the control definition: the point where the flight passes from ARTCC to Oceanic Control.

ETOPS

Entering the GOTA eastbound. Is an RCL required?

Yes, to Gander Oceanic

FOM 22.2.1 states that transition areas (GOTA, NOTA, SOTA, BOTA) are buffers between domestic and oceanic airspace, are not part of the OCAs, and leave aircraft under domestic ATC control - GOTA under Gander or Montreal, NOTA and SOTA under Shannon Domestic, BOTA under Brest Domestic. The section then calls out the GOTA specifically: eastbound flights through the GOTA are to send an RCL message to Gander Oceanic. Timing comes from FOM 22.6.4: Gander wants the RCL 60 to 90 minutes prior to the OEP, and if you depart an airport less than 45 minutes flying time from the OEP you send it 10 minutes prior to start-up. For any other transition-area quirk, FOM 22.2.1 sends you to the FIR card and Operational Notes in Jeppesen FD Pro.

> **FOM 22.2.1 Transition Areas**
>
> *"transition area. Also, eastbound flights through the GOTA are to send an RCL message to Gander Oceanic. For specific transition area rules and exceptions, see the FIR card and Operational Notes"*
>
> This is a named exception. The GOTA is not part of the Gander OCA and stays under Gander or Montreal Center control, yet you still send the RCL to Gander Oceanic. The GOTA also has OEPs in the middle of the airspace, so SLOP begins halfway through it rather than at the far edge.
>
> *Workbook wording: When entering the GOTA eastbound, are you required to send an RCL?*

Airspace ownership vs control

OCEANIC AND NAT

NUUK FIR. Who controls what, and what do you do?

Nuuk owns it; fly Reykjavik OCA north, Gander OCA south

FOM 22.4 makes the ownership-versus-control distinction operational: in North Atlantic operations CPDLC and ADS coverage is based on Oceanic Control Areas, not FIRs, and the logon must be directed to the controlling authority for the relevant OCA regardless of which FIRs you transit. Nuuk FIR is the worked example - its north section lies in the Reykjavik OCA and its south section in the Gander OCA, so CPDLC should sequence and a new SELCAL check is required. Logon is initiated 10 to 25 minutes prior to entry, and once CPDLC is up you still maintain a continuous voice watch on the assigned HF frequency; CPDLC does not replace voice monitoring. Same theme in FOM 22.2.1 for the transition areas, where a domestic center controls airspace adjacent to an OCA.

FOM 22.4 Communication and Surveillance

> *"For example, the north section of the Nuuk FIR is in the Reykjavik OCA, but the south section of the Nuuk FIR is in the Gander OCA. CPDLC should sequence and a new SELCAL check is required."*

The practical action is the SELCAL check: CPDLC coverage in the NAT follows OCAs, not FIRs, so crossing inside one FIR you still change controlling OCA, CPDLC re-sequences, and a new SELCAL check is required on the new HF frequencies.

Workbook wording: NUUK FIR: Owned by Nuuk but controlled by Gander South of N6330 and Reykjavik to the North. What are

OCEANIC AND NAT

Can you cross the NAT-HLA without HF?

No. At least one HF radio is required

FOM 22.4.3 requires two independent LRCS outside VHF coverage in the NAT, one of which must be HF. FOM 22.1.10 lists the NAT HLA equipment as dual IRS, dual long range nav with two MCDUs, dual NDs, one GPS receiver, ADS-C, CPDLC, SATCOM data link, and two HF radios or a single HF radio and SATVOICE. Inmarsat makes this worse at high latitude: FOM 22.4.1 says an Inmarsat CPDLC or SATVOICE system does not qualify as an LRCS north of 80N, so HF is the fallback that keeps you legal up there. If HF fails in flight, notify ATC and Dispatch immediately (22.4.3) and work FOM 22.7 and 22.7.1 - squawk 7600, try SATCOM voice and CPDLC, and relay on 123.45.

FOM 22.4.3 HF Radio Requirements

> *"Operations in the NAT outside of VHF coverage require aircraft to be equipped with two independent Long Range Communications Systems (LRCS), one of which must be HF."*

Cross-check FOM 7.1.8, which lists the only two NAT HLA ETOPS dispatch options: 2 HF radios plus SATDATA/CPDLC, or 1 HF plus SATDATA/CPDLC plus SATVOICE. Both contain HF. FOM 22.1.10's note softens it only after departure: a single HF radio and SATCOM data link.

Workbook wording: Can you Fly across the NAT-HLA without HF radios? If so, how?

INTERNATIONAL THEATERS

GPS jamming reporting requirements?

Report to ATC; file a safety report

FOM 5.1.21.3 says GPS/GNSS disruptions cluster around conflict zones, military operations areas and counter-UAS protection areas, and directs the crew to report any GPS/GNSS malfunction to ATC and submit a safety report for the listed discrepancies. FOM 5.1.21.2 explains the mechanism for the Atlantic and Europe crews: jamming superimposes a stronger interference signal so GNSS position goes unreliable, spoofing broadcasts false GPS signals. The consequence that matters oceanic is loss of performance-based separation - RNP 4 and RNAV/RNP 10 - plus erroneous ADS-B and ADS-C positions being transmitted to controllers. Ground radar and multilateration are not GNSS-dependent and will show controllers your real position, which is how the deviation gets caught.

FOM 5.1.21.3 GPS/GNSS Disruptions

> *"Report any GPS/GNSS malfunction to ATC. Submit a safety report for any of the following discrepancies or anomalies:"*

The safety report is required for any of five listed anomalies: inability to use GPS for navigation, loss of RNAV/RNP capability, unreliable EGPWS triggering, inaccurate position on the ND, and loss of or erroneous ADS-B outputs (FOM 5.1.21.3). Background on the threat is in 5.1.21.2.

Workbook wording: GPS Jamming reporting requirements.

TAXI AND GROUND

How and when do you request an RCL?

ACARS data link; Gander 60-90 min, Shanwick 30-90 min prior OEP

FOM 22.6.1 is the reason the RCL still exists: except for Shanwick, oceanic clearances are no longer issued in North Atlantic OCAs, but the RCL message is still required for every other OCA except New York East. FOM 22.6.2 sets the content - OEP, ETA for the OEP, Mach based on FMS Cost Index, requested flight level, and the highest acceptable flight level as free text MAX FL. If you omit MAX FL, the requested level becomes your ceiling at the OEP. FOM 22.6.4 sets the lead times and adds that a flight departing less than 45 minutes from the Gander OEP sends the RCL 10 minutes prior to start-up, and a flight 30 minutes or less from the Shanwick OEP sends it before departure. Expect only an acknowledgment - RCL RECEIVED BY [ANSP] - per FOM 22.6.5.

> **FOM 22.6.4 Timing for RCL Message**
>
> *"• Gander: 60-90 minutes. Flights departing from airports less than 45 minutes flying time from the OEP should send the RCL 10 minutes prior to start-up. • Shanwick: 30-90 minutes."*
>
> Santa Maria is at least 40 minutes, Bodo at least 20 (FOM 22.6.4). Go to voice only if ACARS is inop, you get RCL REJECTED, or nothing comes back within 15 minutes (FOM 22.6.3). Shanwick telephone fallback is 01294-655100.
>
> *Workbook wording: How do you request an RCL and when?*

OCEANIC AND NAT

What is your oceanic exit point?

Flight specific; the point re-entering domestic airspace

FOM 22.1.6 identifies the Oceanic Exit Point (OXP) as the point of re-entry into domestic airspace, and directs crews to request Normal Speed via CPDLC or voice after it when ATC has assigned a fixed Mach. FOM 5.7.4.7 makes the SLOP boundary explicit: the Captain may apply an offset outbound at the oceanic entry point but must return to centerline at the oceanic exit point. FOM 22.1.11 sets the same bracket for ORCN procedures, which begin no later than the oceanic entry point and end no earlier than the oceanic exit point. In a total communications failure, FOM 22.7 has you maintain the current flight plan to the OXP with no route, level or speed change before it unless the PIC judges one necessary for safety.

> **FOM 22.1.6 Mach Number Technique and Cost Index (ECON)**
>
> *"When ATC assigns a fixed Mach number in Oceanic Airspace, crews should request Normal Speed (via CPDLC or voice) after the Oceanic Exit Point (OXP) upon re-entry into domestic airspace."*
>
> The OXP is taken off your flight plan, not defined in the FOM. Three things key off it: request Normal Speed after it if ATC assigned a fixed Mach (22.1.6 / 23.8), retain the last assigned squawk for 10 minutes past it (22.1.9), and return to centerline from SLOP at it (5.7.4.7).
>
> *Workbook wording: What is your oceanic Exit point?*

OCEANIC AND NAT

Initial call to Gander or Shanwick. Exact words?

Callsign, SELCAL check, and the next OCA

FOM 22.4 publishes the model call for an eastbound flight entering the Gander OCA: Gander Radio, Alaska 123, SELCAL Check, Shanwick Next. The response format shown is Alaska 123, Gander Radio, HF Primary and Secondary, at 30W contact Shanwick Radio HF Primary and Secondary. The rule behind it is that when operating outside VHF coverage, prior to entering each individual OCA the crew must perform a SELCAL check on the designated HF frequencies to establish and confirm voice communication. Establishing CPDLC does not relieve you - FOM 22.4 requires a continuous air-ground voice watch on the assigned HF frequency regardless. This is separate from the RCL, which is an ACARS or voice message covered in FOM 22.6.1 through 22.6.4.

> **FOM 22.4 Communication and Surveillance**
>
> *"Example (Initial contact from an eastbound flight entering GANDER OCA): "Gander Radio, Alaska 123, SELCAL Check, Shanwick Next.""*
>
> Expect the reply to hand you HF primary and secondary frequencies plus the next transfer point, e.g. at 30W contact Shanwick Radio. A new SELCAL check is required before entering each individual OCA, not each FIR (FOM 22.4).
>
> *Workbook wording: What does your initial call to Gander or Shanwick on VHF or HF require? (Exact verbiage (not RCL call))*

TCAS AND TRAFFIC

What transponder setting monitors traffic?

XPDR or above; that enables ADS-B (In)

FCOM 11.10 lists the Transponder/TCAS Mode Selector positions. STBY leaves the transponder inactive. ALT RPTG OFF enables modes A and S, disables altitude reporting, enables ADS-B out but disables ADS-B (In). XPDR enables modes A, C and S, enables altitude reporting in flight, and enables both ADS-B out and ADS-B (In); TA ONLY and TA/RA add the TCAS functions on top. ADS-B (In) is what feeds oceanic traffic to the ND and drives the In Trail Procedure application used to request a step climb across the pond (FCOM 5.40.29, ITP Flight Level Request with ADS-B (In)). Do not confuse the selector position with the code - FOM 22.1.9 governs the code, with Reykjavik and Bermuda radar exceptions where an assigned code is retained.

> **FCOM 11.10 Flight Management, Navigation - Controls and Indicators (Transponder/TCAS Mode Selector)**
>
> > *"• altitude reporting disabled • ADS-B out enabled • ADS-B (In) disabled XPDR (transponder) – • transponder enabled in modes A, C, and S"*
>
> ADS-B (In) is disabled in STBY and in ALT RPTG OFF, so anything below XPDR blinds you to traffic. Separately, FOM 22.1.9 sets the NAT squawk: code 2000 within NAT airspace unless ATC directs otherwise, retaining the last assigned code for 10 minutes after the Oceanic Exit Point.
>
> *Workbook wording: What should you set your transponder to in order to monitor traffic?*

OCEANIC AND NAT

When do you request a SELCAL check?

After CPDLC logon, before the initial gateway fix

When using CPDLC, contact ARINC after a successful CPDLC logon but prior to the initial Gateway Fix in order to obtain a SELCAL check on HF. This establishes a backup means of communication if CPDLC fails.

> **FOM 5.7.3.18 Position Reporting Oceanic (CPDLC)**
>
> > *""Flight Crews must contact ARINC after a successful CPDLC logon, but prior to the initial Gateway Fix in order to obtain a SELCAL check on HF.""*
>
> REV 125.1: if SELCAL is not functioning, one crew member must listen to HF continuously, which is a real workload consideration on a long Pacific crossing.

ETOPS

Is your ETOPS distance 120 or 180?

180 minutes MDT; as dispatched on the Release

FOM 6.2.3 sets the Alaska Airlines authorized Maximum Diversion Time at 180 minutes, computed at one-engine-inoperative cruise speed under standard conditions in still air, and requires the MDT the flight was planned on to be listed in the Dispatch Release. Before changing a planned route inside an ETOPS segment (weather deviations excepted) the crew must coordinate with Dispatch so the aircraft stays inside the Maximum Diversion Distance. The same section warns that a reroute can push you out of the ETOPS 180 airspace entirely. The MDT is a planning construct, not an operational limit on an actual diversion - prevailing winds can make the real diversion longer or shorter than the Release figure.

> **FOM 6.2.3 ETOPS Area of Operation**
>
> > *"The Alaska Airlines' authorized MDT is 180 minutes. For ETOPS up to and including 180 minutes, no"*
>
> MEL-deferred items can knock Hawaiian down to 120 minutes (FOM 6.2.3), so read the Release rather than assuming 180. Table 6.2.3(1) gives the 787 flying distances at .84M/290 KIAS: 410 nm at 60 min, 806 nm at 120 min, 1207 nm at 180 min.
>
> *Workbook wording: What is your ETOPS distance set to? (120 or 180)*

ETOPS

What are your common ETOPS airports?

Those listed on your Flight Plan/Dispatch Release

FOM 6.2.2 defines an ETOPS alternate as an adequate airport listed in FOM 8.1.3 and designated on the Flight Plan/Dispatch Release for use in the event of an ETOPS diversion, and it explicitly says this is a flight-planning definition that does not limit PIC authority in flight. FOM 6.2.1 carries the caution that an adequate airport may not be appropriate for an ETOPS diversion - it may lack rescue and firefighting support or have bad weather. Dispatch cannot list a field as an ETOPS alternate unless forecast weather is at or above the Ops Spec ETOPS alternate minima. FOM 6.4.1 then lets the Flight Crew, in concert with Dispatch, pick a different diversion airport if it is more suitable for prevailing conditions.

> **FOM 6.2.2 ETOPS Alternate Airport**
>
> *"An ETOPS alternate airport is an adequate airport that is listed in 8.1.3 – Authorized Airports and is designated on the Flight Plan/Dispatch Release for use in the event of a diversion during"*
>
> The master list is FOM 8.1.3 - Authorized Airports, where an E in the fleet column means the field may be listed as an ETOPS alternate. Anything coded A, F, R, P or E may also be planned as an adequate airport to define EEP/EXP. Do not memorize a route list; read the Release.

TAXI AND GROUND

Transition level by ATC. When do you set QNH?

In the FOM theater chapters and the Jeppesen pages

A level is flown on standard 29.92 (QNE); an altitude is flown on local QNH; QFE references height above field elevation. Transition altitude is charted and climbing; transition level is descending and often ATC-assigned. Regional specifics are in FOM chapters 19 to 24, for example the Oakland FIR 5500 ft crossover at 20.7.1.

> **FOM Ch 19-24 / Jeppesen**
>
> *Workbook wording: When do you set QNH when transition level states "by ATC"?*

OCEANIC AND NAT

Review the airspace around eastbound oceanic entry

GOTA transition area, then the Gander OCA

FOM 22.2.1 describes the transition areas adjacent to the major OCAs - GOTA, NOTA, SOTA and BOTA - as buffers that are not part of the OCAs, with aircraft in them remaining under domestic ATC: GOTA under Gander or Montreal Centers, NOTA and SOTA under Shannon Domestic, BOTA under Brest Domestic. FOM 22.1.1 bounds the NAT HLA itself as FL285 through FL420 inclusive within the Bodo, Gander, New York Oceanic East, Reykjavik, Santa Maria and Shanwick oceanic control areas, expressly excluding the Shannon and Brest Ocean Transition Areas. SLOP applies only from oceanic entry point to exit point and not in the transition areas (FOM 22.1.5). For the specific rules and exceptions of whatever airspace you are entering, FOM 22.2.1 sends you to the FIR card and Operational Notes in Jeppesen FD Pro.

> **FOM 22.2.1 Transition Areas**
>
> *"Transition areas such as the Gander Oceanic Transition Area (GOTA), Northern Oceanic Transition Area (NOTA), Shannon Oceanic Transition Area (SOTA), and Brest Oceanic Transition Area (BOTA)"*
>
> Two GOTA traps for eastbound entry: OEPs sit in the middle of the airspace so SLOP starts halfway through, and you send the RCL to Gander Oceanic while still talking to a domestic center (FOM 22.2.1). Boundaries are on the High/Low IFR enroute charts in Jeppesen FD Pro (FOM 22.2).
>
> *Workbook wording: Review airspace surrounding Eastbound oceanic entry*

OCEANIC AND NAT

Westbound into BIRD at RATSU. Oceanic clearance needed?

No. Only Shanwick still issues oceanic clearances

FOM 22.6.1 states that except for the Shanwick OCA, oceanic clearances are no longer issued in North Atlantic OCAs, and that for all other OCAs the pre-departure clearance serves as the oceanic clearance while the crew still transmits an RCL message as a notification. Reykjavik is one of the OACs listed in FOM 22.2, so a westbound entering BIRD gets no clearance issued but still owes the RCL. Note that FOM 22.6.4's published lead times cover only Gander, Shanwick, Santa Maria and Bodo, so Reykjavik timing has to come from the FIR card in FD Pro. Reykjavik is also a transponder exception under FOM 22.1.9 - codes it issues are retained throughout the Reykjavik OCA until ATC advises otherwise, rather than reverting to 2000.

FOM 22.6.1 Request for Clearance (RCL) and Oceanic Clearances

> *“Except for the Shanwick OCA, oceanic clearances are no longer issued in North Atlantic OCAs. However, Flight Crews are still required to transmit a Request for Clearance (RCL) message via voice”*

You still send an RCL message as a notification into Reykjavik; the pre-departure clearance serves as the oceanic clearance. RATSU itself is not named in the FOM - get the entry conditions off the Reykjavik FIR card and Operational Notes in Jeppesen FD Pro (FOM 22.2.1). Only New York East needs no RCL.

Workbook wording: Do westbound routes that enter BIRD from Scottish at RATSU need an oceanic clearance?

INTERNATIONAL THEATERS

Arrival and departure considerations on the Jepps

Transition altitude charted; transition level from ATIS or ATC

FOM 23.4 lists the ICAO chart and weather differences that bite on arrival and departure: visibility in meters or kilometers, QNH in hectopascals with some fields defaulting to QFE, and transition altitude/level typically low and country-dependent, with STAR charts that may just say TRANS LEVEL: BY ATC. FOM 23.10 repeats it - transition altitudes are normally charted, transition levels come from the ATIS or the approach controller. FOM 23.16.2 is the other departure/arrival trap: European controllers cancel intermediate SID and STAR altitude restrictions only with explicit phraseology such as cancel level restriction or climb unrestricted, and absent that you fly the published vertical profile. FOM 23.9 adds that terminal speed restrictions are enforced, with instructions like maintain maximum speed or minimum clean speed.

FOM 23.4 Weather (Europe Theater)

> *“• Transition Altitude/Level: Typically, low in Europe and can vary by country. Vigilance is required for correct altimeter settings on arrival and departure. STAR charts in Europe may indicate "TRANS LEVEL: BY ATC."”*

European transition altitudes run lower than US values and vary by country (FOM 22.1.7, 23.10). Altimeter settings come in hectopascals, visibility in meters or kilometers, and wind speed can be in meters per second. Jeppesen FD Pro search term for this is TRANSITION LEVEL.

Workbook wording: Arrival/Departure considerations – Jepps

TAXI AND GROUND

FCO and LHR departures. Who do you call?

Clearance Delivery; they issue TOBT/TSAT and any CTOT

FOM 18.1.25 defines the A-CDM set: TOBT is the airline's ready-for-pushback time, TSAT is ATC's start-up approval time built from TOBT plus any CTOT plus variable taxi time, TTOT is the target takeoff time, and CTOT is the NMOC-allocated slot with a minus 5 / plus 15 minute window. TOBTs and TSATs reach the crew through docking guidance systems, clearance delivery, data link or Operations. FCO adds ARDT, the time you report ready, which must be no later than 5 minutes past TOBT or you request a new slot, and TSE, a free text field in the clearance that is not a fix or part of your route (FOM 23.18.1.1). LHR ties remote holding eligibility to a CTOT that puts more than 30 minutes between TOBT and TSAT (FOM 23.17.1.11). If you cannot make TOBT notify Operations; if you cannot make TTOT/CTOT after pushback notify ATC as soon as possible.

> **FOM 18.1.25 Airport Collaborative Decision Making (A-CDM)**
>
> *"TOBTs and TSATs are communicated through various methods, including docking guidance systems, clearance delivery, data link, or Operations."*
>
> TSAT is set by ATC off your TOBT, any NMOC slot, and a variable taxi time - it is when you can expect pushback/start-up clearance. CTOT is a 20-minute window, minus 5 plus 15. At FCO, ARDT must be no later than TOBT plus 5 or you re-slot (FOM 23.18.1.1).
>
> *Workbook wording: Departure considerations – FCO/LHR. Who are you going to call and what are they giving you?*

TAXI AND GROUND

The ground frequency dance: clearance, ramp, clearance, ground

ATC on #1/L VHF, ramp control on #2/R

FOM 7.1.5 assigns #1/L VHF to primary communications including ATC, Tower, Ground, CTAF and Ramp Control; #2/R to monitoring 121.5 or Company in flight and Company on the ground; and #3/C to ACARS. The note that solves the ground shuffle reads that #2/R VHF may be used for ramp control when transitioning to and from the movement area. Practically that keeps the ATC chain on one radio while ramp lives on the other, which matters at slot-controlled European fields where you bounce between clearance delivery and ramp waiting on TSAT (FOM 18.1.25). Remember FOM 7.1.4 - boom microphone use is required below 18,000 ft, so all of this ground work is on the boom.

> **FOM 7.1.5 VHF Radio Priority**
>
> *"#2/R VHF may be used for ramp control when transitioning to/from the movement area."*
>
> The base rule puts ATC/Tower/Ground/CTAF/Ramp Control on #1/L, 121.5 or Company on #2/R, and ACARS on #3/C. The note lets you park ramp on #2/R so you can hold clearance or ground on #1/L through the handoffs instead of retuning one radio repeatedly.
>
> *Workbook wording: Frequency dance on the ground = Clearance-ramp-clearance-ground*

TAXI AND GROUND

Diversion airport considerations. Benefits and pitfalls?

Planned ETOPS alternates unless a better one exists

FOM 6.4.1 says the ETOPS alternates picked during flight planning are in some cases the optimum diversion airports, but the Flight Crew in concert with Dispatch may select another if it is more suitable for prevailing operational conditions. It then lists the standing crew duties: maintain awareness of suitable airports, have a current diversion plan in mind, track weather and NOTAMs at the diversion airport, be prepared to coordinate with Dispatch, and include the current diversion plan in the brief after a rest period. Two North Atlantic specifics to weigh: FOM 22.2.3 says Alaska does not designate any true-oriented airport as an ETOPS alternate and those should be considered only during emergencies, and the Greenland and Iceland fields have their own restrictions in FOM 22.11 and 22.12. If an engine quits, FOM 6.4.2 and 14 CFR 121.565 force the nearest suitable airport in the Captain's judgment, which can override the pretty answer.

> **FOM 6.4.1 Diversions**
>
> *"The Flight Crew, in concert with Dispatch, may select another diversion airport if they determine it is more suitable based the prevailing operational conditions."*
>
> The pitfall is in FOM 6.2.1's caution: an adequate airport may not be appropriate for an ETOPS diversion because it may lack rescue and firefighting support or be in bad weather. FOM 22.5 adds that NAT diversion stations can be severe in winter and occasionally closed and widely spaced.
>
> *Workbook wording: Diversion airport considerations - Benefits/pitfalls of which ones are used.*